Sinc Approximation of Multidimensional Hilbert Transform and Its Applications*

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February 15, 2021

Abstract

Many science and engineering applications require computing Hilbert transform. Sinc approximation is an efficient algorithm for computing one-dimensional (1D) Hilbert transform. In this paper, we develop Sinc approximation for computing multidimensional Hilbert transform and analyze its convergence rate. We apply our method to two applications: detecting edges of 2D images, and pricing discretely monitored barrier options and calculating survival probabilities in two-asset/three-asset models where the prices follow exponential Lévy processes. Extensive numerical experiments confirm the efficiency of Sinc approximation for computing 2D and 3D Hilbert transforms in these applications.

Key Words: Hilbert transform, Sinc approximation, image analysis, Lévy processes, barrier options, survival probability.

1 Introduction

The Hilbert transform for an n-variable function $f(x_1, \cdots, x_n)$ is defined as

$$\mathcal{H}_n f(\xi_1, \cdots, \xi_n) := \frac{1}{\pi^n} P.V. \int_{\mathbb{R}^n} f(x_1, \cdots, x_n) \prod_{j=1}^{n} \frac{dx_j}{\xi_j - x_j},$$

where $P.V.$ refers to the Cauchy principle value integration (c.f. Eq.(15.26) of King (2009)). Hilbert transform is an important transform that finds itself useful in many areas such as signal processing (King (2009)), image analysis (Kohlmann (1996), Lorenzo-Ginori (2007)), financial engineering (Feng and Linetsky (2008a, 2009) and Feng and Lin (2013b)) and simulation (Chen et al. (2012) and Feng and Lin (2013a)). In these applications, efficiently computing the Hilbert transform is required. For $n = 1$, a well-known method is Sinc approximation, which converges rapidly (Stenger (1993)). This method approximates a one-variable function $f(x)$ with some properties by Sinc expansion as

$$f(x) \approx \sum_{k=-\infty}^{\infty} f(kh)S(k,h)(x), \quad S(k,h)(x) = \frac{\sin(\pi(x-kh)/h)}{\pi(x-kh)/h}, \tag{1.1}$$

*This research was supported by Hong Kong Research Grant Council General Research Fund Grant No. 14203418.

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where $S(k,h)(x)$ is a Sinc function. Utilizing this approximation yields the approximation for the Hilbert transform of f as

$$\mathcal{H}f(\xi) \approx \sum_{k=-\infty}^{\infty} f(kh) \frac{1-\cos[\pi(\xi-kh)/h]}{\pi(\xi-kh)/h}.$$

To compute the infinite series, one can truncate it at an appropriate level. A complete convergence theory of Sinc approximation in the one-dimensional case is developed in Stenger (1993).

In this paper, we generalize Sinc approximation for computing Hilbert transform in an arbitrary dimension and obtain its convergence rate. We then establish the efficiency of this method for $n = 2$ or 3 by considering two applications.

(1) The first application is detecting edges of two-dimensional images, which are represented by two-variable functions. The edge points of an image are given by the local extrema of the two-dimensional (2D) Hilbert transform of the function representing it. We utilize Sinc approximation to compute the 2D Hilbert transform and compare it with the standard discrete Hilbert transform method considered in Kohlmann (1996).

(2) The second application is pricing discretely monitored two-asset or three-asset barrier options and calculating joint survival probability for two or three firms when the asset prices follow exponential Lévy models, which are a popular class of models in finance. In this application, we need to solve a backward recursion problem. We carry out the recursion in the Fourier space for the value function, where Hilbert transform appears in each recursion step and it is calculated by Sinc approximation. We develop efficient implementation of the recursive procedure, analyze how the approximation error propogates as the number of recursion steps grows and compare our method with finite difference, Monte Carlo simulation and Fourier cosine expansion method which are standard methods in the literature for this problem.

1.1 Related Literature on Barrier Option Pricing and Survival Probability Calculation

Barrier options are among the most popular path-dependent options in financial markets and their payoffs depend on whether the price of the underlying asset crosses some pre-specified barriers over the monitoring period. In practice, most barrier options are monitored over a discrete set of dates. Another important financial problem that involves barrier crossing is calculating the survival or default probability of firms. In structural credit risk models, the default of a firm occurs when its asset value falls below some pre-determined default barrier (Black and Cox (1976)). The problem of computing the firm's survival probability can thus be formulated as pricing a down-and-out option written on the firm's asset value with unit payoff.

There exist a large number of works on pricing single-asset barrier options and calculating a single firm's survival probability which we list below.

(1) For specific exponential Lévy models: Wiener-Hopf technique (Fusai et al. (2006)) for the Black-Scholes model and fast Gauss transform (Broadie and Yamamoto (2005)) for models where the return distribution is a mixture of Gaussians.

(2) For general exponential Lévy models: finite difference (Cont and Voltchkova (2005)) and finite element (Feng and Linetsky (2008b)) schemes for solving the associated partial integro-differential equation (PIDE) (these schemes can be adapted by enforcing barrier checking only on monitoring dates), the Laplace transfrom approach based on Spitzer's identity (Petrella and Kou (2004)), the Fourier cosine method (Fang and Oosterlee (2009)), the fast Hilbert transform approach (Feng and Linetsky (2008a)), piecewise polynomial interpolation with Fourier inversion (De Innocentis and Levendorskiĭ (2013)), a combined approach of Hilbert and z-transform (Fusai et al. (2016)) and the local basis method (Kirkby (2017, 2018)).

(3) Outside the class of Lévy processes: numerical quadrature for the CEV diffusion (Fusai and Recchioni (2007)), an eigenfunction expansion approach (Li and Linetsky (2015)) for

Markov models with discrete spectra and Markov chain approximation (Mijatovic and Pistorius (2013), Li and Zhang (2018), Zhang and Li (2019)) for general Markov models.

Despite the extensive literature on the single-asset case, the study for the multi-asset case is rather scarce. In fact, multi-asset discretely monitored barrier options are quite popular products in especially equity and foreign exchange markets. For example, a common type of multi-currency knock-out options (with the two or three currencies being the most common) delivers a payoff that depends on the value of one or all currencies on the maturity date if none of the currencies breach their individual barriers during the monitoring period. Furthermore, knowledge of the joint survival probability is important for managing credit portfolios and valuation adjustments (Lando (2009), Kou and Zhong (2016), Lipton and Savescu (2013, 2014)). Thus, the multi-asset case is practically important and it deserves more research.

There are several standard computational methods that can be used for the multi-asset case. (1) Monte Carlo simulation is free of the curse of dimensionality (e.g., Giles and Xia (2017), Gobet (2000)), but for low dimensions other methods can be significantly faster for obtaining the same level of accuracy. (2) Numerical schemes for solving PIDEs discretize both time and space, and their convergence rate is typically algebraic in the time step and mesh size. This can be too slow when high levels of accuracy are demanded. (3) For the multivariate Black-Scholes model, an efficient method using quadrature rules was developed in Andricopoulos et al. (2007). However, it cannot be applied to general models in which the joint transition density of the underlying assets is unknown.

Our method is applicable to multivariate exponential Lévy models, which are standard choices for modelling asset prices in a variety of markets (Cont and Tankov (2004), Schoutens (2003)) as well as firms' asset values (Schoutens and Cariboni (2010)). This general class of models includes the multivariate Black-Scholes model as well as many classical jump models as special cases. The joint transition density of many multivariate Lévy processes is not known. Our method can be applied as long as the joint characteristic function of the multivariate Lévy process is known, which is given by the Lévy-Khintchine formula, and for many model specifications it has a closed-form expression. In particualr, it can be applied to multivariate Lévy models with different marginals if the joint characteristic function is known.

The convergence speed of our method depends on the type of the Lévy model under consideration. In general, our method converges rapidly when there is a Brownian motion component in the Lévy process, i.e., for the multidimensional Black-Scholes model and jump-diffusions. As these models are often used for fitting the data of multiple assets, our method would perform well in many practical applications. When the process is a pure-jump one, the convergence speed hinges on the decay rate of the joint characteristic function, which for some models, like the Variance Gamma (VG) model, is unfortunately slow. Spectral filters have been successfully applied in Phelan et al. (2018) to accelerate convergence of the Hilbert transform method of Feng and Linetsky (2008a) for the one-dimensional VG model. In this paper, we show that multivariate spectral filters can also hasten convergence of our method for the multidimensional VG model if the horizon of the problem is not too long.

Remark 1.1. After our paper was completed, we learned the work on using 2D Hilbert transform for pricing two-asset barrier options in the dissertation of Lin (2010). There are several major differences between this paper and his work. First, we deal with an arbitrary dimension whereas Lin (2010) only considers the 2D case. Second, Lin (2010) analyzes discrete approximation error for the 2D Hilbert transform of the Fourier transform of some function only, whereas our analysis deals with multidimensional Hilbert transform of general functions. Third, Lin (2010) does not provide analysis of the aggregate error for the recursive algorithm for pricing two-asset barrier options. Fourth, Lin (2010) does not discuss how to implement the approximation formula for computing 2D Hilbert transform. As our discussions show, efficient implementation is key to the performance of the algorithm. Fifth, there are no numerical examples in Lin (2010) to evaluate the performance of the method for different models.

Remark 1.2. Kou and Zhong (2016) and Lipton and Savescu (2014) study continuously monitored first passage problem of 2D and 3D correlated Brownian motions, respectively to obtain joint survival or default probabilities. They deal with the associated PDEs and in the 2D case, Kou and Zhong (2016) found a closed-form solution while for the 3D case, Lipton and Savescu (2014) applied the eigenfunction expansion approach to obtain the Green's function with eigenvalues and eigenfunctions computed by the finite element method. Our focus in this paper is on the discretely monitored problem, but we can use the solution with high monitoring frequencies to approximate the continuously monitored solution if the latter is wanted.

1.2 Some Notations

Throughout the paper bold letters denote column vectors and matrices, while $'$ represents transpose. We use $\circ$ for pointwise multiplication and $*$ for matrix multiplication. For an n-variable function $f(x_1,\cdots,x_n)$, we will simply write it as $f(\boldsymbol{x})$ with $\boldsymbol{x}=(x_1,\cdots,x_n)'$. In addition, $d\boldsymbol{x}$ should be understood as $dx_1\cdots dx_n$. For an n-dimensional vector, $|\cdot|$ denotes its Euclidian norm. The following shorthand notations are also adopted.

$$\mathbb{1}_{(\boldsymbol{l},\infty)}(\boldsymbol{x})=\prod_{p=1}^{n}\mathbb{1}_{(l_p,\infty)}(x_p),\ \boldsymbol{x}^{\boldsymbol{\alpha}}=\prod_{p=1}^{n}x_p^{\alpha_p},$$
$$\boldsymbol{S}/\boldsymbol{K}=(S_1/K_1,\cdots,S_n/K_n)',\ \ln\boldsymbol{S}=(\ln S_1,\cdots,\ln S_n)'.$$

For two vectors, $\boldsymbol{\alpha}\le(<)\boldsymbol{\beta}$ means $\alpha_p\le(<)\beta_p$ for all $1\le p\le n$. We write $\mathbf{0},\mathbf{1}$ as the vector of all 0s, 1s, respectively. In addition, for an n-dimensional vector $\boldsymbol{\xi}=(\xi_1,\cdots,\xi_n)$ and $\boldsymbol{\beta}\in\{0,1\}^n$ (the set of n-dimensional vectors with each entry either 0 or 1), we define $\boldsymbol{\xi}_{\boldsymbol{\beta}}$ to be a n-dimensional vector whose p-th coordinate equals to ξ_p if $\beta_p=1$ and 0 if $\beta_p=0$, and $|\boldsymbol{\xi}|_1=\sum_{p=1}^n|\xi_p|$. In particular, $\boldsymbol{\xi}_{\boldsymbol{\beta}}=\mathbf{0}$ if $\boldsymbol{\beta}=\mathbf{0}$ and $\boldsymbol{\xi}_{\boldsymbol{\beta}}=\boldsymbol{\xi}$ if $\boldsymbol{\beta}=\mathbf{1}$. We also use $\sum$ to denote summation with all dimensions and $\sum^{\boldsymbol{\beta}}$ for summation over all dimensions with $\beta_p=1$. Finally, $L^p(\mathbb{R}^n)$ and $L^p(\mathbb{R}^n,\mathbb{C})$ refer to the L^p space of real-valued and complex-valued functions, respectively.

The rest of the paper is organized as follows. Section 2 develops Sinc approximation of multidimensional Hilbert transform, estimates its convergence rate and shows how to implement it. Section 3 applies the method to the edge detection problem for 2D images. Section 4 develops a recursive algorithm based on Sinc approximation to price barrier options and calculate survival probabilities in multivariate Lévy models. We also estimate its aggregate error and develop an efficient implementation. Section 5 presents various numerical examples and Section 6 concludes. The online companion provides supplementary materials including all proofs.

2 Sinc Approximation of Multidimensional Hilbert Transform

We approximate an n-variable function $f(\boldsymbol{x})$ as

$$f(\boldsymbol{x})\approx\sum_{\boldsymbol{k}\in\mathbb{Z}^n}\left[f(\boldsymbol{kh})\prod_{j=1}^{n}S(k_j,h_j)(x_j)\right],\tag{2.1}$$

where $S(k,h)(\cdot)$ is the Sinc function defined in (1.1), and $\sum_{\boldsymbol{k}\in\mathbb{Z}^n}=\sum_{k_1=-\infty}^{\infty}\cdots\sum_{k_n=-\infty}^{\infty}$. Using this expansion and interchanging integration and summation, we obtain an approximation for the Hilbert transform:

$$\mathcal{H}_nf(\boldsymbol{\xi})\approx\sum_{\boldsymbol{k}\in\mathbb{Z}^n}\left[f(\boldsymbol{kh})\prod_{j=1}^{n}\frac{1-\cos[\pi(\xi_j-k_jh_j)/h_j]}{\pi(\xi_j-k_jh_j)/h_j}\right].\tag{2.2}$$

Calculating a multiple series can be computationally expensive, but for low dimensions (two or three) our method could perform well.

2.1 Convergence Rate of Sinc Approximation

The multimensional Hilbert transform $\mathcal{H}_n f$ is well-defined almost everywhere for $f \in L^p(\mathbb{R}^n)$ with $p \geq 1$ (Section 15.1, King (2009)). To analyze the convergence rate of Sinc approximation, we will consider square-integrable f and assume the behavior of the Fourier transform of f. Recall that the n-dimensional Fourier transform and inverse Fourier transform are defined as

$$\mathcal{F}_n f(\boldsymbol{\xi}) = \int_{\mathbb{R}^n} f(\boldsymbol{x}) e^{i\boldsymbol{\xi}'\boldsymbol{x}} d\boldsymbol{x} \ (\boldsymbol{\xi} \in \mathbb{R}^n), \ \mathcal{F}_n^{-1} f(\boldsymbol{x}) := \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} f(\boldsymbol{\xi}) e^{-i\boldsymbol{\xi}'\boldsymbol{x}} d\boldsymbol{\xi},$$

respectively, provided that these integrals exist. Throughout the paper, we also write $\hat{f}(\boldsymbol{\xi})$ for $\mathcal{F}_n f(\boldsymbol{\xi})$. Let $E(f, \boldsymbol{h})(\boldsymbol{x})$ and $E_H(f, \boldsymbol{h})(\boldsymbol{\xi})$ be the approximation error in (2.1) and (2.2).

Theorem 2.1. *Let $f \in L^2(\mathbb{R}^n)$, and there is a positive constant d such that*

$$|\hat{f}(\boldsymbol{\xi})| = \mathcal{O}(e^{-d|\boldsymbol{\xi}|}), \ |\boldsymbol{\xi}| \to \infty,$$

then as $|\boldsymbol{h}| \to 0$,

$$|E(f, \boldsymbol{h})(\boldsymbol{x})| = \mathcal{O}\left(\prod_{j=1}^{n} e^{-\frac{\pi d}{\sqrt{n} h_j}}\right), \ |E_H(f, \boldsymbol{h})(\boldsymbol{\xi})| = \mathcal{O}\left(\prod_{j=1}^{n} e^{-\frac{\pi d}{\sqrt{n} h_j}}\right).$$

If $|\hat{f}(\boldsymbol{\xi})| = \mathcal{O}\left(|\boldsymbol{\xi}|^{-\alpha}\right), \alpha > 1$, then the convergence order in the above becomes $\mathcal{O}\left(\prod_{j=1}^{n} h_j^{\alpha-1}\right)$.

The theorem shows that convergence rate of Sinc approximation is determined by the decay rate of the Fourier transform of f if f is square-integrable. Exponential (polynomial) decay of the Fourier transform leads to exponential (polynomial) convergence of Sinc approximation. The result generalizes Theorem 1.3.2 in Stenger (2011).

We will apply Theorem 2.1 for the edge dection problem, but for the pricing of barrier options and calculation of survival probabilities, the functions involved do not fulfill the conditions in the theorem. In the following, we consider another class of functions which are analytic in a tube domain and show that Sinc approximation converges exponentially for them.

Let's first recall the class of one-variable analytic functions considered in Stenger (1993). For a positive constant d, define

$$\mathcal{D}_d := \{z \in \mathbb{C} : |\text{Im} z| < d\},$$

which is a strip containing the real axis. Let $\boldsymbol{H}^1(\mathcal{D}_d)$ denote the family of all complex-valued functions analytic in $\mathcal{D}_d$, such that $\int_{-d}^{d} f(x+iy)dy \to 0$ as $x \to \pm\infty$ and

$$\int_{\mathbb{R}} |f(x+id^-)|dx + \int_{\mathbb{R}} |f(x-id^-)|dx := \lim_{y \to d^-} \int_{\mathbb{R}} |f(x+iy)|dx + \lim_{y \to d^-} \int_{\mathbb{R}} |f(x-iy)|dx$$

is finite. Now we consider the multivariate case. We recall Hartogs' theorem which states that a P-variable complex function $f(z_1, \cdots, z_P)$ is analytic in $\mathcal{S}_1 \times \cdots \times \mathcal{S}_P$ if and only if for each $1 \leq p \leq P$, f as a function of z_p is analytic in $\mathcal{S}_p$ for any given value of $z_q \in \mathcal{S}_q$ ($q \neq p$, $1 \leq q \leq P$). We define $\boldsymbol{H}^1(\mathcal{D}_{\boldsymbol{d}})$ ($\boldsymbol{d} > \boldsymbol{0}$), a class of n-variable complex functions, which consists of $f(\boldsymbol{z})$ that is analytic in the tube domain $\mathcal{D}_{\boldsymbol{d}} = \mathcal{D}_{d_1} \times \cdots \times \mathcal{D}_{d_n}$ and for any $\boldsymbol{\beta} \in \{0,1\}^n, \boldsymbol{\beta} \neq 0$,

$$\int_{\mathbb{R}^{|\boldsymbol{\beta}|_1}} \left| f(\boldsymbol{x}_{\boldsymbol{\beta}} \pm i\boldsymbol{d}_{\boldsymbol{\beta}}^- + \boldsymbol{z}_{\boldsymbol{1}-\boldsymbol{\beta}}) \right| d\boldsymbol{x}_{\boldsymbol{\beta}} \leq C, \tag{2.3}$$

$$\lim_{x_j \to \pm\infty, \ \beta_j = 1} \int_{\prod_{\{j:\beta_j=1\}}(-d_j, d_j)} f(\boldsymbol{x}_{\boldsymbol{\beta}} + i\boldsymbol{y}_{\boldsymbol{\beta}} + \boldsymbol{z}_{\boldsymbol{1}-\boldsymbol{\beta}}) d\boldsymbol{y}_{\boldsymbol{\beta}} = 0. \tag{2.4}$$

with $\boldsymbol{x} = (x_1, \cdots, x_n)'$, $x_j = \mathrm{Re}(z_j)$ for $j = 1, \cdots, n$ and C is a positive constant, independent of $\boldsymbol{z} := (z_1, \cdots, z_n)' \in \mathcal{D}_{\boldsymbol{d}}$. In the following discussion, we will also need the integral of f on $\mathbb{R}^n$, which we approximate by the multidimensional rectangle rule below derived from Sinc expansion:

$$\int_{\mathbb{R}^n} f(\boldsymbol{x}) d\boldsymbol{x} \approx \left(\prod_{j=1}^{n} h_j \right) \sum_{\boldsymbol{k} \in \mathbb{Z}^n} f(\boldsymbol{kh}).$$

The approximation error is denoted by $E_I(f, \boldsymbol{h})$. The next theorem provides estimates of these errors, which are counterparts of the 1D estimates in Theorems 3.1.3, 3.2.1 and 3.4.4 of Stenger (1993) for a general dimension.

Theorem 2.2. *Consider $f(\boldsymbol{z}) \in \mathcal{H}^{\boldsymbol{1}}(\mathcal{D}_{\boldsymbol{d}})$. For $z_j = x_j + iy_j \in \mathcal{D}_{d_j}, j = 1, \cdots, n$,*

$$|E(f, \boldsymbol{h})(\boldsymbol{z})| \leq \sum_{\boldsymbol{\beta} \in \{0,1\}^n, \boldsymbol{\beta} \neq \boldsymbol{0}} \left[C_{\boldsymbol{\beta}} \prod_{\{p: \beta_p = 1\}} \frac{e^{-\pi(d_p - |y_p|)/h_p}(1 + e^{-2\pi|y_p|/h_p})}{(d_p - |y_p|)(1 - e^{-2\pi d_p/h_p})} \right], \tag{2.5}$$

$$|E_I(f, \boldsymbol{h})| \leq \sum_{\boldsymbol{\beta} \in \{0,1\}^n, \boldsymbol{\beta} \neq \boldsymbol{0}} \left[C_{\boldsymbol{\beta}} \prod_{\{p: \beta_p = 1\}} \frac{e^{-2\pi d_p/h_p}}{1 - e^{-2\pi d_p/h_p}} \right], \tag{2.6}$$

$$|E_H(f, \boldsymbol{h})(\boldsymbol{\xi})| \leq \sum_{\boldsymbol{\beta} \in \{0,1\}^n, \boldsymbol{\beta} \neq \boldsymbol{0}} \left[C_{\boldsymbol{\beta}} \prod_{\{p: \beta_p = 1\}} \frac{e^{-\pi d_p/h_p}}{d_p(1 - e^{-\pi d_p/h_p})} \right]. \tag{2.7}$$

For any $\boldsymbol{\beta}$, the positive constant $C_{\boldsymbol{\beta}}$ is independent of $\boldsymbol{h}$, $\boldsymbol{z}$ and $\boldsymbol{\xi}$.

2.2 Implementation of Sinc Approximation of Hilbert transform

For an n-variable function $f(\boldsymbol{x})$, we compute its Hilbert transform $\mathcal{H}_n f(\boldsymbol{\xi})$ by Sinc Approximation (2.2). As the series in (2.2) is infinite, we truncate each k_j at level $-M_j$ and M_j and compute $\mathcal{H}_n f(\boldsymbol{\xi})$ for $\boldsymbol{\xi}$ on a grid $\{\boldsymbol{kh}, -M_j \leq k_j \leq M_j, 1 \leq j \leq n\}$.

Define

$$c(k, m) := \frac{1 - (-1)^{k-m}}{\pi(k - m)}, k \neq m, \ c(k, k) := 0. \tag{2.8}$$

Then,

$$\begin{aligned} \mathcal{H}_n f(\boldsymbol{kh}) &\approx \sum_{m_1=-M_1}^{M_1} \cdots \sum_{m_n=-M_n}^{M_n} \left[f(\boldsymbol{mh}) \prod_{j=1}^{n} c(k_j, m_j) \right] \\ &= \sum_{m_n=-M_n}^{M_n} c(k_n, m_n) \cdots \sum_{m_2=-M_2}^{M_2} c(k_2, m_2) \sum_{m_1=-M_1}^{M_1} c(k_1, m_1) f(\boldsymbol{mh}) \end{aligned}$$

For each $q \in \{1, \cdots, n\}$, construct a matrix $\boldsymbol{T}^q$ with $T^q_{km} = c(k, m)$ $(-M_q \leq k_q, m_q \leq M_q)$. Each T^q is a Toeplitz matrix and we can utilize fractional FFT to efficiently implement Toeplitz matrix-vector multiplication which costs $O(M \log_2 M)$ if the size of the matrix is M. We refer readers to Feng and Linetsky (2008a) Appendix B for details on this implementation. Let $\boldsymbol{u_0}(\boldsymbol{kh}) = f(\boldsymbol{kh})$ and we calculate $\boldsymbol{u_q}$ recursively from $q = 1$ to n as follows. For every tuple $(k_1 h_1, \cdots, k_{q-1} h_{q-1}, k_{q+1} h_{q+1}, \cdots, k_n h_n)$, compute

$$\begin{aligned} &\boldsymbol{u_q}(k_1 h_1, \cdots, k_{q-1} h_{q-1}, :, k_{q+1} h_{q+1}, \cdots, k_n h_n) \\ &= \boldsymbol{T}^q * \boldsymbol{u_{q-1}}(k_1 h_1, \cdots, k_{q-1} h_{q-1}, :, k_{q+1} h_{q+1}, \cdots, k_n h_n) \end{aligned}$$

by using fractional FFT for the Toeplitz matrix-vector multiplication.

Suppose $M_1 = \cdots = M_n = M$. The computational cost of the algorithm is $O(nM^n \log_2 M)$ and the space requirement is $O(M^n)$ as we need to store u_q for the current q. As we will show, the algorithm works well for 2D and 3D problems.

3 Edge Detection in 2D Images

A 2D image is represented by a two-variable function $f(x_1, x_2)$. Kohlmann (1996) shows that the edge points of a 2D image correspond to the local extrema of the Hilbert transform of the image function. To compute the 2D Hilbert transform, the Kohlmann method is based on the equation

$$\widehat{\mathcal{H}_2 f}(\xi_1, \xi_2) = (-i\mathrm{sgn}(\xi_1))(-i\mathrm{sgn}(\xi_2))\hat{f}(\xi_1, \xi_2)$$

where the Fourier transform of the Hilbert transform equals the Fourier transform of the original function multiplied with some sign functions. The Kohlmann method employs discretization to calculate $\hat{f}$ via fractional FFT (fast Fourier Transform) and then invert the Fourier transform (again via fractional FFT) to recover the Hilbert transform. We compare this approach with Sinc approximation by way of an example. Consider a function given in Kohlmann (1996) where $f(x_1, x_2) = \mathbb{1}_{(20,40)}(x_1)\mathbb{1}_{(20,40)}(x_2)$ and

$$\mathcal{H}_2 f(\xi_1, \xi_2) = \frac{1}{\pi^2} \ln\left|\frac{\xi_1 - 20}{\xi_1 - 40}\right| \ln\left|\frac{\xi_2 - 20}{\xi_2 - 40}\right|,$$

which has four sigularities: $(20, 20)$, $(20, 40)$, $(40, 20)$ and $(40, 40)$.

This is a toy problem but it suffices to demonstrate the essential difference between the two methods. Moreover, the analytical formula for the Hilbert transform makes it easy to obtain an accurate benchmark. For both methods, we descretize over the region $(0, 50) \times (0, 50)$ for (ξ_1, ξ_2) and set $M = 2^{10}$.

The left plot in Figure 1 shows the value of the Hilbert transform over a range for ξ_1 with $\xi_2 = 19$ when the same M is used for both methods. Our method is much more accurate than the Fourier transform approach especially when ξ_1 is away from the peak and trough. The right plot displays the maximum error of the two methods taken over a bounded range for (ξ_1, ξ_2) (except a small neighborhood around 20 and 40) as the number of terms increases. The plot shows that our method converges uniformly in contrast to the Fourier transform approach, which still converges at each point of the domain, but it requires a different number of terms to reach the same accuracy level at a different point because convergence is not uniform. Uniform convergence is unequivocally desirable, which is an advantage of our approach.

From the right plot, we observe that Sinc approximation for Hilbert transform converges at first order (the slope of the regression line is estimated to be -0.98), which is in line with the theoretical estimate in Theorem 2.1. To see this, note that f is square-integrable and its Fourier transform is given by

$$\hat{f}(\boldsymbol{\xi}) = -\frac{1}{\xi_1 \xi_2}\left(e^{i40\xi_1} - e^{i20\xi_1}\right)\left(e^{i40\xi_2} - e^{i20\xi_2}\right),$$

with $|\hat{f}(\boldsymbol{\xi})| = \mathcal{O}\left(|\boldsymbol{\xi}|^{-2}\right)$. Thus, Theorem 2.1 implies the convergence order of Since approximation is one.

Although in this example Sinc approximation only attains polynomial convergence, it can converge exponentially for functions such as those whose Fourier transform decays exponentially or those that are analytic in a tube domain. By contrast, the Fourier transform approach can only converge polynomially.

4 Barrier Options and Survival Probabilities

The asset price processes $S_t^j, 1 \le j \le n$ are assumed to follow

$$S_t^j = K_j e^{X_t^j}, \text{ with } X_0^j = \ln(S_0^j / K_j). \tag{4.1}$$

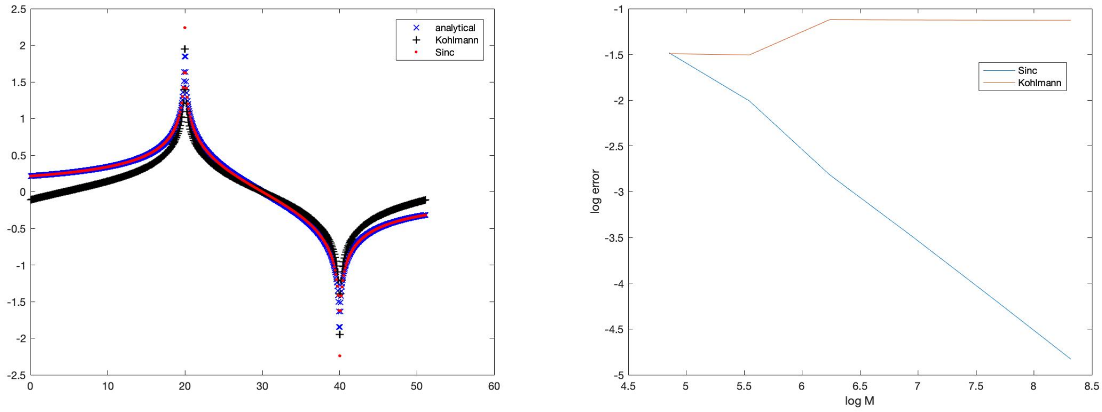


Figure 1: Hilbert transform with $\xi_2 = 19$ computed by the analytical formula, Sinc approximation and K. Kohlmann method (left) and convergence of the maximum error of the two approximation methods (right).

where K_j are positive constants chosen for convenience and $(X_t^1, \cdots, X_t^n)$ is an n-dimensional Lévy process. A review of multidimensional Lévy process can be found in Appendix A. We need the joint characteristic function of $\boldsymbol{X}_t - \boldsymbol{x}$, which is given by

$$\phi_t(\boldsymbol{\xi}) := E\left[e^{i\langle \boldsymbol{\xi}, \boldsymbol{X}_t - \boldsymbol{x}\rangle}\right] = e^{-t\psi(\boldsymbol{\xi})},$$
$$\psi(\boldsymbol{\xi}) = \frac{1}{2}\boldsymbol{\xi}'\boldsymbol{\Sigma}\boldsymbol{\xi} - i\boldsymbol{\mu}'\boldsymbol{\xi} + \int_{\mathbb{R}^2}(1 - e^{i\boldsymbol{\xi}'\boldsymbol{y}} + i\boldsymbol{\xi}'\boldsymbol{y}1_{\{|\boldsymbol{y}|\le 1\}})\Pi(d\boldsymbol{y}),$$

where $\psi(\boldsymbol{\xi})$ is the characteristic exponent of $\boldsymbol{X}$. We also consider

$$\mathcal{I}_{\boldsymbol{X}} := \left\{\boldsymbol{\alpha} \in \mathbb{R}^n : \int_{|\boldsymbol{y}|>1} e^{-\boldsymbol{\alpha}'\boldsymbol{y}}\Pi(d\boldsymbol{y}) < \infty\right\}.$$

Theorem 25.17 in Sato (1999) shows that $\mathcal{I}_{\boldsymbol{X}}$ is a convex set that contains the origin and $\boldsymbol{\alpha} \in \mathcal{I}_{\boldsymbol{X}}$ if and only if $E_{\boldsymbol{x}}[e^{-\boldsymbol{\alpha}'\boldsymbol{X}_t}]$ is finite for all $t > 0$.

4.1 Problem Formulation

We consider pricing a discretely monitored multi-asset down-and-out barrier option monitored over dates $\{t_1, \cdots, t_N = T\}$ ($t_j > 0$ for $1 \le j \le N$ and we set $t_0 = 0$). Our method can deal with arbitrary time increments, but for notational convenience, in the following, we simply assume $t_j = j\Delta$. The terminal payoff function is $h(\boldsymbol{x})$, which is paid out if knock-out does not happen by T. Here, we assume that monitoring of knock-out is done separately for each asset. This case holds for many multi-asset barrier options and the joint survival probability problem.

Throughout this paper, as the discount factor is a constant, we only consider the undiscounted barrier option price at time 0, which is given by

$$E_{\boldsymbol{x}}\left[\prod_{j=1}^{N} \mathbb{1}_{(\boldsymbol{l},\infty)}(\boldsymbol{X}_{j\Delta})h(\boldsymbol{X}_{N\Delta})\right].$$

To keep notations simple, we also assume the barriers are time-independent. In general, they can be time-varying. Our method can be easily extended to handle this case.

Remark 4.1 (Other knock-out types). In general, for each asset, we can have three knock-out situations: down-and-out, up-and-out and double barrier. Thus, for multi-asset barrier options, we have a total of 3^n combinations. We can solve all these problems using the multidimensional Hilbert transform. However, as it would be tedious to write down results for all the cases, we will only present our approach for the case in which all assets are down-and-out. It is also the setting for the calculation of joint survival probabilities in structural credit risk models. For formulas in the other cases, please refer to the thesis of Fan (2018).

Remark 4.2 (Typical payoffs). In practice, typical payoffs for multi-asset barrier options include call/put payoff on a single asset as well as call/put payoff on the sum of two assets, i.e., $\left(\sum_{j=1}^n S_T^j - K\right)^+$ and $\left(K - \sum_{j=1}^n S_T^j\right)^+$. Without loss of generality, we assume the weight of each asset in the portfolio is one. The general positive weight case can be reduced to this case by absorbing the weights into the starting points. The joint survival probability can be thought as a down-and-out option with $h(\boldsymbol{x}) \equiv 1$.

4.2 Fourier Representation of Operators

Consider the transition operator of $\boldsymbol{X}_t$, which is defined as $P_t f(\boldsymbol{x}) = E_{\boldsymbol{x}}[f(\boldsymbol{X}_t)]$. We wish to represent $P_t f(\boldsymbol{x})$ as a Fourier integral to make use of the analytical knowledge of the characteristic function. For the applications considered in this paper, f is not in $L^1(\mathbb{R}^n) := L^1(\mathbb{R}^n, d\boldsymbol{x})$ and neither is $P_t f$. Therefore, we cannot directly compute the Fourier transform of $P_t f$. To overcome this difficulty, we apply the ideas of exponential damping and Esscher transform in a multidimensional setting (see e.g., Cont and Tankov (2004) and Boyarchenko and Levendorskii (2002a)). Previously these ideas were applied to 1D problems.

For a multidimensional vector $\boldsymbol{\alpha}$, define $f_{\boldsymbol{\alpha}}(\boldsymbol{x}) := e^{\boldsymbol{\alpha}'\boldsymbol{x}} f(\boldsymbol{x})$, and let $L^1_{\boldsymbol{\alpha}}(\mathbb{R}^n) := L^1(\mathbb{R}^n, e^{\boldsymbol{\alpha}'\boldsymbol{x}} d\boldsymbol{x})$. The norm of $L^1(\mathbb{R}^n)$ and of $L^1_{\boldsymbol{\alpha}}(\mathbb{R}^n)$ are denoted by $\|\cdot\|_{L^1(\mathbb{R}^n)}$ and $\|\cdot\|_{L^1_{\boldsymbol{\alpha}}(\mathbb{R}^n)}$, respectively. Suppose $\boldsymbol{\alpha} \in \mathcal{I}_{\boldsymbol{X}}$. Let $Z_t^{\boldsymbol{\alpha}} := e^{-\boldsymbol{\alpha}'\boldsymbol{X}_t + t\psi(i\boldsymbol{\alpha})}$. The integrability of $Z_t^{\boldsymbol{\alpha}}$ is guaranteed by Sato (1999), Theorem 25.17. Then, it is easy to see that $Z_t^{\boldsymbol{\alpha}}$ is an exponential martingale. Consider

$$P_t^{\boldsymbol{\alpha}} f(\boldsymbol{x}) := E_{\boldsymbol{x}}\left[(Z_t^{\boldsymbol{\alpha}}/Z_0^{\boldsymbol{\alpha}}) f(\boldsymbol{X}_t)\right] = e^{\boldsymbol{\alpha}'\boldsymbol{x} + t\psi(i\boldsymbol{\alpha})} E_{\boldsymbol{x}}\left[e^{-\boldsymbol{\alpha}'\boldsymbol{X}_t} f(\boldsymbol{X}_t)\right]. \tag{4.2}$$

Here, $Z_t^{\boldsymbol{\alpha}}/Z_0^{\boldsymbol{\alpha}}$ is a unit martingale and it introduces a new probability measure through the multidimensional Esscher transform with $P_t^{\boldsymbol{\alpha}}$ as the transition operator under it. From (4.2),

$$P_t f(\boldsymbol{x}) = e^{-\boldsymbol{\alpha}'\boldsymbol{x} - t\psi(i\boldsymbol{\alpha})} P_t^{\boldsymbol{\alpha}} f_{\boldsymbol{\alpha}}(\boldsymbol{x}). \tag{4.3}$$

As $\{P_t^{\boldsymbol{\alpha}}, t \geq 0\}$ is a strongly continuous contraction semigroup on $L^1(\mathbb{R}^n)$ (Applebaum (2009), Theorem 3.4.2), we immediately obtain the following result.

Proposition 4.1. *For every $\boldsymbol{\alpha} \in \mathcal{I}_{\boldsymbol{X}}$, $\{P_t, t \geq 0\}$ is a strongly continuous semigroup on $L^1_{\boldsymbol{\alpha}}(\mathbb{R}^n)$, and $\{e^{t\psi(i\boldsymbol{\alpha})} P_t, t \geq 0\}$ is a strongly continuous contraction semigroup on $L^1_{\boldsymbol{\alpha}}(\mathbb{R}^n)$. In particular,*

$$\|P_t f\|_{L^1_{\boldsymbol{\alpha}}(\mathbb{R}^n)} \leq e^{-t\psi(i\boldsymbol{\alpha})} \|f\|_{L^1_{\boldsymbol{\alpha}}(\mathbb{R}^n)}. \tag{4.4}$$

A Fourier integral representation for $P_t f(\boldsymbol{x})$ can be derived under an additional condition. We can further discretize the integral to compute the expected value of f.

Proposition 4.2. *Suppose $\boldsymbol{\alpha} \in \mathcal{I}_{\boldsymbol{X}}$, $f_{\boldsymbol{\alpha}}(\boldsymbol{x}) \in L^1(\mathbb{R}^n)$ and for some $t > 0$,*

$$\int_{\mathbb{R}^n} |\phi_t^{\boldsymbol{\alpha}}(-\boldsymbol{\xi}) \hat{f}_{\boldsymbol{\alpha}}(\boldsymbol{\xi})| d\boldsymbol{\xi} < \infty, \tag{4.5}$$

where $\hat{f}_{\boldsymbol{\alpha}}(\boldsymbol{\xi})$ is the multidimensional Fourier transform of $f_{\boldsymbol{\alpha}}(\boldsymbol{x})$ and

$$\phi_t^{\boldsymbol{\alpha}}(\boldsymbol{\xi}) = \frac{\phi_t(\boldsymbol{\xi} + i\boldsymbol{\alpha})}{\phi_t(i\boldsymbol{\alpha})} \tag{4.6}$$

is the characteristic function of X_t under the new probability measure introduced by $Z_t^{\boldsymbol{\alpha}}/Z_0^{\boldsymbol{\alpha}}$. Then, for all $t > 0$,

$$P_t f(\boldsymbol{x}) = \frac{1}{(2\pi)^n} e^{-\boldsymbol{\alpha}'\boldsymbol{x} - t\psi(i\boldsymbol{\alpha})} \int_{\mathbb{R}^n} e^{-i\boldsymbol{\xi}'\boldsymbol{x}} \phi_t^{\boldsymbol{\alpha}}(-\boldsymbol{\xi}) \hat{f}_{\boldsymbol{\alpha}}(\boldsymbol{\xi}) d\boldsymbol{\xi}. \tag{4.7}$$

Remark 4.3. When $f_{\boldsymbol{\alpha}}(\boldsymbol{x}) \in L^1(\mathbb{R}^n)$, $\hat{f}_{\boldsymbol{\alpha}}(\boldsymbol{\xi})$ is bounded. Therefore, $\phi_t^{\boldsymbol{\alpha}}(\boldsymbol{\xi}) \in L^1(\mathbb{R}^n, \mathbb{C})$ is a sufficient condition for (4.5) to hold.

4.3 Important Identities for Multidimensional Hilbert Transform

To price barrier options, we need to calculate $\mathcal{F}_n\left(\mathbb{1}_{(\boldsymbol{l},\infty)}(\boldsymbol{x})f(\boldsymbol{x})\right)(\boldsymbol{\xi})$, which would involve the partial Fourier transform and the partial Hilbert transform defined as follows. For $\boldsymbol{\beta} \in \{0,1\}^n$,

$$\mathcal{F}_{\boldsymbol{\beta}}(f)(\boldsymbol{\xi}_{\boldsymbol{\beta}} + \boldsymbol{x}_{\boldsymbol{1}-\boldsymbol{\beta}}) := \int_{\mathbb{R}^{|\boldsymbol{\beta}|_1}} f(\boldsymbol{x}) \prod_{\{p:\beta_p=1\}} e^{i\xi_p x_p} dx_p,$$
$$\mathcal{H}_{\boldsymbol{\beta}}(f)(\boldsymbol{\xi}_{\boldsymbol{\beta}} + \boldsymbol{x}_{\boldsymbol{1}-\boldsymbol{\beta}}) := \frac{1}{\pi^{|\boldsymbol{\beta}|_1}} P.V. \int_{\mathbb{R}^{|\boldsymbol{\beta}|_1}} f(\boldsymbol{x}) \prod_{\{p:\beta_p=1\}} \frac{dx_p}{\xi_p - x_p}.$$

The operators $\mathcal{F}_{\boldsymbol{0}}$ and $\mathcal{H}_{\boldsymbol{0}}$ are identity maps. The meaning of $\boldsymbol{\xi}_{\boldsymbol{\beta}}$ and $\boldsymbol{x}_{\boldsymbol{1}-\boldsymbol{\beta}}$ are explained in Section 1.2. Their sum indicates that only those dimensions j with $\beta_j = 1$ are transformed in the above.

To calculate $\mathcal{F}_n\left(\mathbb{1}_{(\boldsymbol{l},\infty)}(\boldsymbol{x})f(\boldsymbol{x})\right)(\boldsymbol{\xi})$, we first recall a result from King (2009) (Eq.(15.60)). For $f \in L^p(\mathbb{R}^n)$ with $p > 1$, we have

$$\mathcal{F}_n\left(\prod_{j=1}^n \operatorname{sgn}(x_j) f(\boldsymbol{x})\right)(\boldsymbol{\xi}) = i^n \mathcal{H}_n(\mathcal{F}_n f)(\boldsymbol{\xi}). \tag{4.8}$$

where $\operatorname{sgn}(x)$ is the sign function. In our problem, we need to deal with $f \in L^1(\mathbb{R}^n)$. For this case, an additional condition is needed for (4.8) to hold. We also need to consider a general situation where the sign function is only applied for part of the dimensions. The following gives the result required for our problem.

Proposition 4.3. *For $f \in L^1(\mathbb{R}^n)$, if $\mathcal{F}_n f \in L^1(\mathbb{R}^n, \mathbb{C})$, then for $\boldsymbol{\beta} \in \{0,1\}^n$, we have*

$$\mathcal{F}_n\left(\prod_{\{p:\beta_p=1\}} sgn(x_p) f(\boldsymbol{x})\right)(\boldsymbol{\xi}) = i^{|\boldsymbol{\beta}|_1} \mathcal{H}_{\boldsymbol{\beta}}(\hat{f}(\boldsymbol{\eta}_{\boldsymbol{\beta}} + \boldsymbol{\xi}_{\boldsymbol{1}-\boldsymbol{\beta}}))(\boldsymbol{\xi}). \tag{4.9}$$

Here $\hat{f}$ is the nD Fourier transform of f, i.e., $\mathcal{F}_n f$.

Based on (4.9), we obtain $\mathcal{F}_n\left(\mathbb{1}_{(\boldsymbol{l},\infty)}(\boldsymbol{x})f(\boldsymbol{x})\right)(\boldsymbol{\xi})$ as in (4.10), which is the key identity for calculating the Fourier transform of the value function in the backward recursion (see Section 4.4).

Proposition 4.4. *For $f \in L^1(\mathbb{R}^n)$, if $\mathcal{F}_n f \in L^1(\mathbb{R}^n, \mathbb{C})$, then*

$$\mathcal{F}_n\left(\mathbb{1}_{(\boldsymbol{l},\infty)}(\boldsymbol{x})f(\boldsymbol{x})\right)(\boldsymbol{\xi}) = \frac{1}{2^n} \sum_{\boldsymbol{\beta} \in \{0,1\}^n} i^{|\boldsymbol{\beta}|_1} e^{i\boldsymbol{\xi}_{\boldsymbol{\beta}}' \boldsymbol{l}_{\boldsymbol{\beta}}} \mathcal{H}_{\boldsymbol{\beta}}\left(e^{-i\boldsymbol{\eta}_{\boldsymbol{\beta}}' \boldsymbol{l}_{\boldsymbol{\beta}}} \hat{f}\left(\boldsymbol{\eta}_{\boldsymbol{\beta}} + \boldsymbol{\xi}_{\boldsymbol{1}-\boldsymbol{\beta}}\right)\right)(\boldsymbol{\xi}). \tag{4.10}$$

4.4 Backward Recursion

Let $v^j(\boldsymbol{x})$ denote the barrier option price at time t_j given $\boldsymbol{X}_{t_j} = \boldsymbol{x}$. We can compute each v^j by the following backward recursion:

$$\begin{aligned} v^N(\boldsymbol{x}) &= g(\boldsymbol{x}) := h(\boldsymbol{x})\mathbb{1}_{(\boldsymbol{l},\infty)}(\boldsymbol{x}), \\ v^{j-1}(\boldsymbol{x}) &= P_\Delta v^j(\boldsymbol{x})\mathbb{1}_{(\boldsymbol{l},\infty)}(\boldsymbol{x}), \quad j = N, N-1, \cdots, 2, \\ v^0(\boldsymbol{x}) &= P_\Delta v^1(\boldsymbol{x}). \end{aligned}$$

In order to apply the Hilbert transform method to compute each step in the induction, we need to apply exponential damping to the payoff function. Consider $\boldsymbol{\alpha} \in \mathcal{I}_{\boldsymbol{X}}$ such that $g_{\boldsymbol{\alpha}}(\boldsymbol{x}) := e^{\boldsymbol{\alpha}'\boldsymbol{x}} g(\boldsymbol{x}) \in L^1(\mathbb{R}^n)$. Define $v^j_{\boldsymbol{\alpha}}(\boldsymbol{x}) = e^{\boldsymbol{\alpha}'\boldsymbol{x}} v^j(\boldsymbol{x})$. Applying (4.3), the backward recursion becomes

$$\begin{aligned} v^N_{\boldsymbol{\alpha}}(\boldsymbol{x}) &= g_{\boldsymbol{\alpha}}(\boldsymbol{x}), \\ v^{j-1}_{\boldsymbol{\alpha}}(\boldsymbol{x}) &= e^{-\Delta\psi(i\boldsymbol{\alpha})} P^{\boldsymbol{\alpha}}_\Delta v^j_{\boldsymbol{\alpha}}(\boldsymbol{x})\mathbb{1}_{(\boldsymbol{l},\infty)}(\boldsymbol{x}), \quad j = N, N-1, \cdots, 2, \\ v^0_{\boldsymbol{\alpha}}(\boldsymbol{x}) &= e^{-\Delta\psi(i\boldsymbol{\alpha})} P^{\boldsymbol{\alpha}}_\Delta v^1_{\boldsymbol{\alpha}}(\boldsymbol{x}), \end{aligned} \tag{4.11}$$

where $P^{\boldsymbol{\alpha}}_\Delta$ is defined in (4.2). Let $V(\boldsymbol{S})$ be the discounted barrier option price at time zero as a function of the initial asset prices. Then,

$$V(\boldsymbol{S}) = e^{-rT-\boldsymbol{\alpha}'\ln(\boldsymbol{S}/\boldsymbol{K})} v^0_{\boldsymbol{\alpha}}\left(\ln(\boldsymbol{S}/\boldsymbol{K})\right).$$

As $g_{\boldsymbol{\alpha}} \in L^1(\mathbb{R}^n)$ and $P^{\boldsymbol{\alpha}}_\Delta$ maps from $L^1(\mathbb{R}^n)$ to itself, we have $P^{\boldsymbol{\alpha}}_\Delta v^j_{\boldsymbol{\alpha}}$ $(1 \le j \le N)$ and $v^j_{\boldsymbol{\alpha}}$ $(0 \le j \le N)$ are all in $L^1(\mathbb{R}^n)$. Let $\|\cdot\|_{L^1(\mathbb{R}^n)}$ be the L^1-norm. Using (4.4), we obtain

$$\|v^j_{\boldsymbol{\alpha}}\|_{L^1(\mathbb{R}^n)} \le e^{-\Delta(N-j)\psi(i\boldsymbol{\alpha})}\|g_{\boldsymbol{\alpha}}\|_{L^1(\mathbb{R}^n)}, \; j = 0, 1, \cdots, N. \tag{4.12}$$

In general, we cannot calculate expectations like $P^{\boldsymbol{\alpha}}_\Delta v^j_{\boldsymbol{\alpha}}(\boldsymbol{x})$ directly. To carry out the backward recursion, we follow Feng and Linetsky (2008a) to take Fourier transform on both sides of the recursive equation (4.11). By Proposition 9 in Bertoin (1998), its multidimensional Fourier transform is given by

$$\mathcal{F}_n\left(P^{\boldsymbol{\alpha}}_\Delta v^j_{\boldsymbol{\alpha}}(\boldsymbol{x})\right)(\boldsymbol{\xi}) = \phi^{\boldsymbol{\alpha}}_\Delta(-\boldsymbol{\xi})\hat{v}^j_{\boldsymbol{\alpha}}(\boldsymbol{\xi}).$$

If we further have $\phi^{\boldsymbol{\alpha}}_\Delta(-\boldsymbol{\xi})\hat{v}^j_{\boldsymbol{\alpha}}(\boldsymbol{\xi}) \in L^1(\mathbb{R}^n)$, we can apply (4.10) to calculate $\hat{v}^{j-1}_{\boldsymbol{\alpha}}(\boldsymbol{\xi})$. Consequently, the backward induction in the Fourier space becomes

$$\hat{v}^N_{\boldsymbol{\alpha}}(\boldsymbol{\xi}) = \hat{g}_{\boldsymbol{\alpha}}(\boldsymbol{\xi}), \tag{4.13}$$

$$\begin{aligned} \hat{v}^{j-1}_{\boldsymbol{\alpha}}(\boldsymbol{\xi}) =& \frac{e^{-\Delta\psi(i\boldsymbol{\alpha})}}{2^n} \sum_{\boldsymbol{\beta}\in\{0,1\}^n} i^{|\boldsymbol{\beta}|_1} e^{i\boldsymbol{\xi}'_{\boldsymbol{\beta}}\boldsymbol{l}_{\boldsymbol{\beta}}} \\ & \cdot \mathcal{H}_{\boldsymbol{\beta}}\left(e^{-i\boldsymbol{\eta}'_{\boldsymbol{\beta}}\boldsymbol{l}_{\boldsymbol{\beta}}}\phi^{\boldsymbol{\alpha}}_\Delta\left(-\boldsymbol{\eta}_{\boldsymbol{\beta}} - \boldsymbol{\xi}_{\boldsymbol{1}-\boldsymbol{\beta}}\right)\hat{v}^j_{\boldsymbol{\alpha}}\left(\boldsymbol{\eta}_{\boldsymbol{\beta}} + \boldsymbol{\xi}_{\boldsymbol{1}-\boldsymbol{\beta}}\right)\right)(\boldsymbol{\xi}), j = N, \cdots, 2, \end{aligned} \tag{4.14}$$

$$\hat{v}^0_{\boldsymbol{\alpha}}(\boldsymbol{\xi}) = e^{-\Delta\psi(i\boldsymbol{\alpha})}\phi^{\boldsymbol{\alpha}}_\Delta(-\boldsymbol{\xi})\hat{v}^1_{\boldsymbol{\alpha}}(\boldsymbol{\xi}). \tag{4.15}$$

The following theorem shows that this backward recursion procedure is valid under mild assumptions and recovers $v^0(\boldsymbol{x})$ through Fourier inversion.

Theorem 4.1. *Consider $\boldsymbol{\alpha} \in \mathcal{I}_{\boldsymbol{X}}$ such that $g_{\boldsymbol{\alpha}}(\boldsymbol{x}) \in L^1(\mathbb{R}^n)$. Suppose that*

$$\int_{\mathbb{R}^n} |\phi^{\boldsymbol{\alpha}}_\Delta(-\boldsymbol{\xi})\hat{g}_{\boldsymbol{\alpha}}(\boldsymbol{\xi})| d\boldsymbol{\xi} < \infty. \tag{4.16}$$

Also assume that there is some time $\chi > 0$ such that

$$\|\phi_\chi^{\boldsymbol{\alpha}}(\cdot)\|_{L^1(\mathbb{R}^n)} := \int_{\mathbb{R}^n} |\phi_\chi^{\boldsymbol{\alpha}}(\boldsymbol{\xi})| d\boldsymbol{\xi} < \infty \tag{4.17}$$

Then, for $1 \le j \le N$,

$$\int_{\mathbb{R}^n} |\phi_\Delta^{\boldsymbol{\alpha}}(-\boldsymbol{\xi}) \hat{v}_{\boldsymbol{\alpha}}^j(\boldsymbol{\xi})| d\boldsymbol{\xi} < \infty. \tag{4.18}$$

and (4.14) holds. Finally, $v^0(\boldsymbol{x})$ can be recovered by

$$v^0(\boldsymbol{x}) = \frac{1}{(2\pi)^n} e^{-\boldsymbol{\alpha}'\boldsymbol{x} - \Delta\psi(i\boldsymbol{\alpha})} \int_{\mathbb{R}^n} e^{-i\boldsymbol{\xi}'\boldsymbol{x}} \phi_\Delta^{\boldsymbol{\alpha}}(-\boldsymbol{\xi}) \hat{v}_{\boldsymbol{\alpha}}^1(\boldsymbol{\xi}) d\boldsymbol{\xi}. \tag{4.19}$$

4.5 Discrete Approximations of Two Operators

The Hilbert transforms in (4.14) and the integral in (4.19) have to be evaluated numerically. Introduce the following operators:

$$\mathcal{P}^\Delta g(\boldsymbol{\xi}) = \sum_{\boldsymbol{\beta} \in \{0,1\}^n} i^{|\boldsymbol{\beta}|_1} e^{i\boldsymbol{\xi}_{\boldsymbol{\beta}}'\boldsymbol{l}_{\boldsymbol{\beta}}} \mathcal{H}_{\boldsymbol{\beta}} \left(e^{-i\boldsymbol{\eta}_{\boldsymbol{\beta}}'\boldsymbol{l}_{\boldsymbol{\beta}}} \phi_\Delta^{\boldsymbol{\alpha}}\big(-\boldsymbol{\eta}_{\boldsymbol{\beta}} - \boldsymbol{\xi}_{\boldsymbol{1}-\boldsymbol{\beta}}\big) g\big(\boldsymbol{\eta}_{\boldsymbol{\beta}} + \boldsymbol{\xi}_{\boldsymbol{1}-\boldsymbol{\beta}}\big)\right)(\boldsymbol{\xi}),$$

$$\mathcal{R}^\Delta g(\boldsymbol{x}) = \int_{\mathbb{R}^n} e^{-i\boldsymbol{\xi}'\boldsymbol{x}} \phi_\Delta^{\boldsymbol{\alpha}}(-\boldsymbol{\xi}) g(\boldsymbol{\xi}) d\boldsymbol{\xi}.$$

Employing the general Sinc approximation (2.2) and further truncating the infinite series, we obtain the following approximation for $\mathcal{P}^\Delta$ and $\mathcal{R}^\Delta$:

$$\mathcal{P}^\Delta g(\boldsymbol{\xi}) \approx \mathcal{P}_{\boldsymbol{h},\boldsymbol{M}}^\Delta g(\boldsymbol{\xi}) := \sum_{\boldsymbol{\beta} \in \{0,1\}^n} i^{|\boldsymbol{\beta}|_1} e^{i\boldsymbol{\xi}_{\boldsymbol{\beta}}'\boldsymbol{l}_{\boldsymbol{\beta}}} \cdot \sum_{-M_j \le m_j \le M_j}^{\boldsymbol{\beta}}$$
$$\prod_{\{j:\beta_j=1\}} \frac{1-\cos[\pi(\xi_j - m_j h_j)/h_j]}{\pi(\xi_j - m_j h_j)/h_j} e^{-i(\boldsymbol{m}_{\boldsymbol{\beta}}\boldsymbol{h}_{\boldsymbol{\beta}})'\boldsymbol{l}_{\boldsymbol{\beta}}} \phi_\Delta^{\boldsymbol{\alpha}}(-\boldsymbol{m}_{\boldsymbol{\beta}}\boldsymbol{h}_{\boldsymbol{\beta}} - \boldsymbol{\xi}_{\boldsymbol{1}-\boldsymbol{\beta}}) g(\boldsymbol{m}_{\boldsymbol{\beta}}\boldsymbol{h}_{\boldsymbol{\beta}} + \boldsymbol{\xi}_{\boldsymbol{1}-\boldsymbol{\beta}}), \tag{4.20}$$

$$\mathcal{R}^\Delta g(\boldsymbol{x}) \approx \mathcal{R}_{\boldsymbol{h},\boldsymbol{M}}^\Delta g(\boldsymbol{x}) := \prod_{j=1}^n h_j \sum_{m_1=-M_1}^{M_1} \cdots \sum_{m_n=-M_n}^{M_n} e^{-i(\boldsymbol{m}\boldsymbol{h})'\boldsymbol{x}} \phi_\Delta^{\boldsymbol{\alpha}}(-\boldsymbol{m}\boldsymbol{h}) g(\boldsymbol{m}\boldsymbol{h}). \tag{4.21}$$

The approximation error comes from discretization and trucation. The discretization error can be estimated using Theorem 2.2. To estimate the trucation error, we assume

$$|\phi_t(\boldsymbol{\xi})| \le a e^{-tb|\boldsymbol{\xi}|^\nu} \text{ for some } \nu \in (0,2],\ a, b > 0, \text{ and } \boldsymbol{\xi} \in \mathbb{R}^n, t > 0. \tag{4.22}$$

This condition holds for many Lévy processes, in particualr for all Lévy processes with a Brownian component (see Appendix A for details). The next theorem shows the error estimates.

Theorem 4.2. *Suppose (4.22) holds. Let $\boldsymbol{x} = (x_1, \cdots, x_n)'$ with $x_j = Re(z_j)$ for $1 \le j \le n$. Further assume that for $\boldsymbol{z} \in \mathcal{D}_{\boldsymbol{d}}$, $\forall \boldsymbol{\beta} \in \{0,1\}^n, \boldsymbol{\beta} \neq 0$,*

$$\int_{\mathbb{R}^{|\boldsymbol{\beta}|_1}} \left| e^{-\Delta\psi(\boldsymbol{x}_{\boldsymbol{\beta}} \pm i\boldsymbol{d}_{\boldsymbol{\beta}}^- + \boldsymbol{z}_{\boldsymbol{1}-\boldsymbol{\beta}})} \right| d\boldsymbol{x}_{\boldsymbol{\beta}} \le C, \tag{4.23}$$

$$\lim_{x_j \to \pm\infty,\ \beta_j = 1} \int_{\prod_{\{j:\beta_j=1\}}(-d_j, d_j)} e^{-\Delta\psi(\boldsymbol{x}_{\boldsymbol{\beta}} + i\boldsymbol{y}_{\boldsymbol{\beta}} + \boldsymbol{z}_{\boldsymbol{1}-\boldsymbol{\beta}})} d\boldsymbol{y}_{\boldsymbol{\beta}} = 0, \tag{4.24}$$

where C is a finite constant independent of $\boldsymbol{z}$. Consider $g(\boldsymbol{z}) : \mathbb{C}^n \mapsto \mathbb{C}$, which is analytic and bounded in $\mathcal{D}_{\boldsymbol{d}}$. Then, there exist positive constants $A_{\boldsymbol{\beta}}, B_{\boldsymbol{\beta}}, \boldsymbol{\beta} \in \{0,1\}^n$ that are independent of $\boldsymbol{h}, \boldsymbol{M}$ such that

$$\left\| \mathcal{P}^\Delta g(\boldsymbol{\xi}) - \mathcal{P}_{\boldsymbol{h},\boldsymbol{M}}^\Delta g(\boldsymbol{\xi}) \right\|_{L^\infty(\mathbb{R}^n)}$$

$$\leq \sum_{\boldsymbol{\beta}\in\{0,1\}^n, \boldsymbol{\beta}\neq \mathbf{0}} \left[A_{\boldsymbol{\beta}} \prod_{\{j:\beta_j=1\}} \frac{e^{-\pi d_j/h_j}}{1-e^{-\pi d_j/h_j}} + B_{\boldsymbol{\beta}} \prod_{\{j:\beta_j=1\}} \Gamma\left(1/\nu, \Delta b 2^{\nu/2-1}(M_j h_j)^{\nu}\right)/h_j \right], \tag{4.25}$$

where $\|\cdot\|_{L^\infty(\mathbb{R}^n)}$ *denotes the maximum norm on* $\mathbb{R}^n$ *and* $\Gamma(a,x) := \int_x^\infty y^{a-1}e^{-y}dy$ *is the incomplete Gamma function. In particular, let*

$$h_j = \left(n^{1-\nu/2}\pi d_j/(\Delta b)\right)^{\frac{1}{1+\nu}} M_j^{-\frac{\nu}{1+\nu}}, j = 1, \cdots, n. \tag{4.26}$$

Then there exists a positive constant C*, which is independent of* M_j*, such that*

$$\left\| \mathcal{P}^{\Delta} g(\boldsymbol{\xi}) - \mathcal{P}^{\Delta}_{\boldsymbol{h},\boldsymbol{M}} g(\boldsymbol{\xi}) \right\|_{L^\infty(\mathbb{R}^n)} \leq C \sum_{j=1}^{n} M_j^{\frac{1}{1+\nu}} \exp\left(-(\Delta b n^{\nu/2-1})^{\frac{1}{1+\nu}} (\pi d_j M_j)^{\frac{\nu}{1+\nu}} \right). \tag{4.27}$$

There also exist positive constants $A'_{\boldsymbol{\beta}}, B'_{\boldsymbol{\beta}}, \boldsymbol{\beta} \in \{0,1\}^n$ *that are independent of* $\boldsymbol{h}, \boldsymbol{M}$ *such that*

$$\left\| \mathcal{R}^{\Delta} g(\boldsymbol{x}) - \mathcal{R}^{\Delta}_{\boldsymbol{h},\boldsymbol{M}} g(\boldsymbol{x}) \right\|_{L^\infty(\mathbb{R}^n)}$$
$$\leq \sum_{\boldsymbol{\beta}\in\{0,1\}^n, \boldsymbol{\beta}\neq \mathbf{0}} \left[A'_{\boldsymbol{\beta}} \prod_{\{j:\beta_j=1\}} \frac{e^{-2\pi d_j/h_j}}{1-e^{-2\pi d_j/h_j}} + B'_{\boldsymbol{\beta}} \prod_{\{j:\beta_j=1\}} \Gamma\left(1/\nu, \Delta b 2^{\nu/2-1}(M_j h_j)^{\nu}\right)/h_j \right],$$

In particular, set $\boldsymbol{h}$ *by* (4.26)*, there exists a positive constant* C'*, which is independent of* $\boldsymbol{M}$*, such that*

$$\begin{aligned} &\left\| \mathcal{R}^{\Delta} g(\boldsymbol{x}) - \mathcal{R}^{\Delta}_{\boldsymbol{h},\boldsymbol{M}} g(\boldsymbol{x}) \right\|_{L^\infty(\mathbb{R}^n)} \\ \leq & C' \sum_{j=1}^{n} \max(1, M_j^{\frac{2(1-\nu)}{1+\nu}}) \exp\left(-2(\Delta b n^{\nu/2-1})^{\frac{1}{1+\nu}} (\pi d_j M_j)^{\frac{\nu}{1+\nu}} \right). \end{aligned} \tag{4.28}$$

Remark 4.4. One can further reduce the parameters for the algorithm by equalizing the terms $M_i d_i, 1 \leq i \leq n$, i.e. letting

$$M_i d_i = M_j d_j, i \neq j. \tag{4.29}$$

Thus, we only need to specify M_1, and then determine $M_j, 2 \leq j \leq n$ by (4.29) and $h_j, 1 \leq j \leq n$ by (4.26). This reduces the number of parameters for the algorithm to one.

Remark 4.5. In general, if we set $h_j = c^{\frac{1}{1+\nu}} M_j^{-\frac{\nu}{1+\nu}}$ for some $c > 0$, we would obtain error estimates similar to those in (4.27) and (4.28) with the exponent of M_j unchanged. The choice (4.26) for h_j is derived by setting $e^{-\pi d_j/h_j} = e^{-\Delta b 2^{\nu/2-1}(M_j h_j)^{\nu}}$, where the right-hand side is the leading order for the incomplete Gamma function.

Remark 4.6. Conditions (4.23) and (4.24) can be checked for a given model and are not stringent at all. In particular, they hold for all the models considered in the numerical section.

Remark 4.7. Our method for pricing barrier options and calculating survival probability is applicable to Lévy models with known joint characteristic function and it converges provided that

$$\sum_{m_1=-\infty}^{\infty} \cdots \sum_{m_n=-\infty}^{\infty} |\phi_{\Delta}(-\boldsymbol{m}\boldsymbol{h})| < \infty. \tag{4.30}$$

Thus, our method applies to Lévy processes with different marginals if their joint characteristic functions satisfy (4.30). Condition (4.22) is used to show the method converges exponentially but it is not necessary for achieving exponential convergence. For example, one can use the weaker condition

$$|\phi_t(\boldsymbol{\xi})| \le a e^{-tb\sum_{j=1}^n |\xi_j|^\nu} \text{ for some } \nu \in (0,2],\ a,b>0,\ \text{and } \boldsymbol{\xi}\in\mathbb{R}^n, t>0. \tag{4.31}$$

For specific models, one can even twist the condition (4.22) or (4.31) in accordance with the form of the characteristic function to show exponential convergence.

For Lévy processes with different types of marginals, condition (4.22) may or may not hold. Consider the example where X_t^1 is a Brownian motion and X_t^2 is a jump-diffusion with its Brownian motion component correlated with X_t^1. Obviously they are of different types, but (4.22) holds. As another example, let L_t^1 be an NIG process and L_t^2 be an independent normal tempered stable process with order $\nu \in (0,1)$ (so it's not NIG). We construct X_t^1 and X_t^2 as $X_t^1 = L_t^1$, $X_t^2 = L_t^1 + L_t^2$. Now X^1 and X^2 are of different types and (4.31) doesn't hold. But it is still possible to show exponential convergence in this case (with terms exponentially decaying with different orders in the exponent).

Remark 4.8. The pure-jump VG model is a challenging case for our method as its characteristic function doesn't decay exponentially. However, it still satisfies (4.30), so our method still converges, but at a rate slower than exponential convergence. In Section 5.4, we will apply multidimensional spectral filters to accelerate convergence for this model.

4.6 The Aggregate Error of Backward Recursion

Our algorithm yields an approximation of $\hat{v}_{\boldsymbol{\alpha}}^j$ for each j. Let $\tilde{v}_{\boldsymbol{\alpha}}^N$ be the approximation of $\hat{v}_{\boldsymbol{\alpha}}^N$ in the hyperrectangle $\Omega_{\boldsymbol{h},\boldsymbol{M}} := [-M_1h_1, M_1h_1]\times\cdots\times[-M_nh_n, M_nh_n]$ and we set $\tilde{v}_{\boldsymbol{\alpha}}^N(\boldsymbol{\xi}) = 0$ for $\boldsymbol{\xi}\notin\Omega_{\boldsymbol{h},\boldsymbol{M}}$. Define $\tilde{v}_{\boldsymbol{\alpha}}^j, 1\le j\le N-1$ recursively as

$$\tilde{v}_{\boldsymbol{\alpha}}^j = \frac{e^{-\Delta\psi(i\boldsymbol{\alpha})}}{2^n}\mathcal{P}_{\boldsymbol{h},\boldsymbol{M}}^{\Delta}\tilde{v}_{\boldsymbol{\alpha}}^{j+1}.$$

From the definition, $\tilde{v}_{\boldsymbol{\alpha}}^j(\boldsymbol{\xi}) = 0$ for $\boldsymbol{\xi}\notin\Omega_{\boldsymbol{h},\boldsymbol{M}}$ and $\tilde{v}_{\boldsymbol{\alpha}}^j(\boldsymbol{\xi})$ approximates $\hat{v}_{\boldsymbol{\alpha}}^j(\boldsymbol{\xi})$ on $\Omega_{\boldsymbol{h},\boldsymbol{M}}$. Also introduce

$$\tilde{v}^0 = \frac{1}{(2\pi)^n}e^{-\boldsymbol{\alpha}'\boldsymbol{x}-\Delta\psi(i\boldsymbol{\alpha})}\mathcal{R}_{\boldsymbol{h},\boldsymbol{M}}^{\Delta}\tilde{v}_{\boldsymbol{\alpha}}^1,\ \bar{v}^0 = \frac{1}{(2\pi)^n}e^{-\boldsymbol{\alpha}'\boldsymbol{x}-\Delta\psi(i\boldsymbol{\alpha})}\mathcal{R}^{\Delta}\tilde{v}^1.$$

where $\tilde{v}^0(\boldsymbol{x})$ is our approximation of $v^0(\boldsymbol{x})$.

We assume (4.22) and estimate $|\tilde{v}^0(\boldsymbol{x}) - v^0(\boldsymbol{x})|$ as follows:

$$\begin{aligned}
&|\tilde{v}^0(\boldsymbol{x}) - v^0(\boldsymbol{x})| \le |\bar{v}^0(\boldsymbol{x}) - v^0(\boldsymbol{x})| + |\tilde{v}^0(\boldsymbol{x}) - \bar{v}^0(\boldsymbol{x})| \\
&\le \frac{1}{(2\pi)^n}e^{-\boldsymbol{\alpha}'\boldsymbol{x}-\Delta\psi(i\boldsymbol{\alpha})}\left(\int_{\mathbb{R}^n}\left|\phi_{\Delta}^{\boldsymbol{\alpha}}(-\boldsymbol{\xi})\left(\tilde{v}_{\boldsymbol{\alpha}}^1 - \hat{v}_{\boldsymbol{\alpha}}^1\right)(\boldsymbol{\xi})\right|d\boldsymbol{\xi} + \left|\mathcal{R}^{\Delta}\tilde{\nu}^1(\boldsymbol{x}) - \mathcal{R}_{\boldsymbol{h},\boldsymbol{M}}^{\Delta}\tilde{\nu}^1(\boldsymbol{x})\right|\right) \\
&\le C_1(I_1 + I_2 + I_3),
\end{aligned}$$

where

$$C_1 = \frac{1}{(2\pi)^n}e^{-\boldsymbol{\alpha}'\boldsymbol{x}-\Delta\psi(i\boldsymbol{\alpha})},\ I_1 = \int_{\mathbb{R}^n}|\phi_{\Delta}^{\boldsymbol{\alpha}}(-\boldsymbol{\xi})|\cdot\left|\mathcal{P}_{\boldsymbol{h},\boldsymbol{M}}^{\Delta}\tilde{v}_{\boldsymbol{\alpha}}^2(\boldsymbol{\xi}) - \mathcal{P}^{\Delta}\tilde{v}_{\boldsymbol{\alpha}}^2(\boldsymbol{\xi})\right|d\boldsymbol{\xi},$$
$$I_2 = \int_{\mathbb{R}^n}|\phi_{\Delta}^{\boldsymbol{\alpha}}(-\boldsymbol{\xi})|\cdot\left|\mathcal{P}^{\Delta}\left(\tilde{v}_{\boldsymbol{\alpha}}^2 - \hat{v}_{\boldsymbol{\alpha}}^2\right)(\boldsymbol{\xi})\right|d\boldsymbol{\xi},\ I_3 = \left|\mathcal{R}^{\Delta}\tilde{\nu}^1(\boldsymbol{x}) - \mathcal{R}_{\boldsymbol{h},\boldsymbol{M}}^{\Delta}\tilde{\nu}^1(\boldsymbol{x})\right|.$$

We can estimate I_1 and I_2 recursively using the Holder and Minkowski inequalities (see Appendix D). The result is summarized as follows.

Theorem 4.3. *Suppose* (4.16) *and* (4.22) *hold. Then, we have*

$$|\tilde{v}^0(\boldsymbol{x}) - v^0(\boldsymbol{x})| \le C_N'\left(\|\phi_{\Delta}^{\boldsymbol{\alpha}}(-\cdot)\left(\tilde{v}_{\boldsymbol{\alpha}}^N - \hat{v}_{\boldsymbol{\alpha}}^N\right)(\cdot)1_{\Omega_{\boldsymbol{h},\boldsymbol{M}}}(\cdot)\|_{L^\infty} + I_{\boldsymbol{h},\boldsymbol{M}}\right)$$

$$+\sum_{k=2}^{N} C_k \|\mathcal{P}^{\Delta}_{\boldsymbol{h},\boldsymbol{M}}\tilde{v}^k_{\boldsymbol{\alpha}}(\boldsymbol{\xi}) - \mathcal{P}^{\Delta}\tilde{v}^k_{\boldsymbol{\alpha}}(\boldsymbol{\xi}))\|_{L^\infty}$$
$$+ C_1 \left|\mathcal{R}^{\Delta}\tilde{\nu}^1(\boldsymbol{x}) - \mathcal{R}^{\Delta}_{\boldsymbol{h},\boldsymbol{M}}\tilde{\nu}^1(\boldsymbol{x})\right|, \tag{4.32}$$

where $C_2 = C_1\|\phi^{\boldsymbol{\alpha}}_{\Delta}\|_{L^1}$, $C_k \sim O\left(2^{-n(k-2)}\right)$ *for* $k \geq 3$, C'_N *is bounded in* N, *and*

$$I_{\boldsymbol{h},\boldsymbol{M}} = \int_{\Omega^c_{\boldsymbol{h},\boldsymbol{M}}} a e^{-tb|\boldsymbol{\xi}|^{\nu}} d\boldsymbol{\xi}.$$

The theorem breaks down the aggregate error into three sources: the approximation error for the Fourier transform of the dampened payoff on $\Omega_{\boldsymbol{h},\boldsymbol{M}}$ and truncation (as we set $\tilde{v}^N_{\boldsymbol{\alpha}}$ to be zero outside $\Omega_{\boldsymbol{h},\boldsymbol{M}}$), the discrete approximation error for the operator $\mathcal{P}^{\Delta}$ in each recursion step, and the discrete approximation error for the operator $\mathcal{R}$ in the last step. The second and the third types of errors can be further estimated using (4.27) and (4.28) and they decay exponentially. This means that if the computation of the Fourier transform of the dampend terminal payoff is exact, our algorithm would exhibit exponential convergence. It should be mentioned that since the second type of error builds up as the number of recursions increases, to obtain the same level of accuracy for cases with more monitoring dates we need to reduce the Sinc approximation error in each recursion step and this requires using more terms in the Sinc expansion.

4.7 Implementation of Discrete Approximations

We compute $\hat{v}^j_{\boldsymbol{\alpha}}(\boldsymbol{\xi})$ for $\boldsymbol{\xi}$ on a grid $\{\boldsymbol{kh}, -M_i \leq k_i \leq M_i, 1 \leq i \leq n\}$ for $j = N, \cdots, 1$. Recall $c(k,m)$ defined in (2.8). Using (4.20) and (4.21), we obtain

$$\hat{v}^N_{\boldsymbol{\alpha}}(\boldsymbol{kh}) = \hat{g}_{\boldsymbol{\alpha}}(\boldsymbol{kh}),$$
$$\hat{v}^{j-1}_{\boldsymbol{\alpha}}(\boldsymbol{kh}) \approx \frac{e^{-\Delta\psi(i\boldsymbol{\alpha})}}{2^n} \sum_{\boldsymbol{\beta}\in\{0,1\}^n} i^{|\boldsymbol{\beta}|_1} e^{i(\boldsymbol{k_\beta h_\beta})'\boldsymbol{l_\beta}}$$
$$\cdot \sum_{-M_j\leq m_j\leq M_j}^{\boldsymbol{\beta}} \left(\prod_{\{j:\beta_j=1\}} c(k_j,m_j) e^{-i(\boldsymbol{m_\beta h_\beta})'\boldsymbol{l_\beta}} \phi^{\boldsymbol{\alpha}}_{\Delta}(-\boldsymbol{m_\beta h_\beta} - \boldsymbol{k_{1-\beta} h_{1-\beta}}) \hat{v}^j_{\boldsymbol{\alpha}}(\boldsymbol{m_\beta h_\beta} + \boldsymbol{k_{1-\beta} h_{1-\beta}})\right), \tag{4.33}$$

$$v^0(\boldsymbol{x}) \approx \left(\prod_{j=1}^{n} \frac{h_j}{2\pi}\right) e^{-\boldsymbol{\alpha}'\boldsymbol{x} - \Delta\psi(i\boldsymbol{\alpha})} \sum_{-M_j\leq m_j\leq M_j, 1\leq j\leq n} e^{-i(\boldsymbol{mh})'\boldsymbol{x}} \phi^{\boldsymbol{\alpha}}_{\Delta}(-\boldsymbol{mh}) \hat{v}^1_{\boldsymbol{\alpha}}(\boldsymbol{mh}). \tag{4.34}$$

For any $\boldsymbol{\beta} \in \{0,1\}^n$, introduce the multidimensional discrete Hilbert transform (DHT):

$$\mathcal{H}^{\boldsymbol{h},\boldsymbol{M}}_{\boldsymbol{\beta}} f(\boldsymbol{kh}) = \sum_{-M_j\leq m_j\leq M_j}^{\boldsymbol{\beta}} \prod_{\{j:\beta_j=1\}} c(k_j,m_j) f(\boldsymbol{m_\beta h_\beta} + \boldsymbol{k_{1-\beta} h_{1-\beta}}).$$

The sum in (4.33) involves $2^n - 1$ partial or full DHTs. A naive implementation calculates $\sum^{\boldsymbol{\beta}}$ one by one for $\boldsymbol{\beta} \in \{0,1\}^n$ and for any $\boldsymbol{\beta}$ with k ones a k-fold sum needs to be evaluated. Even if the k-fold sum can be computed with the help of Toeplitz matrix-vector multiplication, the whole procedure is still computationally too intensive.

We propose an efficient way to evaluate (4.33) by recognizing that higher-order partial DHTs can be computed from lower-order ones. For example, the partial DHT for the first two dimensions can be obtained by applying DHT to the second dimension of the partial DHT for only the first dimension. Our implementation proceeds in a recursive way as follows.

Let $\boldsymbol{\phi}^{\boldsymbol{\alpha}}_{\Delta}$(resp. $\hat{\boldsymbol{v}}^j_{\boldsymbol{\alpha}}$) be an n-dimensional matrix with its $\boldsymbol{m}$-th ($-\boldsymbol{M} \leq \boldsymbol{m} \leq \boldsymbol{M}$) entry equal to $\phi^{\boldsymbol{\alpha}}_{\Delta}(-\boldsymbol{mh})$(resp. $\hat{v}^j_{\boldsymbol{\alpha}}(\boldsymbol{mh})$). Recall that $\circ$ denote pointwise multiplication and $*$ denote matrix

multiplication. Now suppose we already have matrices $\boldsymbol{\phi}^{\boldsymbol{\alpha}}_{\Delta}$ and $\hat{\boldsymbol{v}}^{j}_{\boldsymbol{\alpha}}$, the procedure of computing the matrix $\hat{\boldsymbol{v}}^{j-1}_{\boldsymbol{\alpha}}$ via (4.33) consists of the following steps.
(1) Compute the pointwise product $\boldsymbol{v}_1 = \boldsymbol{\phi}^{\boldsymbol{\alpha}}_{\Delta} \circ \hat{\boldsymbol{v}}^{j}_{\boldsymbol{\alpha}}$. This matrix gives the term in (4.33) with $\boldsymbol{\beta} = \boldsymbol{0}$ for all $\boldsymbol{kh}$.
(2) Repeat this step until matrices $\boldsymbol{v}_q, \boldsymbol{u}_q, q = 1, \cdots, n$ are obtained. To calculate $\boldsymbol{u}_q$, for all $-M_p \le k_p \le M_p$, $p = 1, \cdots, q-1, q+1, \cdots, n$, compute

$$
\begin{aligned}
&\boldsymbol{u}_q(k_1h_1, \cdots, k_{q-1}h_{q-1}, :, k_{q+1}h_{q+1}, \cdots, k_nh_n) \\
&= i\boldsymbol{E}(M_q, h_q, l_q) \circ [\boldsymbol{T}^q * (\boldsymbol{E}(M_q, h_q, -l_q) \circ \boldsymbol{v}_q(k_1h_1, \cdots, k_{q-1}h_{q-1}, :, k_{q+1}h_{q+1}, \cdots, k_nh_n))],
\end{aligned} \tag{4.35}
$$

where $\boldsymbol{T}^q$ is a Toeplitz matrix with entry $T^q_{km} = c(k, m)$, $-M_q \le k, m \le M_q$ and $\boldsymbol{E}(M, h, l) = \left(e^{i(-Mh)l}, \cdots, e^{i(Mh)l}\right)'$. The matrix $\boldsymbol{u}_q$ gives the sum of terms in (4.33) with $\boldsymbol{\beta}$ such that $\beta_p = 0$ or 1 for $p = 1, \cdots, q-1$, $\beta_q = 1$ and $\beta_p = 0$ for $p = q+1, \cdots, n$ for all $\boldsymbol{kh}$. Calculate $\boldsymbol{v}_{q+1} = \boldsymbol{v}_q + \boldsymbol{u}_q$. Then $\boldsymbol{v}_{q+1}$ is the sum of terms in (4.33) with $\boldsymbol{\beta}$ such that $\beta_p = 0$ for $p = q+1, \cdots, n$ for all $\boldsymbol{kh}$.
(3) Finally, set $\hat{\boldsymbol{v}}^{j-1}_{\boldsymbol{\alpha}} = \frac{1}{2^n} e^{-\Delta\psi(i\boldsymbol{\alpha})}(\boldsymbol{v}_n + \boldsymbol{u}_n)$.

$$
\begin{tikzcd}
\hat{\boldsymbol{v}}^{j}_{\boldsymbol{\alpha}} \arrow[r, "\circ\boldsymbol{\phi}^{\boldsymbol{\alpha}}_{\Delta}"] & \boldsymbol{v}_1 \arrow[r] \arrow[d, "(4.35)"] & \boldsymbol{v}_2 \arrow[r] \arrow[d, "(4.35)"] & \cdots \arrow[r] \arrow[d] & \boldsymbol{v}_n \arrow[r] \arrow[d, "(4.35)"] & {} \arrow[r, "\cdot\frac{1}{2^n}e^{-\Delta\psi(i\boldsymbol{\alpha})}"] & \hat{\boldsymbol{v}}^{j-1}_{\boldsymbol{\alpha}} \\
& \boldsymbol{u}_1 \arrow[ur] & \boldsymbol{u}_2 \arrow[ur] & \cdots \arrow[ur] & \boldsymbol{u}_n \arrow[ur] & &
\end{tikzcd}
$$

Figure 2: Diagram for computing $\hat{\boldsymbol{v}}^{j-1}_{\boldsymbol{\alpha}}$

The diagram in Figure 2 illustrates the recursive procedure. It only requires calculating (4.35) n times and this is the minimal number that one can obtain. An alternative recursive procedure would require calculating $2^n - 1$ times of equations like (4.35).

Below we summarize our algorithm, termed as *multidimensional fast Hilbert transform*, for pricing barrier options.
Step 1. Choose $\boldsymbol{\alpha} \in \mathcal{I}_{\boldsymbol{X}}$ such that $g_{\boldsymbol{\alpha}} \in L^1(\mathbb{R}^n)$. Select appropriate $M_p, h_p, 1 \le p \le n$. Compute $\phi^{\boldsymbol{\alpha}}_{\Delta}(-\boldsymbol{kh})$ and $\hat{g}_{\boldsymbol{\alpha}}(\boldsymbol{kh})$ for all $-M_p \le k_p \le M_p$, $p = 1, \cdots, n$.
Step 2. For $j = N, N-1, \cdots, 2$, implement steps (1) to (3) above to compute the value of $\hat{v}^{j-1}_{\boldsymbol{\alpha}}(\boldsymbol{kh})$ for all $-M_p \le k_p \le M_p$, $p = 1, \cdots, n$.
Step 3. Obtain $v^0(\boldsymbol{x})$ by (4.34).

Now we analyze the time and space computational complexity of the algorithm. Recall that $M = \max(M_1, \cdots, M_n)$ and assume that all M_ps are simply constant multiples of M. Computing Toeplitz matrix-vector product in (4.35) can be done efficiently using FFT (see Feng and Linetsky (2008a), Appendix B for the implementation detail). The complexity for computing $\hat{v}^{j-1}_{\boldsymbol{\alpha}}(k_1h_1, \cdots, k_{q-1}h_{q-1}, :, k_{q+1}h_{q+1}, \cdots, k_nh_n)$ for each pair $(k_1, \cdots, k_{q-1}, k_{q+1}, \cdots, k_n)$ is $O(M \log_2 M)$ and this has to be repeated $O(M^{n-1})$ times. So the cost of obtaining the matrix $\boldsymbol{u}_q$ for each q is $O(M^n \log_2 M)$ and adding them up gives the cost for $\hat{\boldsymbol{v}}^{j-1}_{\boldsymbol{\alpha}}$, which is $O(nM^n \log_2 M)$. In addition, the cost of computing (4.21) at a given $\boldsymbol{x}$ is $O(M^n)$. Thus, the backward induction procedure requires a total of $O(nNM^n \log_2 M)$ operations, where N is the number of monitoring dates. As we need to store the matrices $\boldsymbol{\phi}^{\boldsymbol{\alpha}}_{\Delta}$ as well as $\hat{\boldsymbol{v}}^{j}_{\boldsymbol{\alpha}}$ for the current j, the space requirement is $O(M^n)$ (we do not need to store $\hat{\boldsymbol{v}}^{l}_{\boldsymbol{\alpha}}$ for $l > j$).

Remark 4.9. Our method generalizes the Feng-Linetsky method for pricing barrier options and calculating survival probabilities in 1D exponential Lévy models to general dimensions. The extension is not trivial for several reasons. First, the nD Fourier transform of a function multiplied by an indicator involves Hilbert transforms of orders from 1 to n (see (4.10)), where the 1D case only involves the 1D Hilbert transform. As our previous discussions show, the sequence in which these Hilbert tranforms of varying orders are computed matters as it directly affects the computational efficiency. Second, while the Fourier transform of 1D dampened payoffs of calls and puts can be obtained in simple closed-form formulas, the Fourier transform for nD dampened payoffs of calls and puts may not admit a closed-form expression and even if there is one it can involve sophisticated special functions which are too time-consuming to compute (see Appendix B.1). In our implementation we reduce the Fourier transform integral to integrals of lower dimensions and calculate them numerically using a quadrature rule. Third, while $\mathcal{I}_{\boldsymbol{X}}$ is an interval in the 1D case and it can be easily dertermined, its structure in multidimensions is hard to figure out. Instead we look for a rectangular region inside $\mathcal{I}_{\boldsymbol{X}}$ (see Appendix B.2). Fourth, Sinc theory is only developed for the 1D case in Stenger (1993, 2011). We extend the theory to multidimensions and finding the right conditions and completing the proof require work.

5 Numerical Results for Barrier Options and Survival Probabilities

We test our method on several representative Lévy models in 2D and 3D for calculating the joint survival probability and pricing down-and-out call options under several monitoring frequencies (annual, monthly, weekly and daily). Computations are done using Matlab on a desktop computer with Intel Core i5-4590 CPU at 3.30 GHz and 8 GB RAM, running Windows 7. Below we denote the barrier for the price of asset j by L_j. In all cases, for the fast Hilbert transform algorithm we only set the value of M_1 for computation and the other M_ps and all h_ps are determined following Remark 4.4. After $\boldsymbol{\alpha}$ is chosen, we calculate $\boldsymbol{d}$ according to (B.9) for the survival probability and (B.10) for the call. We also provide the 99% confidence interval from Monte Carlo simulation with 10^7 samples as a benchmark for each case. For one case, we also show results from the Fourier-cosine method and the finite difference method.

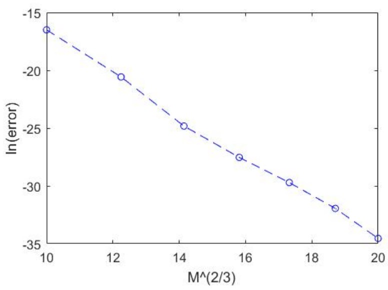
5.1 2D Black-Scholes Model

$\boldsymbol{X}$ follows a 2D Brownian motion with drift vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$. It is easy to see that $\lambda_-^j = -\infty$ and $\lambda_+^j = \infty$ for $j = 1, 2$. Therefore, from (B.8), any rectangular region $(\tilde{\lambda}_-^1, \tilde{\lambda}_+^1) \times (\tilde{\lambda}_-^2, \tilde{\lambda}_+^2)$ with $\tilde{\lambda}_-^j < -1$ and $\tilde{\lambda}_-^j > 0$ $(j = 1, 2)$ is inside $\mathcal{I}_{\boldsymbol{X}}$ and hence α_1, α_2 can be chosen freely from negative numbers. For this model, $\nu = 2$ in (4.22) due to (A.3) and $b = (\sigma_1^2 + \sigma_2^2 - \sqrt{\sigma_1^4 + \sigma_2^4 + (4\rho^2 - 2)\sigma_1^2\sigma_2^2})/2$, which is half of the smallest eigenvalue of Σ.

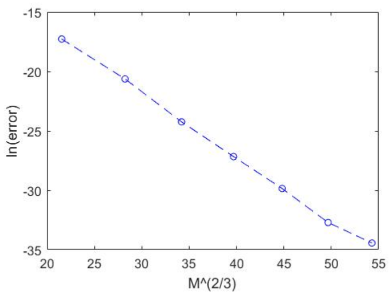
For the survival probability problem, we set $\mu_1 = 0.1$, $\sigma_1 = 0.2$, $\mu_2 = 0.05$, $\sigma_2 = 0.3$, $\rho = 0.5$, $S_0^1 = 100$, $S_0^2 = 80$, $L_1 = 70$, $L_2 = 40$ and $T = 1$. Table 1 shows the one-year joint survival probability under the physical measure for different monitoring frequencies. It can be seen that with only a few terms, we can already achieve very accurate results. In Figure 3 (a), we plot the convergence of the error as M_1 increases for $N = 12$, which clearly corroborates the theoretical estimate in Theorem 4.3 that shows exponential decay in $M_1^{1/2}$.

N	probability	time(s)	MC 99% CI
1	0.9832	0.013	(0.9830, 0.9834)
12	0.9741	0.092	(0.9738, 0.9744)
50	0.9685	0.378	(0.9682, 0.9688)
250	0.9647	8.114	(0.9644, 0.9650)

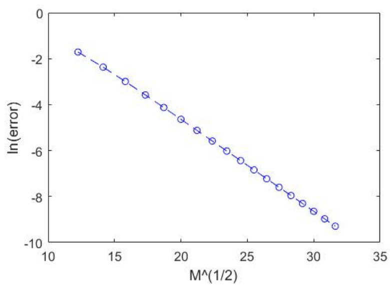
Table 1: One-year joint survival probability under the 2D Black-Scholes model ($L_1 = 70, L_2 = 40$). We set $\alpha_1 = \alpha_2 = -4$, $M_1 = 100$ for $N = 1, 12, 50$ and 200 when $N = 250$. It takes 216, 2526, 10544 and 37284 seconds for $N = 1, 12, 50, 250$, respectively to obtain the confidence interval from Monte Carlo simulation. The fast Hilbert transform results all fall in the intervals.



(a) 2D GBM



(b) 2D Merton's JD



(c) 2D NIG

Figure 3: Convergence of the 2D fast Hilbert transform algorithm for joint survival probability in three models under $N = 12$. In the plots, M refers to the value of M_1 and the values of M_2, h_1, h_2 are set according to Remark 4.4 from M. The results validate the conclusion in Theorem 4.3 which shows the algorithm converges exponentially if there is no approximation error in the terminal step and this is the case for the survival probability problem.

Moreover, to show our algorithm also work with barriers close to initial prices, we test it with two set of parameters $L_1 = 90, L_2 = 75$ (high barriers) and $L_1 = 98, L_2 = 76$ (extremely high barriers) with all other parameters remain unchanged. See Table 2 and Table 3.

N	probability	time(s)	MC 99% CI
1	0.5982	0.010	(0.5974, 0.5982)
12	0.2916	0.069	(0.2911, 0.2919)
50	0.2237	0.269	(0.2233, 0.2239)
250	0.1888	1.335	(0.1883, 0.1889)

Table 2: One-year joint survival probability under the 2D Black-Scholes model with high barriers ($L_1 = 90, L_2 = 75$). We set $\alpha_1 = \alpha_2 = -4$, $M_1 = 100$. It takes 107, 613, 2002 and 8634 seconds for $N = 1, 12, 50, 250$, respectively to obtain the confidence interval from Monte Carlo simulation. The fast Hilbert transform results all fall in the intervals.

N	probability	time(s)	MC 99% CI
1	0.5385	0.008	(0.5379, 0.5387)
12	0.1849	0.069	(0.1846, 0.1852)
50	0.1142	0.280	(0.1137, 0.1142)
250	0.0815	1.413	(0.0814, 0.0818)

Table 3: One-year joint survival probability under the 2D Black-Scholes model with extremely high barriers ($L_1 = 98, L_2 = 75$). We set $\alpha_1 = \alpha_2 = -4$, $M_1 = 100$. It takes 109, 466, 1209 and 4436 seconds for $N = 1, 12, 50, 250$, respectively to obtain the confidence interval from Monte Carlo simulation. The fast Hilbert transform results all fall in the intervals.

For pricing the 2D down-and-out call option, we set $\sigma_1 = 0.5$, $\sigma_2 = 1.2$, $\rho = 0.5$ and $\mu_j = r - q_j - \sigma_j^2/2$ $(j = 1, 2)$ with $r = 0.02$, $q_j = 0$. For the other parameters, $S_0^1 = 20$, $S_0^2 = 30$, $K = 50$, $L_1 = 16$, $L_2 = 24$ and $T = 1$. We compare the performance of the 2D fast Hilbert transform algorithm, the 2D Fourier-cosine method of Ruijter and Oosterlee (2012) and the 2D finite difference scheme using ADI (Duffy (2013)). Ruijter and Oosterlee (2012) does not treat barrier options, but using the idea for 1D barrier option pricing in Fang and Oosterlee (2009), we can easily develop the 2D Fourier-cosine algorithm for pricing 2D barrier call options. This algorithm requires the value of three parameters: L, Q and M. The parameter L controls the length of the truncation interval, while Q is the number of terms for each dimension to calculate the cosine expansion coefficient of the payoff function by the discrete cosine transform and M refers to the number of terms in the Fourier-cosine expansion for each dimension.

The results are listed in Table 4. Both the 2D fast Hilbert transform algorithm and the 2D Fourier-cosine method outperform the ADI finite difference scheme significantly. Comparing the former two, our method fares better than the 2D Fourier-cosine method. For $N = 1, 12$, both methods are very accurate, but our approach is faster. For $N = 50, 250$, our algorithm is notably better in both accuracy and computation time.

5.2 2D Merton's Jump-Diffusion Model with a Common Jump Component

Let $\boldsymbol{X}$ be a 2D jump-diffusion with a common jump component given in Section A. The bivariate jump size vectors are assumed to follow a bivariate normal distribution with mean vector $\boldsymbol{\theta}$ and covariance matrix $\boldsymbol{\Sigma}_J$. This model is a 2D extension of the standard 1D Merton's jump-diffusion model. It is easy to check that $\lambda_-^j = -\infty$ and $\lambda_+^j = \infty$ for $j = 1, 2$. Thus, like the Black-Scholes model, we can choose α_1, α_2 freely from negative numbers. For this model,

N	2D Fast Hilbert Transform price(error)	time(s)	2D Fourier-Cosine price(error)	time(s)	2D FDM-ADI price(error)	time(s)
1	14.8379(0.0024)	0.17	14.8379(0.0024)	0.29	14.8307(-0.0048)	6.45
12	8.3156(0.0006)	0.22	8.3155(0.0005)	0.61	8.3415(-0.0265)	6.58
50	6.4396(0.0003)	0.44	6.4288(-0.0105)	1.71	6.5074(0.0681)	6.41
250	5.4361(0.0008)	1.72	5.3618(-0.0735)	7.42	5.5821(0.1468)	6.39

Table 4: Prices of a one-year 2D down-and-out call option under the 2D Black-Scholes model. For the 2D fast Hilbert transform approach, we set $\alpha_1 = \alpha_2 = -2$, $M_1 = 100$. For the 2D Fourier-cosine method, we set $L = 10$, $Q = 2^{11}$ and, $M = 2^7$ for $N = 1$ and $M = 2^8$ for $N = 12, 50, 250$. For the finite difference method, we use 500 and 400 respectively for the number of discretization in time and in log-price. The benchmarks are $14.8355, 8.3150, 6.4393, 5.4353$ for $N = 1, 12, 50, 250$ respectively and they are obtained by running the 2D fast Hilbert transform and the 2D Fourier-cosine method with a large number of terms until they agree on the digits shown.

we have $\nu = 2$ in (4.22) due to (A.3) and b takes the same value as the Black-Scholes case. We set $\mu_1 = 0.04$, $\sigma_1 = 0.2$, $\mu_2 = 0.07$, $\sigma_2 = 0.3$, $\rho = 0.5$, $\lambda = 1$, $\theta_1 = -0.01$, $\theta_2 = -0.02$, $\sigma_{J1} = 0.15$, $\sigma_{J2} = 0.2$, $\rho_J = 0.4$, $S_0^1 = 100$, $S_0^2 = 80$, $L_1 = 70$, $L_2 = 40$ and $T = 1$. Table 5 displays the results for the one-year joint survival probability under the physical measure and our method also works very well in this case. Figure 3 (b) shows the exponential decay of the error in $M_1^{2/3}$, which again validates the theoretical estimate in Theorem 4.3. The performance of our method for pricing the 2D barrier call option under the Merton's model is similar to the Black-Scholes case and hence is omitted here.

N	probability	time(s)	MC 99% CI
1	0.9252	0.13	(0.9249, 0.9255)
12	0.8904	0.23	(0.8900, 0.8906)
50	0.8763	0.41	(0.8759, 0.8765)
250	0.8678	7.88	(0.8675, 0.8681)

Table 5: One-year joint survival probability under the 2D Merton's jump-diffusion model. We set $\alpha_1 = \alpha_2 = -4$, $M_1 = 100$ for $N = 1, 12, 50$, and 200 for $N = 250$. It takes 533, 5454, 21642 and 108381 seconds for $N = 1, 12, 50, 250$, respectively to obtain the confidence interval from Monte Carlo simulation. The fast Hilbert transform results all fall in the intervals.

5.3 2D Pure-Jump NIG Model

$\boldsymbol{X}$ is a 2D pure-jump NIG process presented in Section A. For the model parameters, we set $\mu_1 = 0.03$, $\mu_2 = 0.05$, $\sigma_1 = 0.2$, $\sigma_2 = 0.3$, $\rho = 0.5$, $c = 1$, $\eta = \pi$. To obtain the risk-neutral parameters we adjust the values of the drift vector using $r = 0.02$ and $q_j = 0$. The formulas for λ_-^j and λ_+^j can be found in Feng and Linetsky (2008a), Table 3.1. We use (B.8) with $n = 2$ to obtain one rectangular region inside $\mathcal{I}_{\boldsymbol{X}}$. It can be checked that $\tilde{\lambda}_-^j = \lambda_-^j/2 < -1$ in this example and $\boldsymbol{\alpha}$ is calculated by (B.9) and (B.10). For $S_0^1, S_0^2, L_1, L_2, K, T$, we use the same values as those in the 2D Black-Scholes model for the two problems. Table 6, Table 7 and Table 8 shows the performance of calculating the one-year risk-neutral joint survival probability with three different sets of barriers and Figure 3 (c) demonstrates the exponential decay of the error in $M_1^{1/2}$, substantiating the theoretical estimate in Theorem 4.3. Comparing the magnitude of the log-error in Figure 3 (a)/(b) to (c) for similar M, we can see that convergence is slower in the pure-jump case than the previous two models in which a Brownian motion component exists and consequently ν attains its maximum value 2. Despite the slower convergence rate, the results show that our method still works reasonably well for the survival probability problem. For the call option price, the results are presented in Table 9 and the performance is still good.

N	probability	time(s)	MC 99% CI
1	0.9541	0.023	(0.9539, 0.9543)
12	0.9383	0.852	(0.9380, 0.9384)
50	0.9319	16.4	(0.9316, 0.9321)
250	0.9291	680	(0.9287, 0.9292)

Table 6: One-year joint survival probability under the 2D pure-jump NIG model ($L_1 = 70, L_2 = 40$). We set $\alpha_1 = -6.4028, \alpha_2 = -4.4092$, $M_1 = 100$ for $N = 1$, 200 for $N = 12$, 500 for $N = 50$ and 2000 for $N = 250$. It takes 265, 2783, 10110 and 47603 seconds for $N = 1, 12, 50, 250$, respectively to obtain the confidence interval from Monte Carlo simulation. The fast Hilbert transform results all fall in the intervals.

N	probability	time(s)	MC 99% CI
1	0.4594	0.026	(0.4591, 0.4599)
12	0.2250	0.130	(0.2249, 0.2255)
50	0.1871	2.213	(0.1867, 0.1873)
250	0.1728	57.17	(0.1722, 0.1728)

Table 7: One-year joint survival probability under the 2D pure-jump NIG model with high barriers ($L_1 = 90, L_2 = 75$). We set $\alpha_1 = -6.4028, \alpha_2 = -4.4092$, $M_1 = 100$ for $N = 1, 12$, 200 for $N = 50$ and 500 for $N = 250$. It takes 183, 1041, 3663 and 17027 seconds for $N = 1, 12, 50, 250$, respectively to obtain the confidence interval from Monte Carlo simulation. The fast Hilbert transform results all fall in the intervals.

N	probability	time(s)	MC 99% CI
1	0.3780	0.016	(0.3778, 0.3786)
12	0.1240	0.530	(0.1238, 0.1244)
50	0.0889	11.40	(0.0887, 0.0891)
250	0.0764	57.14	(0.0761, 0.0765)

Table 8: One-year joint survival probability under the 2D pure-jump NIG model with extremely high barrier ($L_1 = 98, L_2 = 75$). We set $\alpha_1 = \alpha_2 = -4$, $M_1 = 100$ for $N = 1$, 200 for $N = 12$, and 500 for $N = 50, 250$. It takes 186, 718, 2190 and 9541 seconds for $N = 1, 12, 50, 250$, respectively to obtain the confidence interval from Monte Carlo simulation. The fast Hilbert transform results all fall in the intervals.

5.4 2D Pure-Jump VG Model

From Remark 4.8, we know our algorithm does not converge exponentially in this model. To accelerate its converge, we apply 2D spetral filtering and its details can be found in Appendix C. We set the parameters as follows: $S_1 = 100, S_2 = 80, L_1 = 70, L_2 = 40, \mu_1 = 0.03, \mu_2 = 0.05, \sigma_1 = 0.2, \sigma_2 = 0.3, \rho = 0.5, c = 4, \lambda = 4, T = 1/4$. We use the 2D exponential filter with its exponent set to 10 and fix $h_1 = h_2 = 0.5$ and let $\alpha_1 = -6.7060, \alpha_2 = -4.4444$. Table 10

N	price	time(s)	MC 99% CI
1	4.9675	1.23	(4.9578, 4.9718)
12	4.6773	2.02	(4.6723, 4.6863)
50	4.5896	94.3	(4.5790, 4.5930)
250	4.5515	3079	(4.5427, 4.5567)

Table 9: Price of a one-year 2D down-and-out call option under the 2D pure-jump NIG model. We set $\alpha_1 = -6.4028$, $\alpha_2 = -4.4091$, $M_1 = 200$ for $N = 1, 12$, 1000 for $N = 50$ and 4000 for $N = 250$. It takes 275, 2583, 9110 and 43603 seconds for $N = 1, 12, 50, 250$, respectively to obtain the confidence interval from Monte Carlo simulation. The fast Hilbert transform results all fall in the intervals.

clearly shows the spectal filter improves the accuracy of our method with the same $\boldsymbol{M}$ and $\boldsymbol{h}$. Significant improvement can also be observed for maturities equal to 1/12 and 1/2, but not for $T = 1$. The results are in line with those in the 1D case reported in Phelan et al. (2018).

N	probability(HT)	probability(filter)	time(s)	MC 99% CI
1	0.99736	0.99734	0.16	(0.99729, 0.99737)
12	0.99475	0.99680	1.51	(0.99679, 0.99688)
50	0.98648	0.99661	6.34	(0.99672, 0.99682)
250	0.99209	0.99638	31.8	(0.99672, 0.99681)

Table 10: One-quarter joint survival probability under the 2D VG model. We set $M_1 = 500$ and M_2 is determined using equation (4.29) and fix $h_1 = h_2 = 0.5$ for all all cases. It takes 156, 1877, 7881 and 39255 seconds for $N = 1, 12, 50, 250$, respectively to obtain the confidence interval from Monte Carlo simulation.

5.5 3D Black-Scholes Model

We set $\sigma_1 = 0.2$, $\sigma_2 = 0.3$, $\sigma_3 = 0.5$, $\rho_{12} = \rho_{13} = \rho_{23} = 0.5$, $S_0^1 = 100$, $S_0^2 = 80$, $S_0^3 = 100$, $L_1 = 50$, $L_2 = 40$, $L_3 = 60$ and $T = 1$. Like the 2D case, $\boldsymbol{\alpha}$ can be chosen freely from negative numbers. For calculation of survival probability under the physical measure, we set $\mu_1 = 0.1$, $\mu_2 = 0.05$, $\mu_3 = 0.08$. For pricing the call option under the risk-neutral measure, we adjust the drift vector with $r = 0.04$ and $q = 0$, and the strike $K = 280$. The results are shown in Table 11 and 12. For all frequencies, computation time increases compared with the 2D case, but it is still reasonable in practice. The only result that falls outside the confidence interval is the call price with daily monitoring. However, in this case the relative error of our result is small enough to be considered as accurate. The performance of the 3D Merton's jump-diffusion model is very similar to the 3D Black-Scholes model, so it is omitted to save space.

5.6 3D Pure-Jump NIG Model

We set $\mu_1 = 0.03$, $\mu_2 = 0.05$, $\mu_3 = 0.07$, $\sigma_1 = 0.2$, $\sigma_2 = 0.3$, $\sigma_3 = 0.4$, $\rho_{12} = \rho_{13} = \rho_{23} = 0.5$, $c = 1, \lambda = \pi$, $S_0^1 = 100$, $S_0^2 = 80$, $S_0^3 = 100$, $L_1 = 50$, $L_2 = 40$, $L_3 = 60$, $T = 1$ and call option strike $K = 280$. We obtain the drift under the risk-neutral measure with $r = 0.04$ and $q_j = 0$. In this example, we first set $\tilde{\lambda}_\pm^j = \lambda_\pm^j/2$ and verify that they give a rectangular region inside $\mathcal{I}_{\boldsymbol{X}}$. This is better than using (B.8) with $n = 3$. It can be checked that $\tilde{\lambda}_-^j < -1$ in this example and $\boldsymbol{\alpha}$ is calculated by (B.9) and (B.10).

Table 13 and 14 present the results for the survival probability and the call option. It can be seen that to obtain results of similar accuracy, it takes more time under the 3D pure-jump

N	probability	time(s)	MC 99% CI
1	0.8787	0.48	(0.8784, 0.8790)
12	0.8009	3.83	(0.8006, 0.8012)
50	0.7713	24.20	(0.7711, 0.7717)
250	0.7533	989	(0.7530, 0.7536)

Table 11: One-year joint survival probability under the 3D Black-Scholes model. We set $\alpha_1 = \alpha_2 = \alpha_3 = -4$, $M_1 = 50$ for $N = 1, 12$, 60 for $N = 50$ and 150 for $N = 250$. It takes 217, 2332, 9425 and 47036 seconds for $N = 1, 12, 50, 250$, respectively to obtain the confidence interval from Monte Carlo simulation. The fast Hilbert transform results all fall in the intervals.

N	price	time(s)	MC 99% CI
1	36.1119	6.10	(36.0420, 36.1404)
12	35.4911	39.95	(35.4270, 35.5254)
50	35.2137	188	(35.1286, 35.2270)
250	35.1024	1859	(34.9129, 35.0113)

Table 12: One-year call option under the 3D Black-Scholes model. We set $\alpha_1 = \alpha_2 = \alpha_3 = -2$, $M_1 = 50$ for $N = 1$, 100 for $N = 12$, 120 for $N = 50$ and 200 for $N = 250$. It takes 442, 4627, 18679 and 92205 seconds for $N = 1, 12, 50, 250$, respectively to obtain the confidence interval from Monte Carlo simulation. The fast Hilbert transform results all fall in the intervals except when $N = 250$.

NIG model than the Black-Scholes. Although for $N = 250$, our result is not in the confidence interval, it is still quite accurate. The error can be reduced by increasing the number of terms used in the Sinc approximation. In general, obtaining highly accurate results under $N = 250$ for a 3D pure-jump model is computationally demanding as lack of a Brownian component in the model drags down the convergence rate.

N	probability	time(s)	MC 99% CI
1	0.8670	4.54	(0.8668, 0.8674)
12	0.8180	45.37	(0.8177, 0.8184)
50	0.8020	188	(0.8018, 0.8025)
250	0.7896	4218	(0.7946, 0.7953)

Table 13: One-year joint survival probability under the 3D pure-jump NIG model. We set $\alpha_1 = -2.9514, \alpha_2 = -1.9546, \alpha_3 = -1.0707$, $M = 50$ for $N = 1, 12, 50$ and 150 for $N = 250$. It takes 233, 2611, 10699 and 55296 seconds for $N = 1, 12, 50, 250$, respectively to obtain the confidence interval from Monte Carlo simulation. The fast Hilbert transform results fall in the intervals except when $N = 250$.

We end this section with some remarks. First, our implementation was done in Matlab without parallelization. As our algorithm can be parallelized, significant reductions in computation time can be expected using a parallel implementation. Furthermore, switching to programming languages like C for implementation can also bring in significant reductions. Second, the examples considered all have a one-year maturity. For shorter maturities, there are fewer recursions to do, so computation time would be less than the numbers reported here.

N	price	time(s)	MC 99% CI
1	32.6177	32.55	(32.5510, 32.6417)
12	32.3283	114	(32.2605, 32.3512)
50	32.2219	317	(32.1740, 32.2648)
250	32.2991	4380	(32.0990, 32.1897)

Table 14: One-year call option under the 3D pure-jump NIG model. We set $\alpha_1 = -3.4514, \alpha_2 = -2.4546, \alpha_3 = -1.9611$, $M = 60$ for $N = 1$, 100 for $N = 12, 50$, and 150 for $N = 250$. It takes 249, 2658, 10975 and 53760 seconds for $N = 1, 12, 50, 250$, respectively to obtain the confidence interval from Monte Carlo simulation. The fast Hilbert transform results fall in the intervals except when $N = 250$.

6 Conclusion

This paper develops Sinc approximation for computing multidimensional Hilbert transform and analyze its convergence rate. Our results generalize the classical Sinc approximation results for 1D Hilbert transform. We demonstrate the efficiency of the method in two applications for problems with two or three dimensions: detecting edges in 2D images, and pricing barrier options and calculating survivial probabilities in multivariate exponential Lévy models. In the second application, the convergence rate of our algorithm depends on the decay rate of the characteristic function of the Lévy process. When it is a Brownian motion or a jump-diffusion, our method converges rapidly as the characteristic function decays exponentially with the highest possible order. For pure-jump models, its convergence speed varies and we consider two typical examples: the NIG and VG processes. Our method works well for the NIG model. For the VG model, although our method converges quite slowly, applying it together with a spectral filter can accelerate its convergence and deliver satisfactory results if the horizon of the problem is not too long. We also show through examples that our method compares favorably with Monte Carlo simulation, the Fourier-cosine method and finite difference in terms of computational performance. However, due to the curse of dimensionality, we don't expect the method to be tractable for high-dimensional problems.

The multidimensional Sinc approximation developed in this paper can be applied to solve various other problems, such as the valuation of exotic options like double lookback options (He et al. (1998)) and multi-asset Bermudan options (Ruijter and Oosterlee (2012)), valuation adjustments for credit derivatives in structural models (Lipton and Savescu (2014)) as well as stochastic control problems (Ge and Li (2020)) for exponential Lévy models.

Acknowledgements

We thank the Area Editor, the Associate Editor and three reviewers for their comments and suggestions which led to significant improvement in the paper. This research was supported by Hong Kong Research Grant Council General Research Fund Grant No. 14203418.

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Online Companion to "Sinc Approximation of Multidimensional Hilbert Transform and Its Applications"

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February 15, 2021

A Multidimensional Exponential Lévy Models

For excellent expositions on Lévy processes, one can refer to Bertoin (1998) and Sato (1999). Consider an n-dimensional Lévy process $\boldsymbol{X}_t := (X_t^1, \cdots, X_t^n)'$ that starts at $\boldsymbol{x} := (x_1, \cdots, x_n)'$. For $j = 1, \cdots, n$, X^j has Lévy characteristics $(\mu_j, \sigma_j^2, \Pi_j)$, which are the drift, variance rate of the Brownian motion component and Lévy measure, respectively. For X^j, the Lévy-Khintchine theorem shows that the characteristic function $\phi_t^j(\xi) := E_{x_j}[e^{i\xi(X_t^j - x_j)}] = e^{-t\psi^j(\xi)}$, where the characteristic exponent $\psi^j(\xi) = \frac{1}{2}\sigma_j^2\xi^2 - i\mu_j\xi + \int_{\mathbb{R}}(1 - e^{i\xi y} + i\xi y 1_{\{|y|\le 1\}})\Pi_j(dy)$. The Lévy measure Π_j satisfies $\int_{\mathbb{R}}(y^2 \wedge 1)\Pi_j(dy) < \infty$. Let $\mathcal{I}_{X^j} := \{\alpha \in \mathbb{R} : \int_{|y|>1} e^{-\alpha y}\Pi_j(dy) < \infty\}$, which is an interval with endpoints λ_-^j, λ_+^j $(-\infty \le \lambda_-^j \le 0 \le \lambda_+^j \le \infty)$. For specific 1D Lévy processes, the value of λ_-^j and λ_+^j can be found in Feng and Linetsky (2008), Table 3.1. From Theorem 25.17 in Sato (1999), we know that $\alpha \in \mathcal{I}_{X^j}$ if and only if $E_{x_j}[e^{-\alpha X_t^j}]$ is finite for all $t > 0$. In this paper, we require that $-1 \in \mathcal{I}_{X^j}$ so that $E_{x_j}[e^{X_t^j}]$ is finite for all $t > 0$. Following Feng and Linetsky (2008), we further require that $\lambda_-^j < -1$ and $\lambda_+^j > 0$. For the general Lévy process $\boldsymbol{X}$, we denote its characteristics by $(\boldsymbol{\mu}, \boldsymbol{\Sigma}, \Pi)$, with

$$\boldsymbol{\mu} = (\mu_1, \cdots, \mu_n)', \boldsymbol{\Sigma} = (\rho_{ij}\sigma_i\sigma_j)_{1\le i,j\le n}, \int_{\mathbb{R}^n}(|\boldsymbol{y}|^2 \wedge 1)\Pi(d\boldsymbol{y}) < \infty. \tag{A.1}$$

Here, $\boldsymbol{\rho} = (\rho_{ij})_{1\le i,j\le n}$ is the correlation matrix for the n-dimensional Brownian component. The joint characteristic function of $\boldsymbol{X}_t - \boldsymbol{x}$ is given by

$$\phi_t(\boldsymbol{\xi}) := E\left[e^{i\langle \boldsymbol{\xi}, \boldsymbol{X}_t - \boldsymbol{x}\rangle}\right] = e^{-t\psi(\boldsymbol{\xi})},$$
$$\psi(\boldsymbol{\xi}) = \frac{1}{2}\boldsymbol{\xi}'\boldsymbol{\Sigma}\boldsymbol{\xi} - i\boldsymbol{\mu}'\boldsymbol{\xi} + \int_{\mathbb{R}^2}(1 - e^{i\boldsymbol{\xi}'\boldsymbol{y}} + i\boldsymbol{\xi}'\boldsymbol{y}1_{\{|\boldsymbol{y}|\le 1\}})\Pi(d\boldsymbol{y}),$$

where $\psi(\boldsymbol{\xi})$ is the characteristic exponent of $\boldsymbol{X}$. We also consider

$$\mathcal{I}_{\boldsymbol{X}} := \left\{\boldsymbol{\alpha} \in \mathbb{R}^n : \int_{|\boldsymbol{y}|>1} e^{-\boldsymbol{\alpha}'\boldsymbol{y}}\Pi(d\boldsymbol{y}) < \infty\right\}.$$

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Theorem 25.17 in Sato (1999) shows that $\mathcal{I}_{\boldsymbol{X}}$ is a convex set that contains the origin and $\boldsymbol{\alpha} \in \mathcal{I}_{\boldsymbol{X}}$ if and only if $E_{\boldsymbol{x}}[e^{-\boldsymbol{\alpha}'\boldsymbol{X}_t}]$ is finite for all $t > 0$.

The asset price processes $S_t^j, 1 \leq j \leq n$ are assumed to follow

$$S_t^j = K_j e^{X_t^j}, \text{ with } X_0^j = \ln(S_0^j/K_j). \tag{A.2}$$

Here, K_j are positive constants chosen for convenience. If (A.2) is the model under a chosen risk-neutral measure, to ensure no arbitrage, the process $e^{-(r-q_j)t}e^{X_t^j}$ should be a martingale, which is equivalent to

$$\mu_j = r - q_j - \sigma_j^2/2 + \int_{\mathbb{R}} (1 - e^y + y1_{\{|y|\leq 1\}})\Pi_j(dy),$$

where r is the risk-free rate and q_j is the dividend yield for asset j.

Below we list several classes of multivariate Lévy processes that are common in financial modelling and present their characteristic exponents.

Multidimensional Brownian motion: Let $\boldsymbol{X}$ be a multidimensional Brownian motion with drift vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$ given in (A.1). Then $\psi(\boldsymbol{\xi}) = \frac{1}{2}\boldsymbol{\xi}'\boldsymbol{\Sigma}\boldsymbol{\xi} - i\boldsymbol{\mu}'\boldsymbol{\xi}$.

Multidimensional jump-diffusions with common jumps: Let $\boldsymbol{W}$ be a multidimensional Brownian motion with drift vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$ given in (A.1). For $j = 1, \cdots, n$, let $\boldsymbol{X}_t = \boldsymbol{W}_t + \sum_{k=1}^{N_t} \boldsymbol{Y}_k$, where N is a Poisson process with arrival rate λ, and $\boldsymbol{Y}_1, \boldsymbol{Y}_2, \cdots$ are i.i.d. with $\boldsymbol{Y}_k$ follows a given multidimensional distribution F. It is assumed that the jump part is independent of $\boldsymbol{W}$. For this process, $\psi(\boldsymbol{\xi})$ is given by $\psi(\boldsymbol{\xi}) = \frac{1}{2}\boldsymbol{\xi}'\boldsymbol{\Sigma}\boldsymbol{\xi} - i\boldsymbol{\mu}'\boldsymbol{\xi} + \lambda(1 - \int_{\mathbb{R}^n} e^{i\boldsymbol{\xi}'\boldsymbol{y}} F(d\boldsymbol{y}))$. Idiosyncratic jumps modelled by compound Poisson processes can be further added to the model (see Cont and Tankov (2004), Section 5.2 for general discussions).

Multidimensional time-changed Brownian motion: Let $\boldsymbol{W}$ be a multidimensional Brownian motion with drift vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$ given in (A.1). Consider a 1D Lévy subordinator T_t, which is a non-negative Lévy process (this implies that it is also nondecreasing). Its Laplace transform is given by

$$E[e^{-\lambda T_t}] = e^{-\Phi(\lambda)t}, \quad \Phi(\lambda) = \gamma\lambda + \int_{[0,\infty)} (1 - e^{-\lambda s})\nu(ds), \ \mathrm{Re}(\lambda) \geq 0.$$

where $\gamma \geq 0$ is the drift of T and ν is its Lévy measure. Suppose T and $\boldsymbol{W}$ are independent. Define $X_t^j := W_{T_t}^j$. Then $\boldsymbol{X}_t := (X_t^1, \cdots, X_t^n)'$ is a multidimensional Lévy process and $\psi(\boldsymbol{\xi}) = \Phi\left(\frac{1}{2}\boldsymbol{\xi}'\boldsymbol{\Sigma}\boldsymbol{\xi} - i\boldsymbol{\mu}'\boldsymbol{\xi}\right)$. While W^j has continuous sample paths, X^j is a jump-diffusion ($\gamma > 0$) or a pure-jump process ($\gamma = 0$). This class of processes was studied and applied to equity modelling in Luciano and Schoutens (2006) and Leoni and Schoutens (2008). Further generalization through multivariate subordination can be found in Luciano et al. (2016) and Mendoza-Arriaga and Linetsky (2016). A popular choice of Lévy subordinators in the literature is the inverse Gaussian process for which $\nu(ds) = cs^{-3/2}e^{-\eta s}ds$ $(c, \eta > 0)$ and $\Phi(\lambda) = \gamma\lambda + 2c\sqrt{\pi}[\sqrt{\lambda + \eta} - \sqrt{\eta}]$. Using this time change, each X^j is a normal inverse Gaussian (NIG) process and we call $\boldsymbol{X}$ as the multidimensional NIG model. Another popular Lévy subordinator is the Gamma process with $\nu(ds) = cs^{-1}e^{-\eta s}ds$ $(c, \eta > 0)$ and $\Phi(\lambda) = \gamma\lambda + c\ln(1 + \lambda/\eta)$. This time change leads to a multidimensional model with each coordinate a 1D variance gamma (VG) process and the joint process is called the multidimensional VG model. Other choices of Lévy subordinators can be found in Madan and Yor (2008) and Luciano et al. (2016), which can be used to construct multidimensional models with each coordinate a CGMY, a Meixner or a generalized hyperbolic process.

Linear combinations of independent Lévy processes: Ballotta and Bonfiglioli (2016) construct multivariate Lévy models by taking linear combinations of independent Lévy processes. Let $Y_1, \cdots, Y_n, Z$ be independent univariate Lévy processes. Then define $X_{jt} = Y_{jt} + a_j Z_t$ for

$j = 1, \cdots, n$. The process Z models the systematic factor whereas Y_j represents the idiosyncratic factor specific to asset j. This is a flexible and parsimonious family of models as shown in Ballotta and Bonfiglioli (2016). The joint characteristic function of $\boldsymbol{X}$ is given by

$$\phi_t(\boldsymbol{\xi}) = \phi_t^Z\left(\sum_{j=1}^n a_j \xi_j\right) \prod_{j=1}^n \phi_t^{Y_j}(\xi_j),$$

which is in closed-form if the characteristic functions of Y_js and Z are so.

Multidimensional KoBoL processes: This class of pure-jump Lévy processes is proposed in Boyarchenko and Levendorskii (2002). We refer readers to Section 9.1 for details on these processes and their characteristic exponents. The multidimensional NIG and VG models are special cases in this family.

For several results in the paper, we impose condition (4.22) for $\phi_t(\boldsymbol{\xi})$. This condition holds for any jump-diffusion and it can be shown as follows. The Lévy-Itô decomposition shows that $\boldsymbol{X}_t = \boldsymbol{\mu} t + \boldsymbol{B}_t + \boldsymbol{X}_t^l + \lim_{\epsilon \to 0} \tilde{\boldsymbol{X}}_t^\epsilon$, where $\boldsymbol{B}_t$ is a multidimensional driftless Brownian motion with covariance matrix $\boldsymbol{\Sigma}$, $\boldsymbol{X}^l$ is the process associated with the jumps of $\boldsymbol{X}$ with Euclidian norm greater than 1 while $\tilde{\boldsymbol{X}}_t^\epsilon$ is the compensated jump process for jumps of $\boldsymbol{X}$ with Euclidian norm between ϵ and 1, and the four components are independent. Using the decomposition, it is easy to see that, if the Brownian motion component is present (i.e., $\boldsymbol{\Sigma} \neq \boldsymbol{0}$),

$$|\phi_t(\boldsymbol{\xi})| = |e^{i\boldsymbol{\xi}'\boldsymbol{\mu} t}| \cdot |E[e^{i\boldsymbol{\xi}'\boldsymbol{B}_t}]| \cdot |E[e^{i(\boldsymbol{X}_t^l + \lim_{\epsilon\to 0} \tilde{\boldsymbol{X}}_t^\epsilon)}]| \leq |E[e^{i\boldsymbol{\xi}'\boldsymbol{B}_t}]| = e^{-\frac{1}{2} t \boldsymbol{\xi}'\boldsymbol{\Sigma}\boldsymbol{\xi}} \leq e^{-tb|\boldsymbol{\xi}|^2},$$

where b is one half of the smallest eigenvalue of $\boldsymbol{\Sigma}$ (we assume $\boldsymbol{\Sigma}$ is positive definite, so b is positive). Therefore, for any jump-diffusion, condition (4.22) holds with $\nu = 2$. For pure-jump Lévy processes, (4.22) needs to be checked case by case. An example we will analyze is the pure-jump NIG model ($\gamma = 0$), for which,

$$\psi(\boldsymbol{\xi}) = 2c\sqrt{\pi}\left[\sqrt{\boldsymbol{\xi}'\boldsymbol{\Sigma}\boldsymbol{\xi}/2 - i\boldsymbol{\mu}'\boldsymbol{\xi} + \eta} - \sqrt{\eta}\right].$$

It can be checked that for this model, (4.22) is valid with $\nu = 1$ and $b = c\sqrt{2\pi\lambda_m}$, where λ_m is the smallest eigenvalue of $\boldsymbol{\Sigma}$.

B Issues of Discrete Approximations for Barrier Option Pricing

This section provides discrete approximations for (4.13), (4.14) and (4.15). We want to compute $\hat{v}_{\boldsymbol{\alpha}}^k(\boldsymbol{\xi})$ for $\boldsymbol{\xi}$ on a grid $\{(\boldsymbol{kh}) : -M_j \leq k_j \leq M_j, 1 \leq j \leq n\}$ for $k = N, \cdots, 1$. Let $M = \max(M_1, \cdots, M_n)$.

B.1 Computation of $\hat{g}_{\boldsymbol{\alpha}}(\boldsymbol{\xi})$

Unlike the 1D case where $\hat{g}_{\boldsymbol{\alpha}}(\boldsymbol{\xi})$ has an easy-to-compute closed-form formula for typical barrier payoffs, the efficient computation of $\hat{g}_{\boldsymbol{\alpha}}(\boldsymbol{\xi})$ for a general dimension has to be addressed on a case-by-case basis. For the survival probability calculation, $g(\boldsymbol{x}) = \mathbb{1}_{(\boldsymbol{l},\infty)}(\boldsymbol{x})$ and the dampened function $g_{\boldsymbol{\alpha}}(\boldsymbol{x}) \in L^1(\mathbb{R}^n)$ if $\boldsymbol{\alpha} < \boldsymbol{0}$. In this case, it is easy to obtain a closed-form formula for $\hat{g}_{\boldsymbol{\alpha}}(\boldsymbol{\xi})$, which is given by

$$\hat{g}_{\boldsymbol{\alpha}}(\boldsymbol{\xi}) = \prod_{p=1}^n \frac{e^{l_p(\alpha_p + i\xi_p)}}{\alpha_p + i\xi_p}. \tag{B.1}$$

We next show how to calculate $\hat{g}_{\boldsymbol{\alpha}}(\boldsymbol{\xi})$ for the call payoff $(\sum_{p=1}^n S_T^p - K)^+$ (see Remark 4.2) when $n = 2, 3$. The put payoff can be dealt with similarly. We select $K_p = K$ for all p

in (A.2) and for the call $g(\boldsymbol{x})$ can be written as $K(\sum_{p=1}^{n} e^{x_p} - 1)^{+}\mathbb{1}_{(\boldsymbol{l},\infty)}(\boldsymbol{x})$. The dampened function $g_{\boldsymbol{\alpha}}(\boldsymbol{x}) \in L^1(\mathbb{R}^n)$ if $\boldsymbol{\alpha} < -\mathbf{1}$. When $\sum_{p=1}^{n} e^{l_p} \geq 1$, the formula for $g_{\boldsymbol{\alpha}}(\boldsymbol{x})$ can be easily obtained. Below we will only discuss the nontrivial case where $\sum_{p=1}^{n} e^{l_p} < 1$ (detailed derivation is omitted).

(1) $n = 2$:

$$\hat{g}_{\boldsymbol{\alpha}}(\boldsymbol{\xi}) = I_1(\boldsymbol{\xi}) - I_2(\boldsymbol{\xi}),$$

where

$$I_1(\boldsymbol{\xi}) = K e^{(i\xi_1+\alpha_1)l_1+(i\xi_2+\alpha_2)\ln(1-e^{l_1})}\cdot \left[\frac{e^{l_1}}{(i\xi_1+\alpha_1+1)(i\xi_2+\alpha_2)} + \frac{e^{\ln(1-e^{l_1})}}{(i\xi_1+\alpha_1)(i\xi_2+\alpha_2+1)} - \frac{1}{(i\xi_1+\alpha_1)(i\xi_2+\alpha_2)}\right], \tag{B.2}$$

$$I_2(\boldsymbol{\xi}) = K\int_{l_2}^{\ln(1-e^{l_1})} h_{\boldsymbol{\xi}}(x_2)dx_2, \tag{B.3}$$

and

$$h_{\boldsymbol{\xi}}(x_2) = \frac{e^{(\alpha_2+i\xi_2)x_2}(1-e^{x_2})^{\alpha_1+i\xi_1+1}}{\alpha_1+i\xi_1+1} + \frac{e^{(\alpha_2+i\xi_2+1)x_2}(1-e^{x_2})^{\alpha_1+i\xi_1}}{\alpha_1+i\xi_1} - \frac{e^{(\alpha_2+i\xi_2)x_2}(1-e^{x_2})^{\alpha_1+i\xi_1}}{\alpha_1+i\xi_1}.$$

The integral in (B.3) can be calculated in closed-form using the Gauss hypergeometric function ${}_2F_1$ with complex-valued arguments. However, we find that computing ${}_2F_1$ is time-consuming, so we propose to calculate I_2 by discretizing the 1D integral. For simplicity, we use the rectangle rule on $[l_2, \ln(1-e^{l_1})]$ by dividing it into $\tilde{M}$ equal subintervals and obtain

$$I_2(\boldsymbol{\xi}) \approx K\tilde{h}\sum_{j=1}^{\tilde{M}} h_{\boldsymbol{\xi}}(\tilde{x}_j), \tilde{h} = \frac{\ln(1-e^{l_1}) - l_2}{\tilde{M}}, \tilde{x}_j = l_2 + (j-0.5)\tilde{h}. \tag{B.4}$$

In our implementation, we set $\tilde{M}$ sufficiently large to obtain good accuracy and $\tilde{M} \geq 2M+1$. To efficiently compute the sum in (B.4), we utilize the fractional FFT to calculate the sum for all ξ_2 for every ξ_1. This approach has a total time complexity of $O(M\tilde{M}\log_2\tilde{M})$ as opposed to $O(M\tilde{M}^2)$ for the implementation without fractional FFT. Of course, higher-order numerical quadrature rules can also be applied to discretize the integral.

(2) $n = 3$:

$$\hat{g}_{\boldsymbol{\alpha}}(\boldsymbol{\xi}) = I_1(\boldsymbol{\xi}) - I_2(\boldsymbol{\xi}),$$

where

$$\begin{aligned} I_1(\boldsymbol{\xi}) =& K\Bigg[\frac{e^{(\alpha_1+1+i\xi_1)l_1}}{\alpha_1+1+i\xi_1}\frac{e^{(\alpha_2+i\xi_2)l_2}}{\alpha_2+i\xi_2}\frac{e^{(\alpha_3+i\xi_3)l_3}}{\alpha_3+i\xi_3} + \frac{e^{(\alpha_1+i\xi_1)l_1}}{\alpha_1+i\xi_1}\frac{e^{(\alpha_2+1+i\xi_2)l_2}}{\alpha_2+1+i\xi_2}\frac{e^{(\alpha_3+i\xi_3)l_3}}{\alpha_3+i\xi_3} \\ &+ \frac{e^{(\alpha_1+i\xi_1)l_1}}{\alpha_1+i\xi_1}\frac{e^{(\alpha_2+i\xi_2)l_2}}{\alpha_2+i\xi_2}\frac{e^{(\alpha_3+1+i\xi_3)l_3}}{\alpha_3+1+i\xi_3} - \frac{e^{(\alpha_1+i\xi_1)l_1}}{\alpha_1+i\xi_1}\frac{e^{(\alpha_2+i\xi_2)l_2}}{\alpha_2+i\xi_2}\frac{e^{(\alpha_3+i\xi_3)l_3}}{\alpha_3+i\xi_3}\Bigg], \end{aligned} \tag{B.5}$$

$$I_2(\boldsymbol{\xi}) = K\int_{l_1}^{\ln(1-e^{l_3}-e^{l_2})}\int_{l_2}^{\ln(1-e^{l_3}-e^{x_1})} h_{\boldsymbol{\xi}}(x_1,x_2)dx_2dx_1. \tag{B.6}$$

and

$$\begin{aligned} h_{\boldsymbol{\xi}}(x_1,x_2) =& e^{(\alpha_1+i\xi_1)x_1+(\alpha_2+i\xi_2)x_2} \\ &\times\left[-\frac{(1-e^{x_1}-e^{x_2})^{\alpha_3+1+i\xi_3}}{(\alpha_3+i\xi_3)(\alpha_3+1+i\xi_3)} + \frac{(1-e^{x_1}-e^{x_2})e^{(\alpha_3+i\xi_3)l_3}}{\alpha_3+i\xi_3} - \frac{e^{(\alpha_3+1+i\xi_3)l_3}}{\alpha_3+1+i\xi_3}\right]. \end{aligned}$$

To obtain (B.6), we again apply the rectangle rule. The grid is as follows. For x_1, divide $[l_1, \ln(1-e^{l_3}-e^{l_2})]$ equally, i.e.,

$$x_1^k = l_1 + (k-0.5)\Delta x_1, 1 \le k \le \tilde{M}, \Delta x_1 = \frac{\ln(1-e^{l_3}-e^{l_2})-l_1}{\tilde{M}}.$$

For each x_1^k, divide $[l_2, \ln(1-e^{l_3}-e^{x_1^k})]$ equally, i.e.,

$$x_2^{k,j} = l_2 + (j-0.5)\Delta x_2^k, 1 \le j \le \tilde{M}, \Delta x_2^k = \frac{\ln(1-e^{l_3}-e^{x_1^k})-l_2}{\tilde{M}}.$$

Then, we have

$$I_2(\boldsymbol{\xi}) \approx \sum_{k=1}^{\tilde{M}} \Delta x_1 \left(\sum_{j=1}^{\tilde{M}} \Delta x_2^k \cdot h(x_1^k, x_2^{k,j}) e^{i x_2^{k,j} \xi_2} \right) e^{i x_1^k \xi_1}. \tag{B.7}$$

Like the 2D case, we set $\tilde{M}$ sufficiently large to obtain good accuracy and $\tilde{M} \ge 2M+1$. To efficiently compute the above double sum, we apply the fractional FFT to compute the inner sum for all ξ_2 and all x_1^k, and then apply the fractional FFT again to compute the outer sum for all ξ_1. This procedure is repeated for every ξ_3. The total time complexity is $O(M\tilde{M}^2 \log_2 \tilde{M})$.

B.2 Validity of Discrete Approximations, Choice of $\boldsymbol{\alpha}$ and Analytic Strips

In order to implement the discrete approximations of the two operators in Section 4.5 for the recursion (4.14) and the integral (4.19), we need to show that each term $\phi_\Delta^{\boldsymbol{\alpha}}(-\boldsymbol{z})\hat{v}_{\boldsymbol{\alpha}}^k(\boldsymbol{z})$, $k = N, \cdots, 1$ verifies the conditions in Theorem 4.2. To this end, we assume the following.

(1) Conditions (4.23) and (4.24) hold.

(2) Suppose that for every $j = 1, \cdots, n$ there exist $\tilde{\lambda}_-^j < 0$ and $\tilde{\lambda}_+^j > 0$ such that $(\tilde{\lambda}_-^1, \tilde{\lambda}_+^1) \times \cdots \times (\tilde{\lambda}_-^n, \tilde{\lambda}_+^n) \subset \mathcal{I}_{\boldsymbol{x}}$.

(3) For the chosen $\boldsymbol{\alpha} \in \mathcal{I}_{\boldsymbol{X}}$, $\hat{g}_{\boldsymbol{\alpha}}(\boldsymbol{z})$ is analytic in some tube domain containing $\mathbb{R}^n$.

Under (2), $\phi_\Delta(\boldsymbol{z})$ is analytic in $\mathcal{S}_1 \times \cdots \times \mathcal{S}_n$, where $\mathcal{S}_j := \{z_j \in \mathbb{C} : \mathrm{Im}(z_j) \in (\tilde{\lambda}_-^j, \tilde{\lambda}_+^j)\}$ $(j = 1, \cdots, n)$. From (4.6), $\phi_\Delta^{\boldsymbol{\alpha}}(-\boldsymbol{z})$ is analytic in $\mathcal{S}_1^{\alpha_1} \times \cdots \times \mathcal{S}_n^{\alpha_n}$, where $\mathcal{S}_j^{\alpha_j} := \{z_j \in \mathbb{C} : \mathrm{Im}(z_j) \in (\alpha_j - \tilde{\lambda}_+^j, \alpha_j - \tilde{\lambda}_-^j)\}$. Together with (3), we know that $\phi_\Delta^{\boldsymbol{\alpha}}(-\boldsymbol{z})\hat{g}_{\boldsymbol{\alpha}}(\boldsymbol{z})$ is analytic in some tube domain $\mathcal{D}_{\boldsymbol{d}} = \mathcal{D}_{d_1} \times \cdots \times \mathcal{D}_{d_n}$ containing $\mathbb{R}^n$. We defer the discussions on the choice of $\boldsymbol{\alpha}$ and its impact on $\boldsymbol{d}$ for later (see (B.8) and (B.9)). Additionally, we assume that

(4) For the chosen $\boldsymbol{\alpha} \in \mathcal{I}_{\boldsymbol{X}}$, $g_{\boldsymbol{\alpha}-\boldsymbol{y}} \in L^1(\mathbb{R}^n)$ if $|y_j| < d_j$ for all $j = 1, \cdots, n$.

If a function $f(\boldsymbol{z})$ is analytic in $\mathcal{D}_{\boldsymbol{d}}$, then $\mathcal{H}_{\boldsymbol{\beta}} f$ can be analytically continued for any $\boldsymbol{\beta} \in \{0,1\}^n$ so that it is analytic in $\mathcal{D}_{\boldsymbol{d}}$. Using this observation and the recursion (4.14), we can conclude that all $\hat{v}_{\boldsymbol{\alpha}}^k(\boldsymbol{z})$ are also analytic in $\mathcal{D}_{\boldsymbol{d}}$. Furthermore, all $\hat{v}_{\boldsymbol{\alpha}}^k(\boldsymbol{z})$ are bounded in $\mathcal{D}_{\boldsymbol{d}}$, which we now explain. First, using (4.12), it is easy to see that

$$\|\hat{v}_{\boldsymbol{\alpha}}^k(\cdot)\|_{L^\infty(\mathbb{R}^n)} \le e^{-\Delta(N-k)\psi(i\boldsymbol{\alpha})} \|g_{\boldsymbol{\alpha}}(\cdot)\|_{L^1(\mathbb{R}^n)}, \ k = 1, \cdots, N.$$

This inequality further implies that

$$\|\hat{v}_{\boldsymbol{\alpha}}^k(\cdot + i\boldsymbol{y})\|_{L^\infty(\mathbb{R}^n)} \le e^{-\Delta(N-k)\psi(i(\boldsymbol{\alpha}-\boldsymbol{y}))} \|g_{\boldsymbol{\alpha}-\boldsymbol{y}}(\cdot)\|_{L^1(\mathbb{R}^n)}, \ k = 1, \cdots, N,$$

if $|y_j| < d_j$ for $j = 1, \cdots, n$. Boundedness of $\hat{v}_{\boldsymbol{\alpha}}^k(\boldsymbol{z})$ for $\boldsymbol{z} \in \mathcal{D}_{\boldsymbol{d}}$ then follows.

Now we conclude that under (1) to (4), the discrete approximations (4.20) and (4.21) can be applied to compute the recursion (4.14) and the integral (4.19).

We next discuss verification of (1) to (4).

For (1), it can be done given the explicit expression of the characteristic function of a given model (see Remark 4.6).

In (2), we require a rectangular region inside $\mathcal{I}_{\boldsymbol{X}}$, which is always possible. Unlike the 1D case, for a multidimensional Lévy process, it is difficult to characterize $\mathcal{I}_{\boldsymbol{X}}$ explicitly and it may not be rectangular. To find such a region inside $\mathcal{I}_{\boldsymbol{X}}$, we apply Hölder's inequality with $p = \frac{n}{n-1}, q = n$ and use induction to obtain

$$\begin{aligned} E_{\boldsymbol{x}}[e^{-\boldsymbol{\alpha}'\boldsymbol{X}_t}] &\leq \left(E_{(x_1,\cdots,x_{n-1})}\left[e^{-\frac{n}{n-1}\sum_{j=1}^{n-1}\alpha_j X_t^j}\right]\right)^{\frac{n-1}{n}} \left(E_{x_n}\left[e^{-n\alpha_n X_t^n}\right]\right)^{\frac{1}{n}} \\ &\leq \prod_{j=1}^{n}\left(E_{x_j}\left[e^{-n\alpha_j X_t^j}\right]\right)^{\frac{1}{n}}. \end{aligned}$$

Recall that each $\mathcal{I}_{X^j}$ is an interval with endpoints λ_-^j, λ_+^j. Thus,

$$\left(\lambda_-^1/n, \lambda_+^1/n\right) \times \cdots \times \left(\lambda_-^n/n, \lambda_+^n/n\right) \subset \mathcal{I}_{\boldsymbol{X}},$$

and we simply set $\tilde{\lambda}_\pm^j = \lambda_\pm^j/n$. Alternatively, one can use trial-and-error, i.e., propose a rectangular region and verify that it is inside $\mathcal{I}_{\boldsymbol{X}}$.

The requirement in (3) can be checked from the explicit expression of $\hat{g}_{\boldsymbol{\alpha}}(\boldsymbol{z})$ if it is available. Otherwise we replace $\hat{g}_{\boldsymbol{\alpha}}(\boldsymbol{z})$ by its approximation and analyze its analyticity.

For (4), it can be verified directly after $\boldsymbol{\alpha}$ is selected and $\boldsymbol{d}$ is determined.

The value of $\boldsymbol{\alpha}$ is critical to the convergence speed of the algorithm as it determines $\boldsymbol{d}$, the vector of half-widths of the analytic strips, which affects the convergence rate (see (4.27) and (4.28)). To show how to select $\boldsymbol{\alpha}$, it is best to work through some examples.

- Joint survival probability: For $\boldsymbol{\alpha} < \boldsymbol{0}$, the poles of (B.1) are given by $(i\alpha_1, \cdots, i\alpha_n)$. Thus, $\phi_\Delta^{\boldsymbol{\alpha}}(-\boldsymbol{z})\hat{g}_{\boldsymbol{\alpha}}(\boldsymbol{z})$ is analytic in $\{\boldsymbol{z} \in \mathbb{C}^n : \mathrm{Im}(Z_j) \in (\alpha_j, \alpha_j - \tilde{\lambda}_-^j),\ j = 1, \cdots, n\}$. To maximize every d_j, we set $(\alpha_j, \alpha_j - \tilde{\lambda}_-^j)$ to be symmetric around the real axis, which gives

$$\alpha_j = \tilde{\lambda}_-^j/2,\ d_j = -\alpha_j. \tag{B.8}$$

- 2D/3D call: For $\boldsymbol{\alpha} < -\boldsymbol{1}$, we can determine the poles of the approximation of $\hat{g}_{\boldsymbol{\alpha}}(\boldsymbol{\xi})$ from (B.2), (B.4) for the 2D case and from (B.5), (B.7) for the 3D case. Suppose $\tilde{\lambda}_-^j < -1$ for each j. Then, $\phi_\Delta^{\boldsymbol{\alpha}}(-\boldsymbol{z})\hat{g}_{\boldsymbol{\alpha}}(\boldsymbol{z})$ is analytic in $\{\boldsymbol{z} \in \mathbb{C}^n : \mathrm{Im}(z_j) \in (\alpha_j + 1, \alpha_j - \tilde{\lambda}_-^j),\ j = 1, \cdots, n\}$ for $n = 2, 3$. To maximize every d_j, we set $(\alpha_j + 1, \alpha_j - \tilde{\lambda}_-^j)$ to be symmetric around the real axis, which gives

$$\alpha_j = (\tilde{\lambda}_-^j - 1)/2,\ d_j = -\alpha_j - 1. \tag{B.9}$$

In both examples, using the choice of $\boldsymbol{\alpha}$, one can verify that (4) holds.

C Multidimensional Spectral Filters

When the characteristic function does not decay exponetially, we propose to apply spectral filters to accelerate the convergence of our method. For 1D exponential Lévy models, 1D spectral filters have been applied by Ruijter et al. (2015) for accerlerating the Fourier cosine method and by Phelan et al. (2018) for speeding up the fast Hilbert transform algorithm of Feng and Linetsky (2008). Here, we consider multidimensional spectral filters and present a short but self-contained discussion for the convenience of readers.

First recall the definition of a 1D filter of oder p defined in Vandeven (1991) and Gottlieb and Shu (1997). It is a function denoted by $\sigma(\eta)$ supported on $\eta \in [-1, 1]$ with the following properties:

$$\begin{cases} (a) & \sigma(0) = 0, \sigma^{(l)}(0) = 0, \ \forall l < p, \\ (b) & \sigma(\eta) = 0 \text{ for } |\eta| = 1, \\ (c) & \sigma(\eta) \in C^{p-1}. \end{cases} \tag{C.1}$$

A commonly used 1D filter is the exponential filter, which taks the following form:

$$\sigma(\eta) = e^{-C\eta^p},$$

where p is even and positive. This function does not meet the requirement (b) in Eq. (C.1) as it is not exactly zero when $|\eta| = 1$. However, we can select $C < \epsilon \log 10$, where $10^{-\epsilon}$ is machine precision so that the tiny difference can be neglected. An advantage of the exponential filter is that it has a simple form and the order of the filter is given by the parameter p.

For multidimensional filters, we consider

$$\sigma(\boldsymbol{\eta}) = \prod_{j=1}^{n} \sigma_j(\eta_j)$$

where $\sigma_j, j = 1, \cdots, n$ are 1D filters. For simplicity, they are from the same filter class and in our implementation each σ_j is a 1D exponential filter.

To use a multidimensional spectral filter in the computation of barrier option prices and survival probabilities, we simply replace the characteristic function $\phi_\Delta^{\boldsymbol{\alpha}}(\boldsymbol{\xi})$ in the formulas for $\mathcal{P}_{\boldsymbol{h},\boldsymbol{M}}^{\Delta}$ and $\mathcal{R}_{\boldsymbol{h},\boldsymbol{M}}^{\Delta}$ with $\phi_\Delta^{\boldsymbol{\alpha}}(\boldsymbol{\xi})\sigma(\boldsymbol{\xi})$.

D Proofs

Theorem 2.1. We first analyze $E(f, \boldsymbol{h})(\boldsymbol{x})$. Since $f \in L^2$, we can represent f as $\mathcal{F}^{-1}\hat{f}$.

$$\begin{aligned} & f(\boldsymbol{x}) - \sum_{\boldsymbol{k}\in\mathbb{Z}^n} \left(f(\boldsymbol{kh}) \prod_{i=1}^{n} S(k_i, h)(x_i) \right) \\ =& \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} \hat{f}(\boldsymbol{t}) \left(e^{-i\boldsymbol{x}'\boldsymbol{t}} - \sum_{\boldsymbol{k}\in\mathbb{Z}^n} \left(\prod_{j=1}^{n} S(k_j, h)(x_j) e^{-ik_j h t_j} \right) \right) d\boldsymbol{t} \\ =& \frac{1}{(2\pi)^n} \int_{\{\boldsymbol{t}:|t_j|>\pi/h \ \forall j\}} \hat{f}(\boldsymbol{t}) \left(e^{-i\boldsymbol{x}'\boldsymbol{t}} - \sum_{\boldsymbol{k}\in\mathbb{Z}^n} \left(\prod_{j=1}^{n} S(k_j, h)(x_j) e^{-ik_j h t_j} \right) \right) d\boldsymbol{t} \end{aligned}$$

We used the following relationship in the above derivation:

$$\sum_{k\in\mathbb{Z}} S(k, h)(x) e^{ikht} = \begin{cases} e^{ixt}, & \text{if } -\pi/h < t < \pi/h, \\ e^{ix(t-2m\pi/h)}, & \text{if } (2m-1)\pi/h < t < (2m+1)\pi/h, \\ \cos(\pi x/h), & \text{if } t = (2m \pm 1)\pi/h, \end{cases}$$

which shows the term in the parenthesis is zero in the region $\{\boldsymbol{t} : |t_j| \le \pi/h \ \forall j\}$, and bounded by 2 outside the region. Thus, we have

$$\|E(f, \boldsymbol{h})(\boldsymbol{x})\| \le \frac{1}{(2\pi)^n} \int_{\{\boldsymbol{t}:|t_j|>\pi/h_j \ \ \forall j\}} 2|\hat{f}(\boldsymbol{t})| d\boldsymbol{t} \le \frac{C^n}{\pi^n} \int_{\{\boldsymbol{t}:|t_j|>\pi/h_j \ \ \forall j\}} e^{-d|\boldsymbol{t}|} d\boldsymbol{t}$$

$$\le \frac{C^n}{\pi^n}\int_{\{\boldsymbol{t}:|t_j|>\pi/h_j\ \forall j\}} e^{-\frac{d}{\sqrt{n}}(|t_1|+\cdots+|t_n|)}d\boldsymbol{t} \le \frac{C^n}{\pi^n}\prod_{j=1}^n \int_{|t_j|>\pi/h_j} e^{-\frac{d}{\sqrt{n}}|t_j|}dt_j$$
$$\le \left(\frac{C\sqrt{n}}{\pi d}\right)^n \prod_{j=1}^n e^{-\frac{d\pi}{\sqrt{n}h_j}}.$$

For the multidimensional Sinc approximation on $\mathcal{H}_n f$, using the representation

$$\mathcal{H}_n f(\boldsymbol{x}) = \frac{1}{(2\pi)^n}\int_{\mathbb{R}^n}\left(\prod_{j=1}^n -\mathrm{isgn}(t_j)\right)\hat{f}(\boldsymbol{t})e^{-i\boldsymbol{x}'\boldsymbol{t}}d\boldsymbol{t}$$

and the fact that

$$\sum_{k\in\mathbb{Z}}\frac{\cos[\pi(x-kh)/h]-1}{\pi(x-kh)/h}e^{-ikh\xi} = \frac{P.V.}{\pi}\int_{\mathbb{R}}\frac{S(k,h)(t)e^{-ikh\xi}}{t-x}dt$$
$$= \frac{P.V.}{\pi}\int_{\mathbb{R}}\frac{e^{-it\xi'}}{t-x}dt = -\mathrm{isgn}(\xi')e^{-ix\xi'}, \xi' = \begin{cases}\xi, & -\pi/h<\xi<\pi/h,\\ \xi-(2m\pi/h), & \text{otherwise.}\end{cases}$$

we have

$$\mathcal{H}_n f(x) - \sum_{k\in\mathbb{Z}}\left(f(\boldsymbol{kh})\prod_{j=1}^n\frac{1-\cos[\pi(x-k_jh)/h]}{\pi(x-k_jh)/h}\right)$$
$$=\frac{1}{(2\pi)^n}\int_{\mathbb{R}^n}\hat{f}(\boldsymbol{t})\left[\prod_{j=1}^n -\mathrm{isgn}(t_j)e^{-ix_jt_j} - \prod_{j=1}^n\frac{1-\cos[\pi(x-k_jh)/h]}{\pi(x-k_jh)/h}e^{-ikht_j}\right]d\boldsymbol{t}$$
$$=\frac{1}{(2\pi)^n}\int_{\{\boldsymbol{t}:|t_j|>\pi/h_j\}}\hat{f}(\boldsymbol{t})\left[\prod_{j=1}^n -\mathrm{isgn}(t_j)e^{-ix_jt_j} - \prod_{j=1}^n\frac{1-\cos[\pi(x-k_jh)/h]}{\pi(x-k_jh)/h}e^{-ikht_j}\right]d\boldsymbol{t}.$$

The term in the square bracket is bounded by 2 and one can apply similar arguments to obtain the estimate for $E_H(f,\boldsymbol{h})(\boldsymbol{\xi})$. The error estimates for the case where $\hat{f}$ decays polynomially can be derived analogously. ◻

To prove Theorem 2.2, we need the following lemma.

Lemma D.1. *Assume* (2.3) *holds. For any* $\boldsymbol{\beta}\in\{0,1\}^n$, $\boldsymbol{\beta}\neq\boldsymbol{0}$ *and* $\boldsymbol{\alpha}\in\{-1,1\}^n$,

$$\left|\frac{\prod_{\{p:\beta_p=1\}}\sin(\pi z_p/h_p)}{(2\pi i)^{|\boldsymbol{\beta}|_1}}\int_{\mathbb{R}^{|\boldsymbol{\beta}|_1}}\frac{f(\boldsymbol{\xi_\beta}+i\boldsymbol{\alpha_\beta d_\beta^-}+\boldsymbol{z_{1-\beta}})d\boldsymbol{\xi_\beta}}{\prod_{\{j:\beta_j=1\}}(\xi_j+\alpha_jd_j^- - z_j)\sin(\pi\zeta_j/h_j)}\right|$$
$$\le C\prod_{\{j:\beta_j=1\}}\frac{e^{-\pi(d_j-|y_j|)/h_j}(1+e^{-2\pi|y_j|/h_j})}{(d_j-|y_j|)(1-e^{-2\pi d_j/h_j})}.$$

Proof of Lemma D.1. Applying the following two inequalities

$$\sinh(\pi|y|/h)\le|\sin(\pi(x\pm iy)/h)|\le\cosh(\pi|y|/h), \tag{D.1}$$
$$|t-z\pm id|\ge d-|y|\ (z=x+iy,\ |y|<d,\ t\in\mathbb{R}), \tag{D.2}$$

we obtain that

$$\left|\frac{\prod_{\{p:\beta_p=1\}}\sin(\pi z_p/h_p)}{(2\pi i)^{|\boldsymbol{\beta}|_1}}\int_{\mathbb{R}^{|\boldsymbol{\beta}|_1}}\frac{f(\boldsymbol{\xi_\beta}+i\boldsymbol{\alpha_\beta d_\beta^-}+\boldsymbol{z_{1-\beta}})d\boldsymbol{\xi_\beta}}{\prod_{\{j:\beta_j=1\}}(\xi_j+\alpha_jd_j^- - z_j)\sin(\pi\zeta_j/h_j)}\right|$$

$$\leq \prod_{\{j:\beta_j=1\}} \frac{1}{2\pi(d_j-|y_j|)} \cdot \frac{\cosh(\pi|y_j|/h_j)}{\sinh(\pi|d_j|/h_j)} \cdot \int_{\mathbb{R}^{|\boldsymbol{\beta}|_1}} \left| f(\boldsymbol{\xi}_{\boldsymbol{\beta}} + i\boldsymbol{\alpha}_{\boldsymbol{\beta}}\boldsymbol{d}_{\boldsymbol{\beta}}^- + \boldsymbol{z}_{\mathbf{1}-\boldsymbol{\beta}}) \right| d\boldsymbol{\xi}_{\boldsymbol{\beta}}.$$

Simplifying and using (2.3) give us the desired bound. $\square$

Theorem 2.2. (1) We prove the error bounds by induction. The case with $n=1$ is proved in Stenger (1993). Suppose our error bounds hold for all dimensions that is less than n. For $j=1,\cdots,n$, let $0<\delta_j<d_j$ and define

$$\mathcal{D}_j(m_j,\delta_j) = \left\{ z_j \in \mathbb{C} : |\mathrm{Re} z_j| < \left(m_j + \frac{1}{2}\right) h_j,\ |\mathrm{Im} z_j| < \delta_j \right\}.$$

Fix $z_j = x_j + iy_j$ in $\mathcal{D}_{d_j}$, and set $\zeta_j = \xi_j + i\eta_j$ $(j=1,\cdots,n)$. If m_j is sufficiently large and δ_j is sufficiently close to d_j, then $z_j \in \mathcal{D}_j(m_j,\delta_j)$. And let

$$E(\boldsymbol{m},\boldsymbol{\delta},f)(\boldsymbol{z}) = f(\boldsymbol{z}) - \sum_{k_1=-m_1}^{m_1} \cdots \sum_{k_n=-m_n}^{m_n} f(\boldsymbol{kh}) \prod_{j=1}^{n} S(k_j,h_j)(z_j) \tag{D.3}$$

We first show by induction that

$$\frac{\prod_{p=1}^n \sin(\pi z_p/h_p)}{(2\pi i)^n} \int_{\partial\mathcal{D}_1(m_1,\delta_1)} \cdots \int_{\partial\mathcal{D}_n(m_n,\delta_n)} \frac{f(\boldsymbol{\zeta})d\boldsymbol{\zeta}}{\prod_{j=1}^n (\zeta_j - z_j)\sin(\pi\zeta_j/h_j)}$$
$$= \sum_{\boldsymbol{\beta}\in\{0,1\}^n} (-1)^{|\boldsymbol{\beta}|_1} \sum_{-M_p\leq k_p\leq M_p}^{\boldsymbol{\beta}} \left[f(\boldsymbol{k}_{\boldsymbol{\beta}}\boldsymbol{h}_{\boldsymbol{\beta}} + \boldsymbol{z}_{1-\boldsymbol{\beta}}) \cdot \prod_{\{j:\beta_j=1\}} S(k_j,h_j)(z_j) \right], \tag{D.4}$$

For $n=1$, the result is given by Cauchy's residue theorem. Now suppose the equality holds for the $n-1$ case, then

$$\frac{\prod_{p=1}^n \sin(\pi z_p/h_p)}{(2\pi i)^n} \int_{\partial\mathcal{D}_1(m_1,\delta_1)} \cdots \int_{\partial\mathcal{D}_n(m_n,\delta_n)} \frac{f(\boldsymbol{\zeta})d\boldsymbol{\zeta}}{\prod_{j=1}^n (\zeta_j - z_j)\sin(\pi\zeta_j/h_j)}$$
$$= \frac{\prod_{p=1}^{n-1} \sin(\pi z_p/h_p)}{(2\pi i)^{n-1}} \int_{\partial\mathcal{D}_1(m_1,\delta_1)} \cdots \int_{\partial\mathcal{D}_{n-1}(m_{n-1},\delta_{n-1})} \frac{d\zeta_1\cdots d\zeta_{n-1}}{\prod_{j=1}^{n-1} (\zeta_j - z_j)\sin(\pi\zeta_j/h_j)}$$
$$\cdot \frac{\sin(\pi z_n/h_n)}{2\pi i} \int_{\partial\mathcal{D}_n(m_n,\delta_n)} \frac{f(\zeta_1,\cdots,\zeta_{n-1},\zeta_n)d\zeta_n}{(\zeta_n - z_n)\sin(\pi\zeta_n/h_n)}$$
$$= \frac{\prod_{j=1}^{n-1} \sin(\pi z_j/h_j)}{(2\pi i)^{n-1}} \int_{\partial\mathcal{D}_1(m_1,\delta_1)} \cdots \int_{\partial\mathcal{D}_{n-1}(m_{n-1},\delta_{n-1})} \frac{d\zeta_1\cdots d\zeta_{n-1}}{\prod_{j=1}^{n-1} (\zeta_j - z_j)\sin(\pi\zeta_j/h_j)}$$
$$\cdot \left[f(\zeta_1,\cdots,\zeta_{n-1},z_n) - \sum_{k_n=-m_n}^{m_n} \frac{f(\zeta_1,\cdots,\zeta_{n-1},k_n h_n)(-1)^{k_n}}{\pi(z_n - k_n h_n)/h_n} \sin(\pi z_n/h_n) \right]$$
$$= \sum_{\boldsymbol{\beta}'\in\{0,1\}^{n-1}} (-1)^{|\boldsymbol{\beta}'|_1} \sum_{-M_p\leq k_p'\leq M_p}^{\boldsymbol{\beta}'} \left[f(\boldsymbol{k}'_{\boldsymbol{\beta}'}\boldsymbol{h}'_{\boldsymbol{\beta}'} + \boldsymbol{z}'_{\mathbf{1}-\boldsymbol{\beta}'}, z_n) \cdot \prod_{\{j:\beta_j'=1\}} S(k_j,h_j)(z_j) \right]$$
$$- \sum_{k_n=-m_n}^{m_n} S(k_n,h_n)(z_n) \sum_{\boldsymbol{\beta}'\in\{0,1\}^{n-1}} (-1)^{|\boldsymbol{\beta}'|_1} \sum_{-M_p\leq k_p'\leq M_p}^{\boldsymbol{\beta}'} \left[f(\boldsymbol{k}'_{\boldsymbol{\beta}'}\boldsymbol{h}'_{\boldsymbol{\beta}'} + \boldsymbol{z}'_{\mathbf{1}-\boldsymbol{\beta}'}, k_n h_n) \prod_{\{j:\beta_j'=1\}} S(k_j,h_j)(z_j) \right]$$
$$= \sum_{\boldsymbol{\beta}\in\{0,1\}^n} (-1)^{|\boldsymbol{\beta}|_1} \left[\sum_{-M_p\leq k_p\leq M_p}^{\boldsymbol{\beta}} \left(f(\boldsymbol{k}_{\boldsymbol{\beta}}\boldsymbol{h}_{\boldsymbol{\beta}} + \boldsymbol{z}_{\mathbf{1}-\boldsymbol{\beta}}) \prod_{\{j:\beta_j=1\}} S(k_j,h_j)(z_j) \right) \right]$$

The third equality follows from induction, the fourth equality is obtained from rewriting $\boldsymbol{z} = (\boldsymbol{z}', z_n), \boldsymbol{k} = (\boldsymbol{k}', k_n), \boldsymbol{h} = (\boldsymbol{h}', h_n)$. Thus by induction, we obtain (D.4).

Noting that the number of positive and negative terms in $\{(-1)^{|\boldsymbol{\beta}|_1} : \boldsymbol{\beta} \in \{0,1\}^n\}$ are the same, we can rewrite the right side of (D.4) as

$$\sum_{\boldsymbol{\beta}\in\{0,1\}^n} (-1)^{|\boldsymbol{\beta}|_1+1} \left[f(\boldsymbol{z}) - \sum_{-M_p\le k_p\le M_p}^{\boldsymbol{\beta}} \left(f(\boldsymbol{k_\beta h_\beta} + \boldsymbol{z_{1-\beta}}) \prod_{\{j:\beta_j=1\}} S(k_j,h_j)(z_j) \right) \right] \tag{D.5}$$

Thus using (D.5), the term $E(\boldsymbol{m},\boldsymbol{\delta},f)(\boldsymbol{z})$ that is defined in (D.3) can be written as

$$E(\boldsymbol{m},\boldsymbol{\delta},f)(\boldsymbol{z}) = (-1)^{n+1} \frac{\prod_{p=1}^n \sin(\pi z_p/h_p)}{(2\pi i)^n} \int_{\partial\mathcal{D}_1(m_1,\delta_1)} \cdots \int_{\partial\mathcal{D}_n(m_n,\delta_n)} \frac{f(\boldsymbol{\zeta})d\boldsymbol{\zeta}}{\prod_{j=1}^n (\zeta_j - z_j)\sin(\pi\zeta_j/h_j)}$$

$$+ \sum_{\boldsymbol{\beta}\in\{0,1\}^n, \boldsymbol{\beta}\neq\boldsymbol{1}} (-1)^{|\boldsymbol{\beta}|_1+n+1} \left[f(\boldsymbol{z}) - \sum_{-M_p\le k_p\le M_p}^{\boldsymbol{\beta}} \left(f(\boldsymbol{k_\beta h_\beta} + \boldsymbol{z_{1-\beta}}) \cdot \prod_{\{j:\beta_j=1\}} S(k_j,h_j)(z_j) \right) \right] \tag{D.6}$$

We now derive the limit of the integral in (D.6) by taking m_j to ∞ for all j. The integral can be split into contributions from the product of horizontal segments only, those from the product of vertical segments only and those from the mixed product of horizontal and vertical segments. We observe that whenever there is a vertical segment involved in the integration region, the integral vanishes in the limit. We illustrate this result by considering an integral of this type below, where a vertical segment is used for the n-th dimension and horizontal segments are used in all other dimensions. Here $\zeta_j = \xi_j + id_j, 1 \le j \le n-1, \zeta_n := \left(m_n + \frac{1}{2}\right) h_n + i\eta_n$, and let $\tilde{\boldsymbol{\xi}} = (\xi_1, \cdots, \xi_{n-1})', \tilde{\boldsymbol{d}} = (d_1, \cdots, d_{n-1})'$.

$$\begin{aligned}
&\left| \int_{-\delta_n}^{\delta_n} \int_{-(m_1+\frac{1}{2})h_1}^{(m_1+\frac{1}{2})h_1} \cdots \int_{-(m_{n-1}+\frac{1}{2})h_{n-1}}^{(m_{n-1}+\frac{1}{2})h_{n-1}} \frac{f\left(\tilde{\boldsymbol{\xi}} + i\tilde{\boldsymbol{d}}^-, \left(m_n+\frac{1}{2}\right)h_n + i\eta_n\right) d\tilde{\boldsymbol{\xi}} d\eta_n}{\prod_{j=1}^{n-1}(\zeta_j - z_j)\sin(\pi\zeta_j/h_j)\cdot(\zeta_n - z_n)\sin(\pi\zeta_n/h_n)} \right| \\
&\le \int_{-\delta_n}^{\delta_n} \left| \int_{-(m_1+\frac{1}{2})h_1}^{(m_1+\frac{1}{2})h_1} \cdots \int_{-(m_{n-1}+\frac{1}{2})h_{n-1}}^{(m_{n-1}+\frac{1}{2})h_{n-1}} \frac{f(\tilde{\boldsymbol{\xi}} + i\tilde{\boldsymbol{d}}^-, \left(m_n+\frac{1}{2}\right)h_n + i\eta_n) d\tilde{\boldsymbol{\xi}}}{\prod_{j=1}^{n-1}(\xi_j + id_j - z_j)\sin(\pi(\xi_j+id_j)/h_j)} \right| d\eta_n \\
&\quad \cdot \frac{1}{\left|\left(m_n+\frac{1}{2}\right)h_n - x_n\right|} \\
&\to 0 \text{ as } m_j \to \infty \text{ for } j = 1, \cdots, n.
\end{aligned}$$

The inequaltiy is obtained as follows. Along $\{(m_n+\frac{1}{2})h_n + i\eta_n : -\delta_n \le \eta_n \le \delta_n\}$, we have

$$\begin{aligned}
|\zeta_n - z_n| &\ge \left| \left(m_n + \frac{1}{2}\right) h_n - x_n \right|, \\
|\sin(\pi\zeta_n/h_n)| &= |(-1)^n \cosh(\pi\eta_n/h_n)| \ge 1.
\end{aligned}$$

In addition, the $(n-1)$-dimensional integral can be bounded using Lemma D.1.

Putting these limiting results together, when $m_j \to \infty$, $\delta_j \to d_j$ for all j, the first part of the RHS of (D.6) becomes

$$\begin{aligned}
&\frac{\prod_{p=1}^n \sin(\pi z_p/h_p)}{(2\pi i)^n} \int_{\partial\mathcal{D}_1(m_1,\delta_1)} \cdots \int_{\partial\mathcal{D}_n(m_n,\delta_n)} \frac{f(\boldsymbol{\zeta})d\boldsymbol{\zeta}}{\prod_{j=1}^n(\zeta_j - z_j)\sin(\pi\zeta_j/h_j)} \\
&= \frac{\prod_{p=1}^n \sin(\pi z_p/h_p)}{(2\pi i)^n} \sum_{\boldsymbol{\alpha}\in\{-1,1\}^n} (-1)^{\sum_{\{j:\alpha_j=1\}} 1} \int_{\mathbb{R}^n} \frac{f(\boldsymbol{\xi} + i\boldsymbol{\alpha d}^-)d\boldsymbol{\xi}}{\prod_{j=1}^n(\xi_j + i\alpha_j d_j^- - z_j)\sin(\pi\zeta_j/h_j)}.
\end{aligned} \tag{D.7}$$

The RHS of (D.7) can be bounded as

$$C \prod_{j=1}^{n} \frac{e^{-\pi(d_j-|y_j|)/h_j}(1+e^{-2\pi|y_j|/h_j})}{(d_j-|y_j|)(1-e^{-2\pi d_j/h_j})} \tag{D.8}$$

using Lemma D.1. For the second part of the RHS of (D.6), applying the induction assumption together with (2.3) and (2.4), we obtain for each $\boldsymbol{\beta} \in \{0,1\}^n, \boldsymbol{\beta} \neq \mathbf{0}, \mathbf{1}$,

$$\begin{aligned} &\left| f(\boldsymbol{z}) - \sum_{-M_j \le k_j \le M_j}^{\boldsymbol{\beta}} \left(f(\boldsymbol{k}_{\boldsymbol{\beta}} \boldsymbol{h}_{\boldsymbol{\beta}} + \boldsymbol{z}_{\mathbf{1}-\boldsymbol{\beta}}) \cdot \prod_{\{j:\beta_j=1\}} S(k_j,h_j)(z_j) \right) \right| \\ \le &\sum_{\boldsymbol{\alpha}\in\{0,1\}^n, \boldsymbol{\alpha} \le \boldsymbol{\beta}, \boldsymbol{\alpha} \neq 0} \left[C_{\boldsymbol{\alpha}} \prod_{\{p:\alpha_p=1\}} \frac{e^{-\pi(d_p-|y_p|)/h_p}(1+e^{-2\pi|y_p|/h_p})}{(d_p-|y_p|)(1-e^{-2\pi d_p/h_p})} \right] \end{aligned} \tag{D.9}$$

Here $\boldsymbol{\alpha} \le \boldsymbol{\beta}$ means $\alpha_j \le \beta_j$ for all $1 \le j \le n$. Adding (D.8) and (D.9) over $\boldsymbol{\beta} \neq \mathbf{0}, \mathbf{1}$ yields (2.5).
(2) We have $E_I(f, \boldsymbol{h}) = \int_{\mathbb{R}^n} E(f, \boldsymbol{h})(\boldsymbol{x}) d\boldsymbol{x}$. Using (D.7) with $z_i = x_i$, interchanging the order of integration and applying the identities

$$\frac{1}{2\pi i} \int_{\mathbb{R}} \frac{\sin(\pi x/h)}{t-x-id} dx = \frac{i}{2} e^{-\pi(d+it)/h}, \quad \frac{1}{2\pi i} \int_{\mathbb{R}} \frac{\sin(\pi x/h)}{x-t-id} dx = \frac{i}{2} e^{-\pi(d-it)/h},$$

and (D.1), we obtain (2.6).
(3) We have $E_H(f, \boldsymbol{h})(\boldsymbol{\xi}) = \frac{1}{\pi^n} \int_{\mathbb{R}^n} \frac{E(f,\boldsymbol{h})(\boldsymbol{x})}{\prod_{i=1}^n (\xi_i - x_i)} d\boldsymbol{x}$. Thus (2.7) can be derived by using (D.7) with $z_i = x_i$, interchanging the order of integration and applying the identity

$$\frac{1}{\pi} \int_{\mathbb{R}} \frac{\sin(\pi x/h)}{(\omega - x)(z-x)} dx = \frac{e^{i\pi\omega \mathrm{sgn}(\mathrm{Im}(\omega))/h} - e^{i\pi z \mathrm{sgn}(\mathrm{Im}(z))/h}}{\omega - z}$$

together with (D.1) and (D.2). □

Proposition 4.2. It holds that

$$E_{\boldsymbol{x}}[f(\boldsymbol{X}_t)] = E_{\boldsymbol{x}}[e^{-\boldsymbol{\alpha}'\boldsymbol{X}_t} f_{\boldsymbol{\alpha}}(\boldsymbol{X}_t)] = e^{-\boldsymbol{\alpha}'\boldsymbol{x} - t\psi(i\boldsymbol{\alpha})} E_{\boldsymbol{x}}^{(\boldsymbol{\alpha})}[f_{\boldsymbol{\alpha}}(\boldsymbol{X}_t)].$$

Take the Fourier transform of $E_{\boldsymbol{x}}^{(\boldsymbol{\alpha})}[f_{\boldsymbol{\alpha}}(\boldsymbol{X}_t)]$ as a function of $\boldsymbol{x}$ and use Proposition 9 in Bertoin (1998), we obtain

$$\mathcal{F}_n \left(E_{\boldsymbol{x}}^{(\boldsymbol{\alpha})}[f_{\boldsymbol{\alpha}}(\boldsymbol{X}_t)] \right)(\boldsymbol{\xi}) = \phi_t^{\boldsymbol{\alpha}}(-\boldsymbol{\xi}) \hat{f}_{\boldsymbol{\alpha}}(\boldsymbol{\xi}).$$

Due to the integrability condition (4.5), we can invert the Fourier transform and obtain (4.7). □

Proposition 4.3. Given $\boldsymbol{\beta} \in \{0,1\}^n$, let $k_{\boldsymbol{\beta}}(\boldsymbol{\xi}) = \prod_{\{j:\beta_j=1\}} 1/(\pi\xi_j)$. The partial Hilbert transform $\mathcal{H}_{\boldsymbol{\beta}}$ of $\hat{f}$ is the convolution of $\hat{f}$ and $k_{\boldsymbol{\beta}}$, i.e. $\mathcal{H}_{\boldsymbol{\beta}} \hat{f}(\boldsymbol{\xi}) = (\hat{f} * k_{\boldsymbol{\beta}})(\boldsymbol{\xi})$ (see Eq.(15.39) in King (2009)). The convolution theorem shows that

$$\mathcal{F}_{\boldsymbol{\beta}}^{-1}\left(\mathcal{H}_{\boldsymbol{\beta}} \hat{f}(\boldsymbol{\xi})\right)(\boldsymbol{x}_{\boldsymbol{\beta}} + \boldsymbol{\xi}_{\mathbf{1}-\boldsymbol{\beta}}) = \mathcal{F}_{\boldsymbol{\beta}}^{-1}(\hat{f}(\boldsymbol{\xi}))(\boldsymbol{x}_{\boldsymbol{\beta}} + \boldsymbol{\xi}_{\mathbf{1}-\boldsymbol{\beta}}) \cdot (2\pi)^{|\boldsymbol{\beta}|_1} \mathcal{F}_{\boldsymbol{\beta}}^{-1}(k_{\boldsymbol{\beta}}(\boldsymbol{\xi}))(\boldsymbol{x}_{\boldsymbol{\beta}} + \boldsymbol{\xi}_{\mathbf{1}-\boldsymbol{\beta}}), \tag{D.10}$$

where

$$\mathcal{F}_{\boldsymbol{\beta}}^{-1}(k_{\boldsymbol{\beta}}(\boldsymbol{\xi}))(\boldsymbol{x}_{\boldsymbol{\beta}} + \boldsymbol{\xi}_{\mathbf{1}-\boldsymbol{\beta}}) = \left(\frac{1}{2\pi}\right)^{|\boldsymbol{\beta}|_1} \int_{\mathbb{R}^{|\boldsymbol{\beta}|_1}} \prod_{\{j:\beta_j=1\}} \frac{e^{-ix_j\xi_j} d\xi_j}{\pi\xi_j} = \left(\frac{-i}{2\pi}\right)^{|\boldsymbol{\beta}|_1} \prod_{\{j:\beta_j=1\}} \mathrm{sgn}(x_j).$$

The function $k_{\boldsymbol{\beta}}(\boldsymbol{\xi})$ is not in $L^1(\mathbb{R}^n)$, so $\mathcal{F}_{\boldsymbol{\beta}}^{-1}(k_{\boldsymbol{\beta}}(\boldsymbol{\xi}))(\boldsymbol{x}_{\boldsymbol{\beta}}+\boldsymbol{\xi}_{\mathbf{1}-\boldsymbol{\beta}})$ must be interpreted as a Cauchy principal integral. We further apply Fourier inversion to those dimensions p with $\beta_p = 0$ on both sides of (D.10) and obtain that

$$\mathcal{F}_n^{-1}\left(\mathcal{H}_{\boldsymbol{\beta}}\hat{f}(\boldsymbol{\xi})\right)(\boldsymbol{x}) = i^{|\boldsymbol{\beta}|_1} f(\boldsymbol{x}) \prod_{\{j:\beta_j=1\}} \operatorname{sgn}(x_j).$$

Here, we use the identity that $\mathcal{F}_{\mathbf{1}-\boldsymbol{\beta}}^{-1}(\mathcal{F}_{\boldsymbol{\beta}}^{-1}(\hat{f}(\boldsymbol{\xi}))(\boldsymbol{x}_{\boldsymbol{\beta}}+\boldsymbol{\xi}_{\mathbf{1}-\boldsymbol{\beta}}))(\boldsymbol{x}) = \mathcal{F}_n^{-1}\hat{f}(\boldsymbol{x}) = f(\boldsymbol{x})$ as $\hat{f} \in L^1(\mathbb{R}^n,\mathbb{C})$. Further applying $\mathcal{F}_n$ on both sides of the above equation gives us (4.9). $\square$

Proposition 4.4. Consider the translation operators

$$\left(\mathcal{T}_{l_j}^{(j)} f\right)(\boldsymbol{x}) := f(x_1,\cdots,x_{j-1},x_j-l_j,x_{j+1},\cdots,x_n)$$

We can write

$$\mathbb{1}_{(\boldsymbol{l},\infty)}(\boldsymbol{x})f(\boldsymbol{x}) = \frac{1}{2^n}\prod_{j=1}^n\left(1+\mathcal{T}_{l_j}^{(j)}\operatorname{sgn}(x_j)\right)f(\boldsymbol{x}) = \frac{1}{2^n}\sum_{\boldsymbol{\beta}\in\{0,1\}^n} f(\boldsymbol{x})\cdot\mathcal{T}_{\boldsymbol{l}}^{\boldsymbol{\beta}}\left(\prod_{\{p:\beta_p=1\}}\operatorname{sgn}(x_p)\right)$$

where $\mathcal{T}_{\boldsymbol{l}}^{\boldsymbol{\beta}} = \prod_{\{p:\beta_p=1\}}\mathcal{T}_{l_p}^{(p)}$. Furthermore, we have

$$f(\boldsymbol{x})\cdot\mathcal{T}_{\boldsymbol{l}}^{\boldsymbol{\beta}}\left(\prod_{\{p:\beta_p=1\}}\operatorname{sgn}(x_p)\right) = \mathcal{T}_{\boldsymbol{l}}^{\boldsymbol{\beta}}\left(\prod_{\{p:\beta_p=1\}}\operatorname{sgn}(x_p)\cdot\mathcal{T}_{-\boldsymbol{l}}^{\boldsymbol{\beta}}f(\boldsymbol{x})\right)$$

Then, by (4.9) and the property of Fourier transform w.r.t. translation, we obtain

$$\mathcal{F}_n\left(\mathcal{T}_{\boldsymbol{l}}^{\boldsymbol{\beta}}\left(\prod_{\{p:\beta_p=1\}}\operatorname{sgn}(x_p)\cdot\mathcal{T}_{-\boldsymbol{l}}^{\boldsymbol{\beta}}f(\boldsymbol{x})\right)\right)(\boldsymbol{\xi}) = i^{|\boldsymbol{\beta}|}e^{i\boldsymbol{\xi}_{\boldsymbol{\beta}}'\boldsymbol{l}_{\boldsymbol{\beta}}}\mathcal{H}_{\boldsymbol{\beta}}\left(e^{-i\boldsymbol{\eta}_{\boldsymbol{\beta}}'\boldsymbol{l}_{\boldsymbol{\beta}}}\hat{f}\left(\boldsymbol{\eta}_{\boldsymbol{\beta}}+\boldsymbol{\xi}_{\mathbf{1}-\boldsymbol{\beta}}\right)\right)(\boldsymbol{\xi}).$$

Applying the $\mathcal{F}_n$ to $\mathbb{1}_{(\boldsymbol{l},\infty)}(\boldsymbol{x})f(\boldsymbol{x})$ and using the above results gives us (4.10). $\square$

Theorem 4.1. First, notice that if (4.17) holds, then for any $t \geq \chi$,

$$\int_{\mathbb{R}^n}\left|\phi_t^{\boldsymbol{\alpha}}(\boldsymbol{\xi})\right|d\boldsymbol{\xi} = \int_{\mathbb{R}^n}\left|\phi_\chi^{\boldsymbol{\alpha}}(\boldsymbol{\xi})\phi_{t-\chi}^{\boldsymbol{\alpha}}(\boldsymbol{\xi})\right|d\boldsymbol{\xi} \leq \int_{\mathbb{R}^n}\left|\phi_\chi^{\boldsymbol{\alpha}}(\boldsymbol{\xi})\right|d\boldsymbol{\xi} < \infty.$$

We prove that (4.18) holds for every j. For $j = N$, it is given by the assumption (4.16). Now assume that the claim (4.18) holds for j to N and we prove below that it also holds for $j-1$. Using the recursion (4.14), we have

$$\int_{\mathbb{R}^n}\left|\phi_\Delta^{\boldsymbol{\alpha}}(-\boldsymbol{\xi})\hat{v}_{\boldsymbol{\alpha}}^{j-1}(\boldsymbol{\xi})\right|d\boldsymbol{\xi} \leq \frac{e^{-\Delta\psi(i\boldsymbol{\alpha})}}{2^n}\sum_{\boldsymbol{\beta}\in\{0,1\}^n} I_{\boldsymbol{\beta}},$$

where

$$I_{\boldsymbol{\beta}} := \int_{\mathbb{R}^n}\left|\phi_\Delta^{\boldsymbol{\alpha}}(-\boldsymbol{\xi})\mathcal{H}_{\boldsymbol{\beta}}\left(e^{-i\boldsymbol{\eta}_{\boldsymbol{\beta}}'\boldsymbol{l}_{\boldsymbol{\beta}}}\phi_\Delta^{\boldsymbol{\alpha}}\big(-\boldsymbol{\eta}_{\boldsymbol{\beta}}-\boldsymbol{\xi}_{\mathbf{1}-\boldsymbol{\beta}}\big)\hat{v}_{\boldsymbol{\alpha}}^j\big(\boldsymbol{\eta}_{\boldsymbol{\beta}}+\boldsymbol{\xi}_{\mathbf{1}-\boldsymbol{\beta}}\big)\right)(\boldsymbol{\xi})\right|d\boldsymbol{\xi}$$

As $|\phi_\Delta^{\boldsymbol{\alpha}}(-\boldsymbol{\xi})|$ is bounded over $\boldsymbol{\xi}$, $I_{\mathbf{0}}$ is finite by the induction assumption. Now pick $p > 1$ such that $p\Delta \geq \chi$ and set $q = p/(p-1)$. Note that $(\phi_\Delta^{\boldsymbol{\alpha}}(-\boldsymbol{\xi}))^p = \phi_{p\Delta}^{\boldsymbol{\alpha}}(-\boldsymbol{\xi}) \in L^1(\mathbb{R}^n)$, and this implies $\phi_\Delta^{\boldsymbol{\alpha}}(-\boldsymbol{\xi}) \in L^p(\mathbb{R}^n)$. The Calderón-Zygmund inequality (see Eq.(15.115) in King (2009)) says that, if $f \in L^p(\mathbb{R}^n)$ with $p > 1$, then

$$\|\mathcal{H}_n f\|_p = \left\|\prod_{k=1}^n\mathcal{H}_{(k)}f\right\|_p = \left\|\mathcal{H}_{(1)}\prod_{k=2}^n\mathcal{H}_{(k)}f\right\|_p$$

$$\leq C_p^{(1)} \left\| \prod_{k=2}^{n} \mathcal{H}_{(k)} f \right\|_p \leq \cdots \leq C_p^{(n)} \|f\|_p$$

for some constants $C_p^{(k)}, 1 \leq k \leq n$. Here, we write $\mathcal{H}_{(k)}$ as the partial Hilbert transform with $\boldsymbol{\beta} = (0, \cdots, 0, 1, 0, \cdots, 0)$, where only $\beta_k = 1$. Thus for $I_{\mathbf{1}}$, by Hölder's inequality,

$$I_{\mathbf{1}} \leq \|\phi_\Delta^{\boldsymbol{\alpha}}(-\cdot)\|_p \cdot \left\| \mathcal{H}_n \left(e^{-i\boldsymbol{\eta}'\boldsymbol{l}} \phi_\Delta^{\boldsymbol{\alpha}}(-\boldsymbol{\eta}) \hat{v}_{\boldsymbol{\alpha}}^j(\boldsymbol{\eta}) \right)(\cdot) \right\|_q \leq C \|\phi_\Delta^{\boldsymbol{\alpha}}(-\cdot)\|_p \cdot \|\phi_\Delta^{\boldsymbol{\alpha}}(-\cdot)\hat{v}_{\boldsymbol{\alpha}}^j(\cdot)\|_q.$$

Using the boundedness of $\hat{v}_{\boldsymbol{\alpha}}^j$, we have

$$\|\phi_\Delta^{\boldsymbol{\alpha}}(-\cdot)\hat{v}_{\boldsymbol{\alpha}}^j(\cdot)\|_q \leq \|\hat{v}_{\boldsymbol{\alpha}}^j(\cdot)\|_\infty^{q-1} \|\phi_\Delta^{\boldsymbol{\alpha}}(-\cdot)\hat{v}_{\boldsymbol{\alpha}}^j(\cdot)\|_1$$

which is finite by the induction assumption. This shows $I_{\mathbf{1}}$ is finite. The finiteness of $\{I_{\boldsymbol{\beta}} : \boldsymbol{\beta} \neq \mathbf{0}, \mathbf{1}\}$ can be proved similarly. Finally, (4.19) follows from Proposition 4.2. □

Theorem 4.2. (1) For $\mathcal{P}^\Delta g(\boldsymbol{\xi})$, the total approximation error consists of the error for the terms involving several partial Hilbert transforms and the term with the n-dimensional Hilbert transform. We can directly apply Theorem 2.2 to estimate the approximation error for the partial Hilbert transforms under (4.23) and the boundedness of g. Below we only analyze the error for the n-dimensional Hilbert transform. We have

$$\left|\mathcal{P}^\Delta g(\boldsymbol{\xi}) - \mathcal{P}_{h,M}^\Delta g(\boldsymbol{\xi})\right| \leq \left|\mathcal{P}^\Delta g(\boldsymbol{\xi}) - \mathcal{P}_{h,\infty}^\Delta g(\boldsymbol{\xi})\right| + \left|\mathcal{P}_{h,\infty}^\Delta g(\boldsymbol{\xi}) - \mathcal{P}_{h,M}^\Delta g(\boldsymbol{\xi})\right|.$$

The first term can be estimated by applying (2.7) as our assumption on characteristic function (4.23) and the boundedness of g allows us to verify the conditions in Theorem 2.2. For the second term, it is bounded by

$$\begin{aligned}
&\|g\|_{L^\infty(\mathbb{R}^n)} \sum_{|m_j|>M_j, 1\leq j\leq n} |\phi_\Delta(-\boldsymbol{m}\boldsymbol{h})| \\
\leq& a \cdot 2^n \|g\|_{L^\infty(\mathbb{R}^n)} \left(\prod_{j=1}^n h_j^{-1}\right) \sum_{m_j>M_j, 1\leq j\leq n} \left(e^{-\Delta b \left(\sum_{j=1}^n m_j^2 h_j^2\right)^{\nu/2}} \cdot \prod_{j=1}^n h_j \right) \\
\leq& a \cdot 2^n \|g\|_{L^\infty(\mathbb{R}^n)} \left(\prod_{j=1}^n h_j^{-1}\right) \sum_{m_j>M_j, 1\leq j\leq n} e^{-\Delta b n^{\nu/2-1} \sum_{j=1}^n (m_j h_j)^\nu} \cdot \prod_{j=1}^n h_j \\
\leq& a \cdot 2^n \|g\|_{L^\infty(\mathbb{R}^n)} \left(\prod_{j=1}^n h_j^{-1}\right) \cdot \prod_{j=1}^n \int_{M_j h_j}^\infty e^{-\Delta b n^{\nu/2-1} x_j^\nu} dx_j \\
\leq& a \cdot 2^n \|g\|_{L^\infty(\mathbb{R}^n)} \prod_{j=1}^n \Gamma\left(1/\nu, \Delta b n^{\nu/2-1} (M_j h_j)^\nu\right) / h_j,
\end{aligned} \tag{D.11}$$

where we used $|(1-\cos(x))/x| \leq 1$, (4.22), $\left(\sum_{j=1}^n m_j^2 h_j^2\right)^{\nu/2} \geq n^{\nu/2-1} \sum_{j=1}^n (m_j h_j)^\nu$ (due to the concavity of $x^{\nu/2}$ as $\nu \in (0, 2]$). Combining Eq.(6.20) in Feng and Linetsky (2008), (2.7) and (D.11), we arrive at (4.25). Now we set h_j according to (4.26) and h_j is bounded as $M_j \geq 1$. Therefore, for some constant $C > 0$,

$$\frac{e^{-\pi d_j / h_j}}{1 - e^{-\pi d_j / h_j}} \leq C e^{-\pi d_j / h_j}. \tag{D.12}$$

where the constant C is independent of j. As $\Gamma(a, x) \sim x^{a-1} e^{-x}$ for x large, we can bound $\Gamma(a, x)$ as a constant times $x^{a-1} e^{-x}$ for all x bounded away from 0. Using this estimate, (4.26), (D.12) and that the term $M_j^{\frac{2}{1+\nu}} \exp\left(-2cM_j^{\frac{\nu}{1+\nu}}\right) \le AM_j^{\frac{1}{1+\nu}} \exp\left(-cM_j^{\frac{\nu}{1+\nu}}\right)$ for some constant $A > 0$ when $c > 0$, we obtain (4.27).

(2) The proof for $\mathcal{R}^\Delta g(\boldsymbol{x})$ is similar to the proof for $\mathcal{P}^\Delta g(\boldsymbol{\xi})$, so the detail is omitted. □

Theorem 4.3. (1) From the proof of Theorem 4.1, we have that for every $q > 1$,

$$\|\mathcal{H}_{\boldsymbol{\beta}} f\|_{L^q} \le C_q^{\boldsymbol{\beta}} \|f\|_{L^q},$$

for some constant $C_q^{\boldsymbol{\beta}} > 0$. Now we use this result to bound $\|\mathcal{P}^\Delta g\|_{L^q}$ for $g \in L^q$. Using the expression of $\mathcal{P}^\Delta g$ yields

$$\begin{aligned}
\left\|\mathcal{P}^\Delta g(\boldsymbol{\xi})\right\|_{L^q} &= \left\| \sum_{\boldsymbol{\beta}\in\{0,1\}^n} i^{|\boldsymbol{\beta}|_1} e^{i\boldsymbol{\xi}_{\boldsymbol{\beta}}' \boldsymbol{l}_{\boldsymbol{\beta}}} \mathcal{H}_{\boldsymbol{\beta}} \left(e^{-i\boldsymbol{\eta}_{\boldsymbol{\beta}}' \boldsymbol{l}_{\boldsymbol{\beta}}} \phi_\Delta^{\boldsymbol{\alpha}} \big(-\boldsymbol{\eta}_{\boldsymbol{\beta}} - \boldsymbol{\xi}_{\boldsymbol{1}-\boldsymbol{\beta}} \big) g\big(\boldsymbol{\eta}_{\boldsymbol{\beta}} + \boldsymbol{\xi}_{\boldsymbol{1}-\boldsymbol{\beta}}\big) \right)(\boldsymbol{\xi}) \right\|_{L^q} \\
&\le \sum_{\boldsymbol{\beta}\in\{0,1\}^n} \left\| \mathcal{H}_{\boldsymbol{\beta}} \left(e^{-i\boldsymbol{\eta}_{\boldsymbol{\beta}}' \boldsymbol{l}_{\boldsymbol{\beta}}} \phi_\Delta^{\boldsymbol{\alpha}} \big(-\boldsymbol{\eta}_{\boldsymbol{\beta}} - \boldsymbol{\xi}_{\boldsymbol{1}-\boldsymbol{\beta}} \big) g\big(\boldsymbol{\eta}_{\boldsymbol{\beta}} + \boldsymbol{\xi}_{\boldsymbol{1}-\boldsymbol{\beta}}\big) \right)(\boldsymbol{\xi}) \right\|_{L^q} \\
&\le \sum_{\boldsymbol{\beta}\in\{0,1\}^n} C_q^{\boldsymbol{\beta}} \left\| e^{-i\boldsymbol{\eta}_{\boldsymbol{\beta}}' \boldsymbol{l}_{\boldsymbol{\beta}}} \phi_\Delta^{\boldsymbol{\alpha}}(\boldsymbol{\xi}) g(\boldsymbol{\xi}) \right\|_{L^q} \\
&\le \sum_{\boldsymbol{\beta}\in\{0,1\}^n} C_q^{\boldsymbol{\beta}} \left\| \phi_\Delta^{\boldsymbol{\alpha}}(\boldsymbol{\xi}) g(\boldsymbol{\xi}) \right\|_{L^q} \\
&\le C_q \left\| \phi_\Delta^{\boldsymbol{\alpha}}(\boldsymbol{\xi}) g(\boldsymbol{\xi}) \right\|_{L^q}
\end{aligned}$$

where we set $C_q = 2^n \max_{\boldsymbol{\beta}\in\{0,1\}^n} C_q^{\boldsymbol{\beta}}$.

(2) Set

$$q_1 = 1, \quad \frac{1}{p_k} + \frac{1}{q_k} = \frac{1}{q_{k-1}}, \ k = 2, \cdots, N.$$

Let

$$C_{norm} = \prod_{i=1}^n 2M_i h_i, \ C_3 = C_1 \cdot 2^n C_{q_2},$$

$$C_k = \frac{C_3 C_{norm}}{2^{n(k-2)}} \prod_{i=2}^k \|\phi_\Delta^{\boldsymbol{\alpha}}(-\cdot)\|_{L^{p_i}}, k \ge 4.$$

For $i = 1, \cdots, N-1$, we have

$$\begin{aligned}
\left|\tilde{v}_{\boldsymbol{\alpha}}^i(\boldsymbol{\xi}) - \hat{v}_{\boldsymbol{\alpha}}^i(\boldsymbol{\xi})\right| &= \frac{e^{-\Delta\psi(i\boldsymbol{\alpha})}}{2^n} \left| \mathcal{P}_{\boldsymbol{h},\boldsymbol{M}}^\Delta \tilde{v}_{\boldsymbol{\alpha}}^{i+1}(\boldsymbol{\xi}) - \mathcal{P}^\Delta \hat{v}_{\boldsymbol{\alpha}}^{i+1}(\boldsymbol{\xi}) \right| \\
\le \frac{e^{-\Delta\psi(i\boldsymbol{\alpha})}}{2^n} &\left(\left| \mathcal{P}_{\boldsymbol{h},\boldsymbol{M}}^\Delta \tilde{v}_{\boldsymbol{\alpha}}^{i+1}(\boldsymbol{\xi}) - \mathcal{P}^\Delta \tilde{v}_{\boldsymbol{\alpha}}^{i+1}(\boldsymbol{\xi}) \right| + \left| \mathcal{P}^\Delta \left(\tilde{v}_{\boldsymbol{\alpha}}^{i+1} - \hat{v}_{\boldsymbol{\alpha}}^{i+1} \right)(\boldsymbol{\xi}) \right| \right)
\end{aligned}$$

Now we estimate I_1 and I_2.

$$\begin{aligned}
I_1 &= C_1 \int_{\mathbb{R}^n} |\phi_\Delta^{\boldsymbol{\alpha}}(-\boldsymbol{\xi})| \cdot \left| \mathcal{P}_{\boldsymbol{h},\boldsymbol{M}}^\Delta \tilde{v}_{\boldsymbol{\alpha}}^2(\boldsymbol{\xi}) - \mathcal{P}^\Delta \tilde{v}_{\boldsymbol{\alpha}}^2(\boldsymbol{\xi}) \right| d\boldsymbol{\xi} \\
&\le C_1 \int_{\mathbb{R}^n} |\phi_\Delta^{\boldsymbol{\alpha}}(-\boldsymbol{\xi})|\, d\boldsymbol{\xi} \cdot \|\mathcal{P}_{\boldsymbol{h},\boldsymbol{M}}^\Delta \tilde{v}_{\boldsymbol{\alpha}}^2 - \mathcal{P}^\Delta \tilde{v}_{\boldsymbol{\alpha}}^2\|_{L^\infty} \\
&\le C_2 \|\mathcal{P}_{\boldsymbol{h},\boldsymbol{M}}^\Delta \tilde{v}_{\boldsymbol{\alpha}}^2 - \mathcal{P}^\Delta \tilde{v}_{\boldsymbol{\alpha}}^2\|_{L^\infty}
\end{aligned}$$

where $C_2 = C_1 \|\phi_\Delta^{\boldsymbol{\alpha}}\|_{L^1}$ and

$$I_2 = C_1 \int_{\mathbb{R}^n} |\phi_\Delta^{\boldsymbol{\alpha}}(-\boldsymbol{\xi})| \cdot \left| \mathcal{P}^\Delta \left(\tilde{v}_{\boldsymbol{\alpha}}^2 - \hat{v}_{\boldsymbol{\alpha}}^2 \right)(\boldsymbol{\xi}) \right| d\boldsymbol{\xi}$$

$$\leq C_1 \|\phi_\Delta^{\boldsymbol{\alpha}}(-\cdot)\|_{L^{p_2}} \cdot \|\mathcal{P}^\Delta \left(\tilde{v}_{\boldsymbol{\alpha}}^2 - \hat{v}_{\boldsymbol{\alpha}}^2\right)\|_{L^{q_2}}$$
$$\leq C_3 \|\phi_\Delta^{\boldsymbol{\alpha}}(-\cdot)\|_{L^{p_2}} \cdot \|\phi_\Delta^{\boldsymbol{\alpha}}(-\cdot)\left(\tilde{v}_{\boldsymbol{\alpha}}^2 - \hat{v}_{\boldsymbol{\alpha}}^2\right)(\cdot)\|_{L^{q_2}}$$

where $C_3 = C_1 \cdot 2^n C_{q_2}$. Applying the Holder inquality and the Minkowski inequality, we obtain

$$\begin{aligned}
&\|\phi_\Delta^{\boldsymbol{\alpha}}(-\cdot)\left(\tilde{v}_{\boldsymbol{\alpha}}^2 - \hat{v}_{\boldsymbol{\alpha}}^2\right)(\cdot)\|_{L^{q_2}} \\
\leq &\|\phi_\Delta^{\boldsymbol{\alpha}}(-\cdot)\|_{L^{p_3}} \cdot \|\tilde{v}_{\boldsymbol{\alpha}}^2 - \hat{v}_{\boldsymbol{\alpha}}^2\|_{L^{q_3}} \\
\leq &\frac{e^{-\Delta\psi(i\boldsymbol{\alpha})}}{2^n} \|\phi_\Delta^{\boldsymbol{\alpha}}(-\cdot)\|_{L^{p_3}} \cdot \|\mathcal{P}_{\boldsymbol{h},\boldsymbol{M}}^\Delta \tilde{v}_{\boldsymbol{\alpha}}^3 - \mathcal{P}^\Delta \hat{v}_{\boldsymbol{\alpha}}^3\|_{L^{q_3}} \\
\leq &\frac{e^{-\Delta\psi(i\boldsymbol{\alpha})}}{2^n} \|\phi_\Delta^{\boldsymbol{\alpha}}(-\cdot)\|_{L^{p_3}} \left(\|\mathcal{P}_{\boldsymbol{h},\boldsymbol{M}}^\Delta \tilde{v}_{\boldsymbol{\alpha}}^3 - \mathcal{P}^\Delta \tilde{v}_{\boldsymbol{\alpha}}^3\|_{L^{q_3}} + \|\mathcal{P}^\Delta \left(\tilde{v}_{\boldsymbol{\alpha}}^3 - \hat{v}_{\boldsymbol{\alpha}}^3\right)(\boldsymbol{\xi})\|_{L^{q_3}}\right) \\
\leq &\frac{\|\phi_\Delta^{\boldsymbol{\alpha}}(-\cdot)\|_{L^{p_3}}}{2^n} \|\mathcal{P}_{\boldsymbol{h},\boldsymbol{M}}^\Delta \tilde{v}_{\boldsymbol{\alpha}}^3 - \mathcal{P}^\Delta \tilde{v}_{\boldsymbol{\alpha}}^3\|_{L^{q_3}} + C_{q_3} \|\phi_\Delta^{\boldsymbol{\alpha}}(-\cdot)\|_{L^{p_3}} \|\phi_\Delta^{\boldsymbol{\alpha}}(-\cdot)\left(\tilde{v}_{\boldsymbol{\alpha}}^3 - \hat{v}_{\boldsymbol{\alpha}}^3\right)(\cdot)\|_{L^{q_3}} \\
\leq &\cdots \leq \sum_{k=3}^N \frac{1}{2^{n(k-2)}} \prod_{i=3}^k \|\phi_\Delta^{\boldsymbol{\alpha}}(-\cdot)\|_{L^{p_i}} \|\mathcal{P}_{\boldsymbol{h},\boldsymbol{M}}^\Delta \tilde{v}_{\boldsymbol{\alpha}}^k - \mathcal{P}^\Delta \tilde{v}_{\boldsymbol{\alpha}}^k\|_{L^{q_k}} \\
+ &\prod_{k=3}^N C_{q_k} \|\phi_\Delta^{\boldsymbol{\alpha}}(-\cdot)\|_{L^{p_k}} \|\phi_\Delta^{\boldsymbol{\alpha}}(-\cdot)\left(\tilde{v}_{\boldsymbol{\alpha}}^N - \hat{v}_{\boldsymbol{\alpha}}^N\right)(\cdot)\|_{L^{q_N}}.
\end{aligned} \tag{D.13}$$

We point out the inequality

$$\|f\|_{L^p} \leq C_{norm} \|f\|_{L^\infty}, \tag{D.14}$$

which holds for $p > 1$ and f supported in the bounded hyperrectangle. We have

$$\begin{aligned}
&\|\phi_\Delta^{\boldsymbol{\alpha}}(-\cdot)\left(\tilde{v}_{\boldsymbol{\alpha}}^N - \hat{v}_{\boldsymbol{\alpha}}^N\right)(\cdot)\|_{L^{q_N}} \\
&\leq \|\phi_\Delta^{\boldsymbol{\alpha}}(-\cdot)\left(\tilde{v}_{\boldsymbol{\alpha}}^N - \hat{v}_{\boldsymbol{\alpha}}^N\right)(\cdot) 1_{\Omega_{\boldsymbol{h},\boldsymbol{M}}}(\cdot)\|_{L^{q_N}} + \|\phi_\Delta^{\boldsymbol{\alpha}}(-\cdot)\left(\tilde{v}_{\boldsymbol{\alpha}}^N - \hat{v}_{\boldsymbol{\alpha}}^N\right)(\cdot) 1_{\Omega_{\boldsymbol{h},\boldsymbol{M}}^c}(\cdot)\|_{L^{q_N}} \\
&\leq C_{norm} \|\phi_\Delta^{\boldsymbol{\alpha}}(-\cdot)\left(\tilde{v}_{\boldsymbol{\alpha}}^N - \hat{v}_{\boldsymbol{\alpha}}^N\right)(\cdot) 1_{\Omega_{\boldsymbol{h},\boldsymbol{M}}}(\cdot)\|_{L^\infty} + \|\hat{g}_{\boldsymbol{\alpha}}\|_{L^\infty} \|\phi_\Delta^{\boldsymbol{\alpha}}(-\cdot) 1_{\Omega_{\boldsymbol{h},\boldsymbol{M}}^c}(\cdot)\|_{L^{q_N}} \\
&\leq C_{norm} \|\phi_\Delta^{\boldsymbol{\alpha}}(-\cdot)\left(\tilde{v}_{\boldsymbol{\alpha}}^N - \hat{v}_{\boldsymbol{\alpha}}^N\right)(\cdot) 1_{\Omega_{\boldsymbol{h},\boldsymbol{M}}}(\cdot)\|_{L^\infty} + \|\hat{g}_{\boldsymbol{\alpha}}\|_{L^\infty} I_{\boldsymbol{h},\boldsymbol{M}}.
\end{aligned} \tag{D.15}$$

Let

$$C'_N = C_3 \max\{C_{norm}, \|\hat{g}_{\boldsymbol{\alpha}}\|_{L^\infty}\} \|\phi_\Delta^{\boldsymbol{\alpha}}(-\cdot)\|_{L^{p_2}} \prod_{k=3}^N C_{q_k} \|\phi_\Delta^{\boldsymbol{\alpha}}(-\cdot)\|_{L^{p_k}}.$$

Using (D.14) and (D.15), we can bound the RHS of (D.13) by the RHS of (4.32).

Next we provide estimates for the constants. First, we have

$$\begin{aligned}
\|\phi_\Delta^{\boldsymbol{\alpha}}(-\cdot)\|_{L^p} &= \left(\int_{\mathbb{R}^n} |\phi_\Delta^{\boldsymbol{\alpha}}(-\boldsymbol{\xi})|^p d\boldsymbol{\xi}\right)^{\frac{1}{p}} \leq \left(\int_{\mathbb{R}^n} a^p e^{-p\Delta b|\boldsymbol{\xi}|^\nu} d\boldsymbol{\xi}\right)^{\frac{1}{p}} \\
&\leq (p\Delta b)^{-\frac{n}{\nu p}} \left(\int_{\mathbb{R}^n} a^p e^{-|\boldsymbol{\eta}|^\nu} d\boldsymbol{\eta}\right)^{\frac{1}{p}}, \boldsymbol{\eta} = (p\Delta b)^{\frac{1}{\nu}} \boldsymbol{\xi} \\
&= A^{\frac{1}{p}} p^{-\frac{n}{\nu p}}, \quad A = \frac{\int_{\mathbb{R}^n} a^p e^{-|\boldsymbol{\eta}|^\nu} d\boldsymbol{\eta}}{(\Delta b)^{\frac{n}{\nu}}}
\end{aligned}$$

and the constant C_p is given by

$$C_p^{\frac{1}{n}} = \begin{cases} \tan(\pi/2p), & 1 < p \leq 2, \\ \cot(\pi/2p), & 2 \leq p < \infty. \end{cases}$$

Then, with $q_k = 2 - \frac{1}{2^k}$, we obtain (assume $A > 1$, otherwise replace $A^{\frac{1}{p}}$ with 1)

$$p_k = \frac{q_{k-1} q_k}{q_k - q_{k-1}} = 2^k \cdot \left(2 - \frac{1}{2^k}\right) \cdot \left(2 - \frac{1}{2^{k-1}}\right) \geq 2^k, \quad p_k^{-\frac{1}{p_k}} \leq 2^{-\frac{k}{2^k}},$$

$$\tan\left(\frac{\pi}{2q_k}\right) \approx 1 + 2\left(\frac{\pi}{2q_k} - \frac{\pi}{4}\right) = 1 + \frac{\pi}{2^{k+1}\left(2 - \frac{1}{2^k}\right)} \leq 1 + \frac{\pi}{2^{k+1}}.$$

and

$$\prod_{i=2}^{k} \|\phi_{\Delta}^{\boldsymbol{\alpha}}(-\cdot)\|_{L^{p_i}} \leq \prod_{i=2}^{k}\left(2^{-\frac{i}{2^i}}\right)^{\frac{n}{\nu}} A^{\frac{1}{2^i}} \leq \sqrt{A}, \quad 3 \leq k \leq N,$$
$$\prod_{k=3}^{N} C_{q_k}\|\phi_{\Delta}^{\boldsymbol{\alpha}}(-\cdot)\|_{L^{p_k}} \leq \prod_{k=3}^{N}\left(2^{-\frac{k}{2^k}}\right)^{\frac{n}{\nu}} \cdot A^{\frac{1}{2^k}} \cdot \left(1 + \frac{\pi}{2^{k+1}}\right)^n \leq A^{\frac{1}{4}} e^n.$$

Thus $C_k \sim O\left(\frac{1}{2^{n(k-2)}}\right)$ and $C'_N \sim O(1)$. □

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